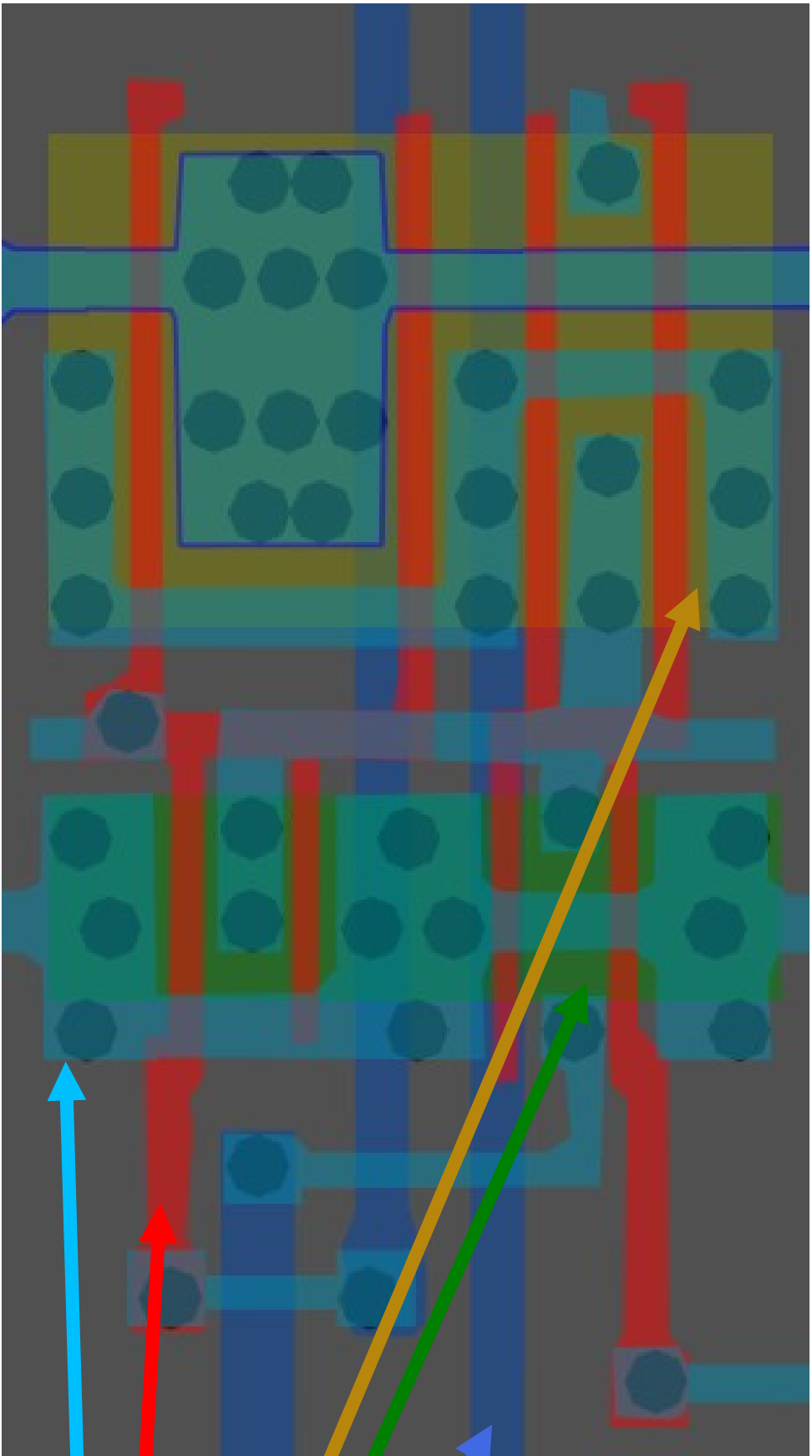
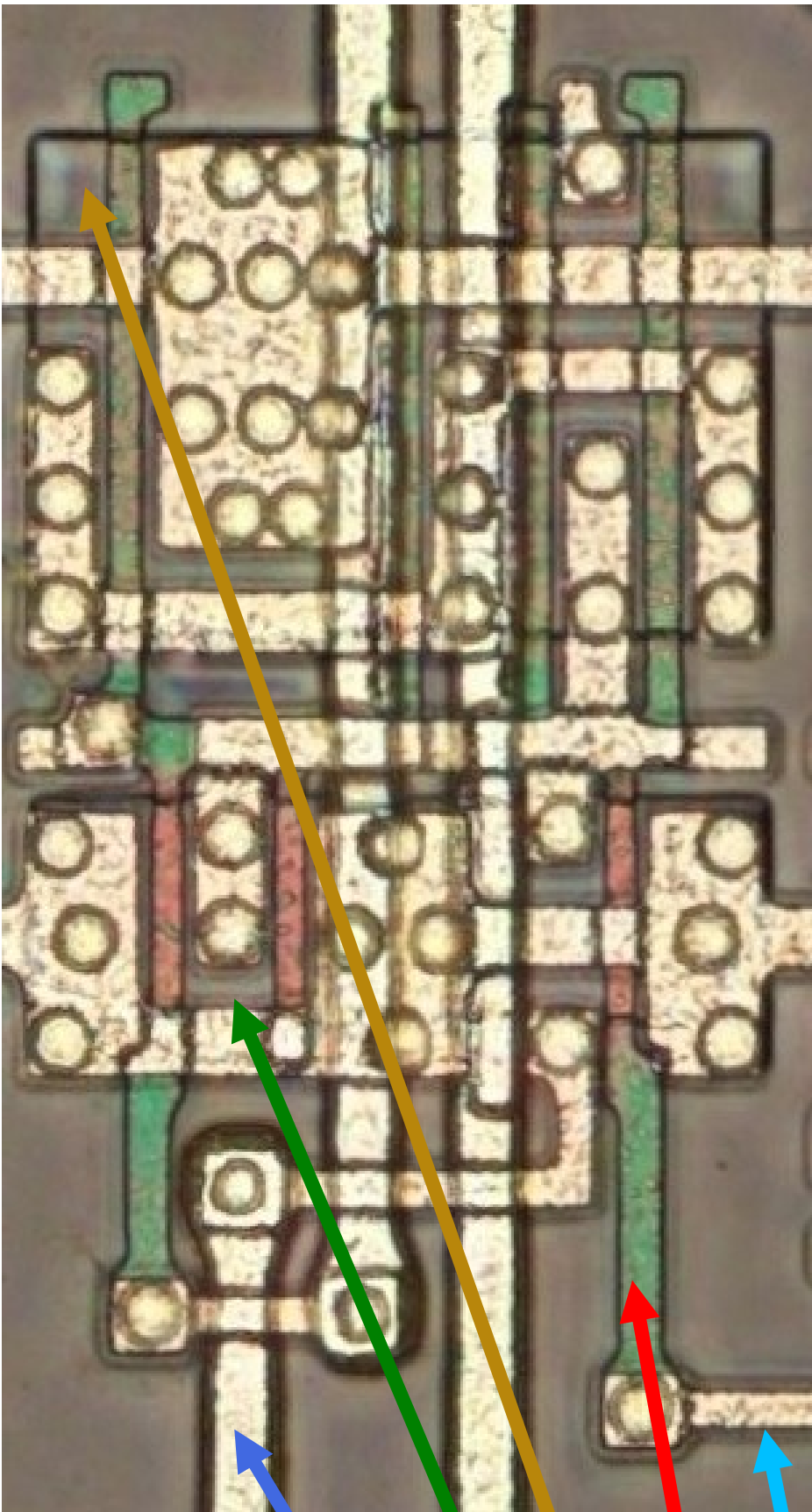


Document subject to change (for now)



Metal1

polysilicon

p-diffuse

n-diffuse

metal2

Meet the family..

Roland

- MB87136a (LA32 synth engine/DSP)
- MB87126a (Reverb DSP chip, fujitsu standard cells)
- MB87419 (DPCM decoder called LP-1)
- MB87420 (DPCM decoder called LP-2)(Normally paired with MB87419)
- MB87424a(TVF filter chip)

Inferred rules for this 1986/87, 1.8 micron standard cell process called 'AV-OEBB'

(It is entirely possible the
node size was something
other than 1.8 micron)

AV-OEBB Series

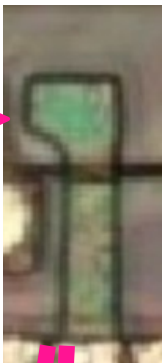
1.8u CMOS standard cell

Fujitsu Micro-electronics	AV-OEBB Si-gate CMOS 1.8 μm	P1 6 μm M1 7 μm M2 9 μm	100	n/s	1.2	140 digital	RAM, ROM, PLA, multipliers, ALU	•	•	•	•	•	None
---------------------------	--	--	-----	-----	-----	-------------	------------------------------------	---	---	---	---	---	------

- Cells can be flipped and mirrored with minimal adjustments.
- Gate height can extend up/down y-axis for the purposes of routing.
- Cell via routing seems unique to each cell type. Good for finding them easily
- These polysilicon ends are where m1 can be connected

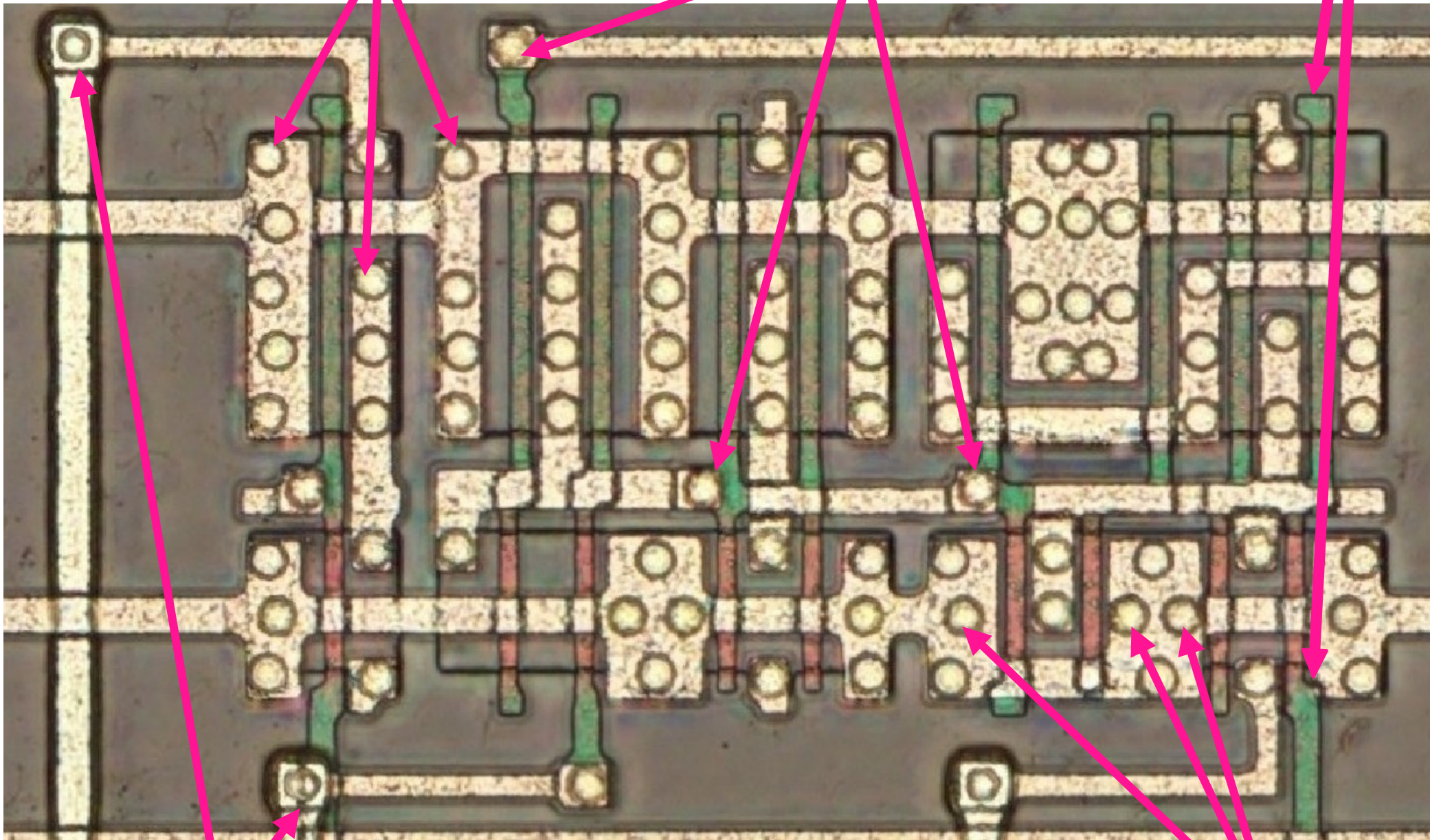
Tip: To understand whether the diffuse is PMOS or NMOS, you need to identify the polysilicon gate and its color:

The gate is green in PMOS, but red in NMOS.



m1 to P-MOS diffuse
via:

m1 to polysilicon
via:

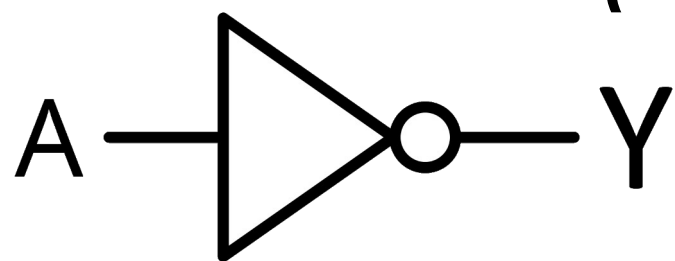


m1 to m2 via:
Generally has smaller vias

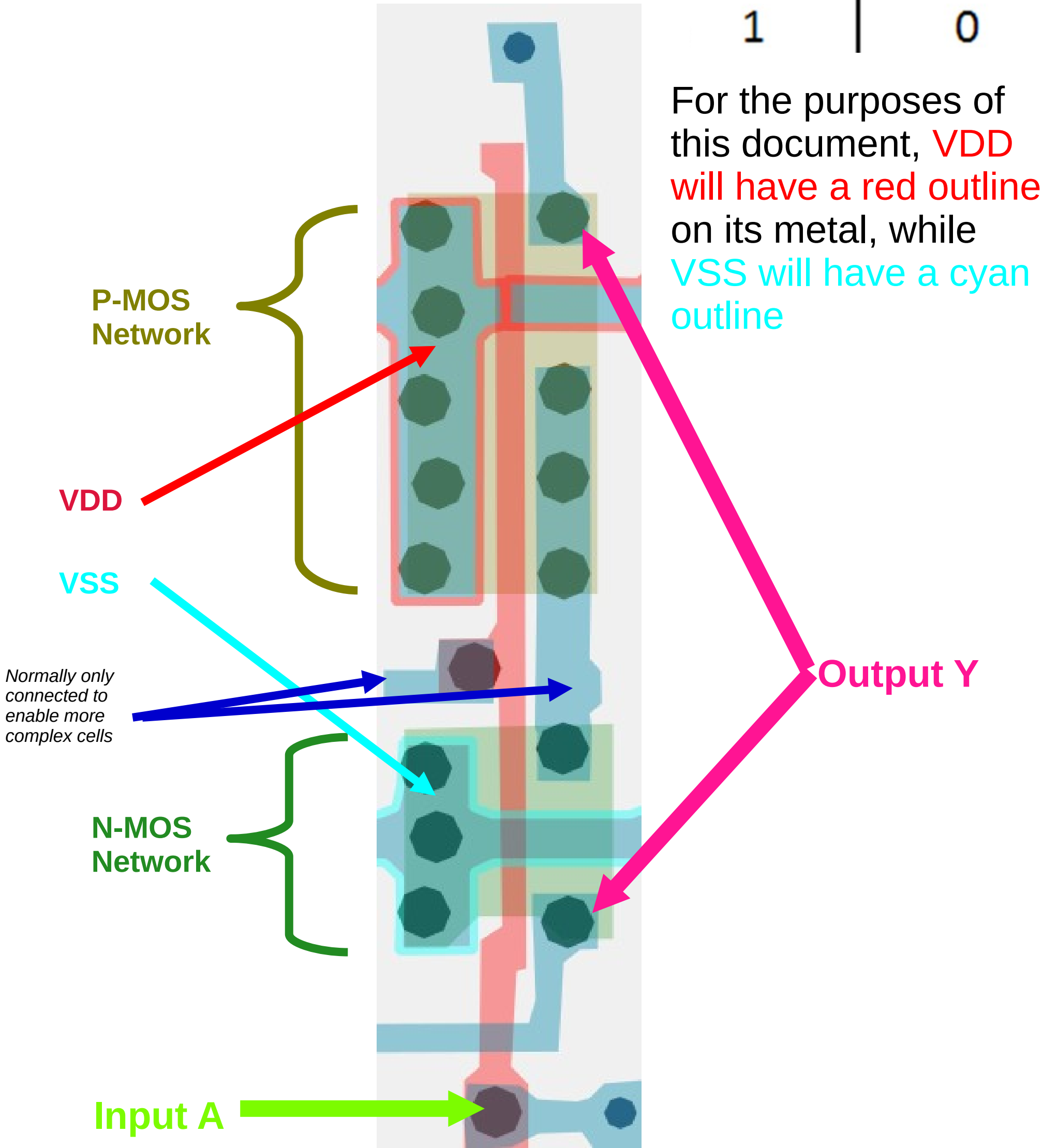
m1 to N-MOS
diffuse via:

How it all works

(Inverter/NOT example)



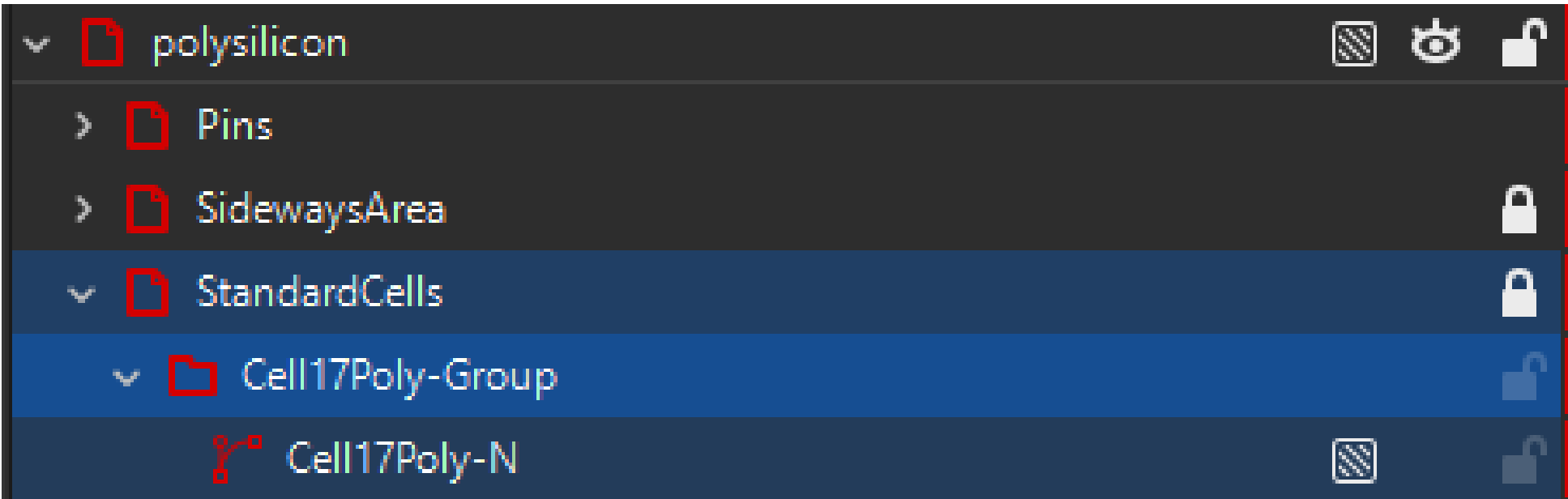
Input	Output
A	Y
0	1
1	0



Object naming and how to understand

I have done my best to make object naming as simple as possible while keeping object names practical.

- PDi – P-Diffusion
- NDi – N-Diffusion
- Poly – Polysilicon



Cell 17 in standard cell area, Polysilicon layer, sub-object N.

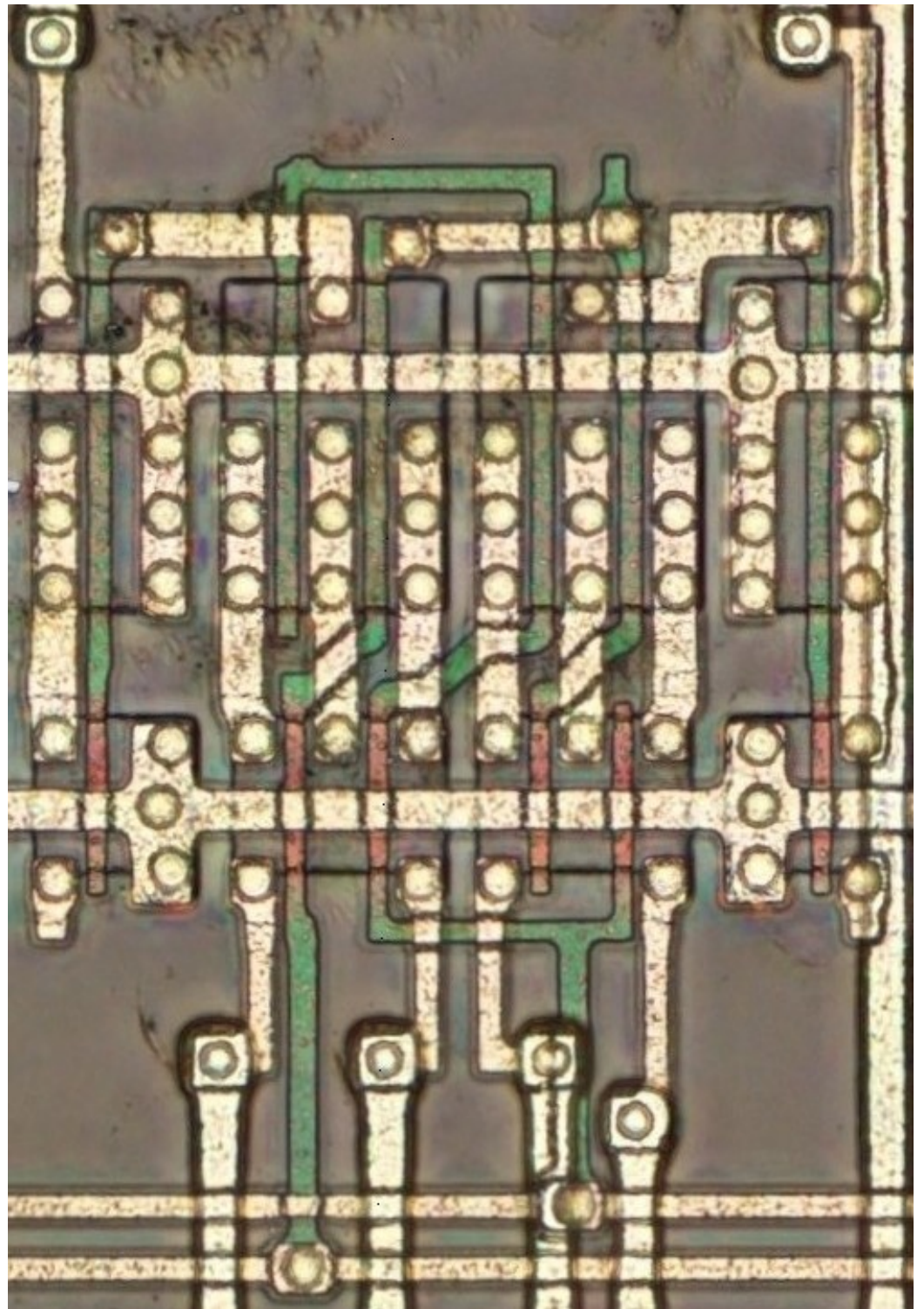
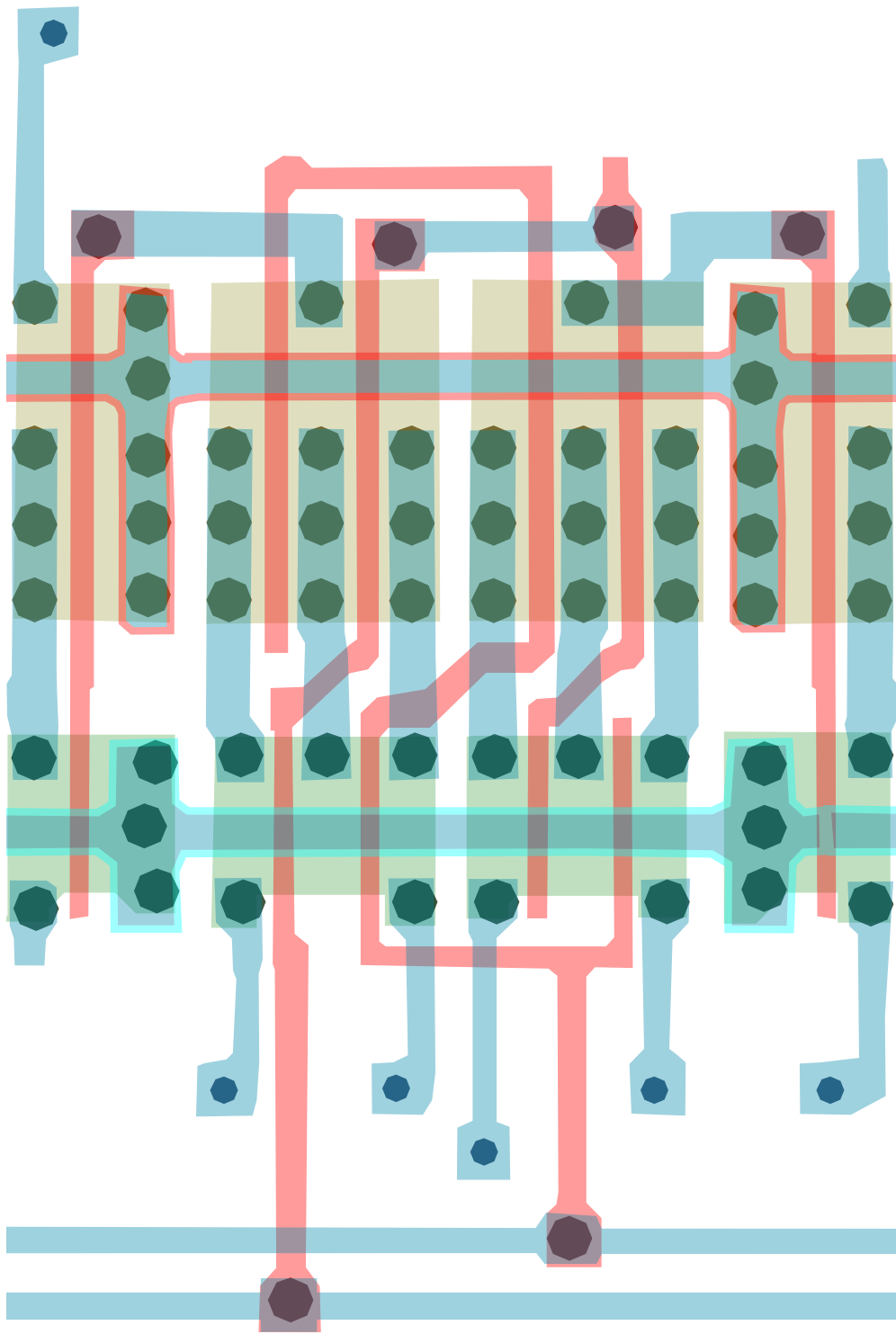
Cell number index

Green shaded cells mean they have been verified

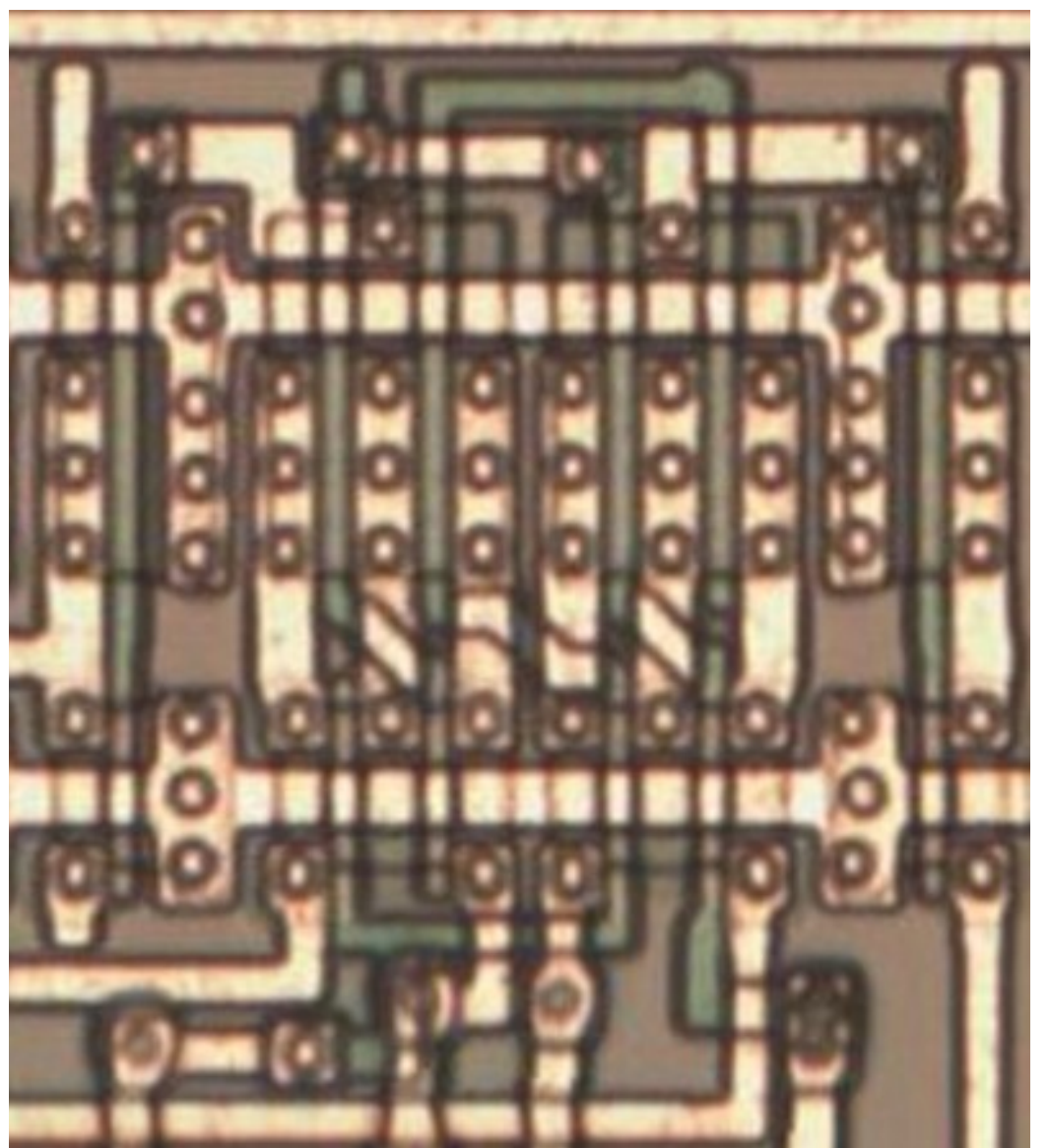
1	2	3a	3b	4	5	6
7	8	9	10	11	12	13a
13b	13c	13d	13e	13f	14	15
16	17	18	19	20	21a	21b
24a	24b	25	26	27a	27b	27c
27d	28	29	30	31	32	33a
33b	34	35	36a	36b		

#f321a5ff

Cell type 1

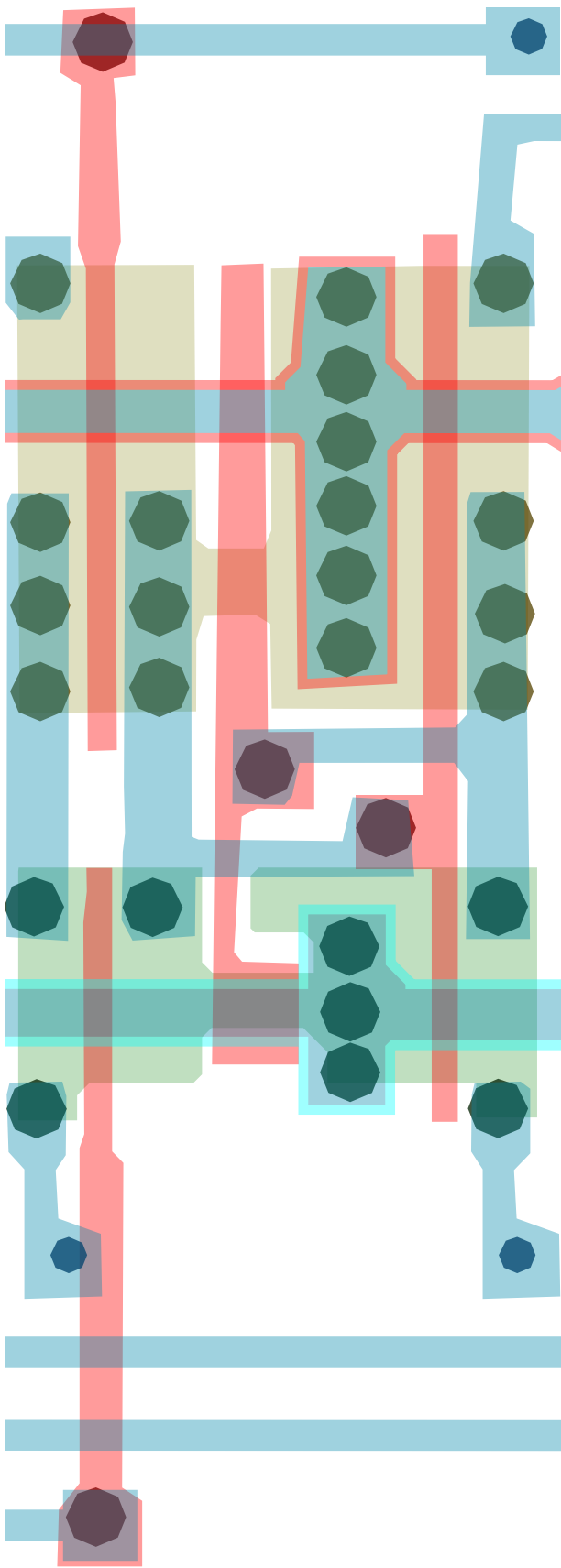


Same cell on
LA32
(mirrored)
(MB87136A)

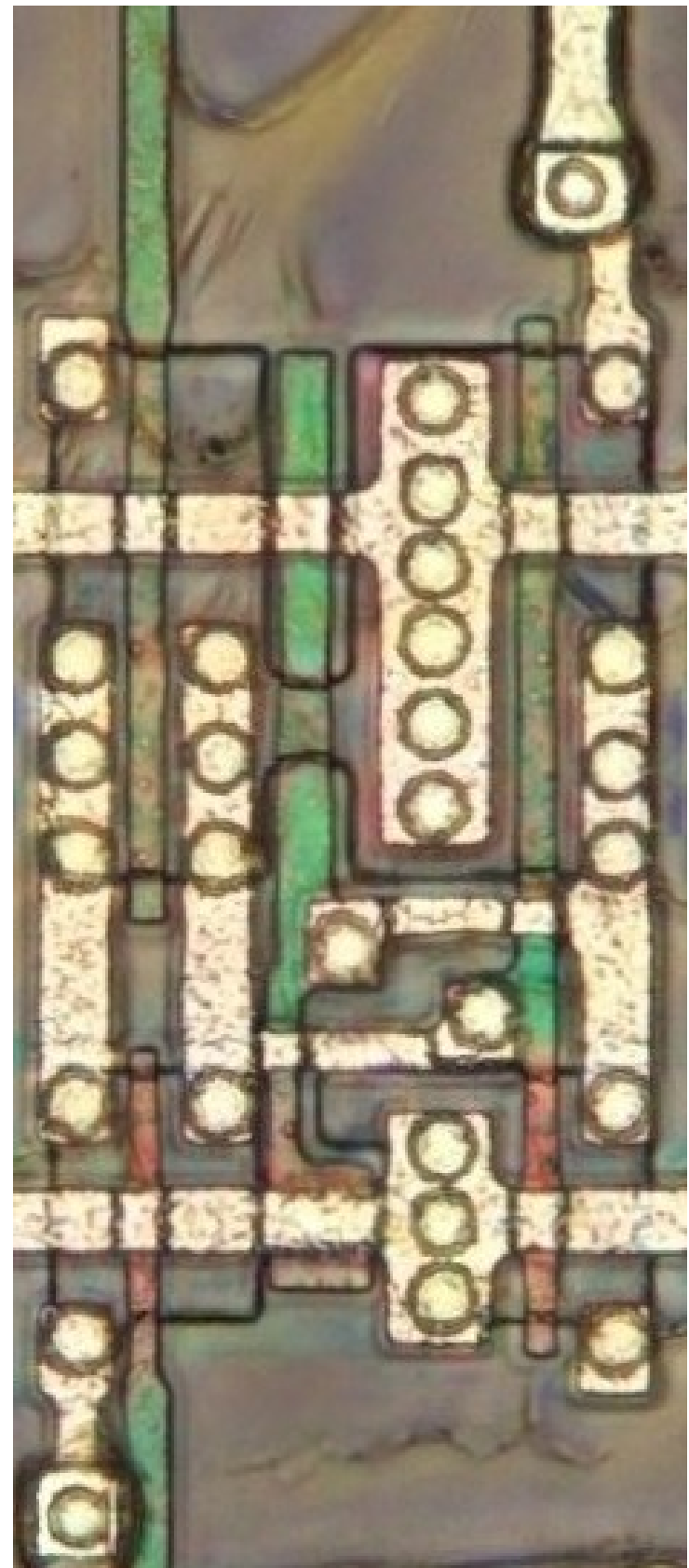
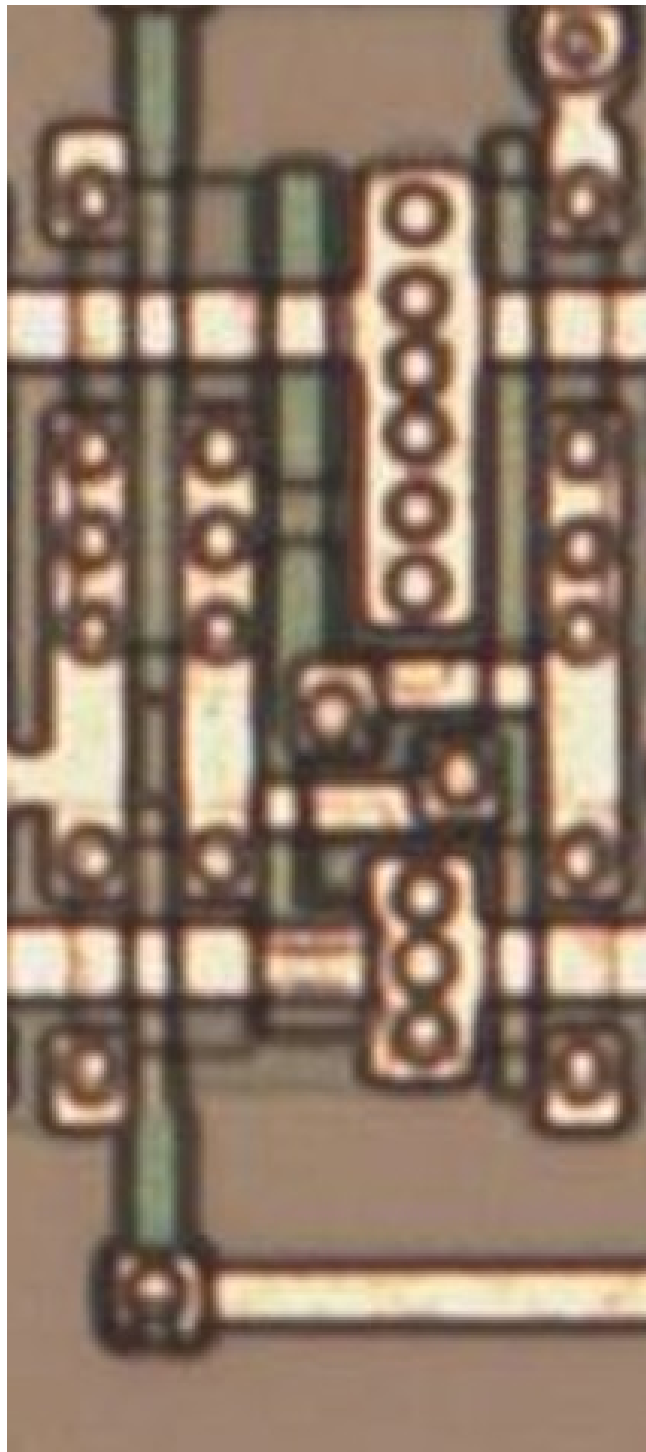


#f34f00ff

Cell 2



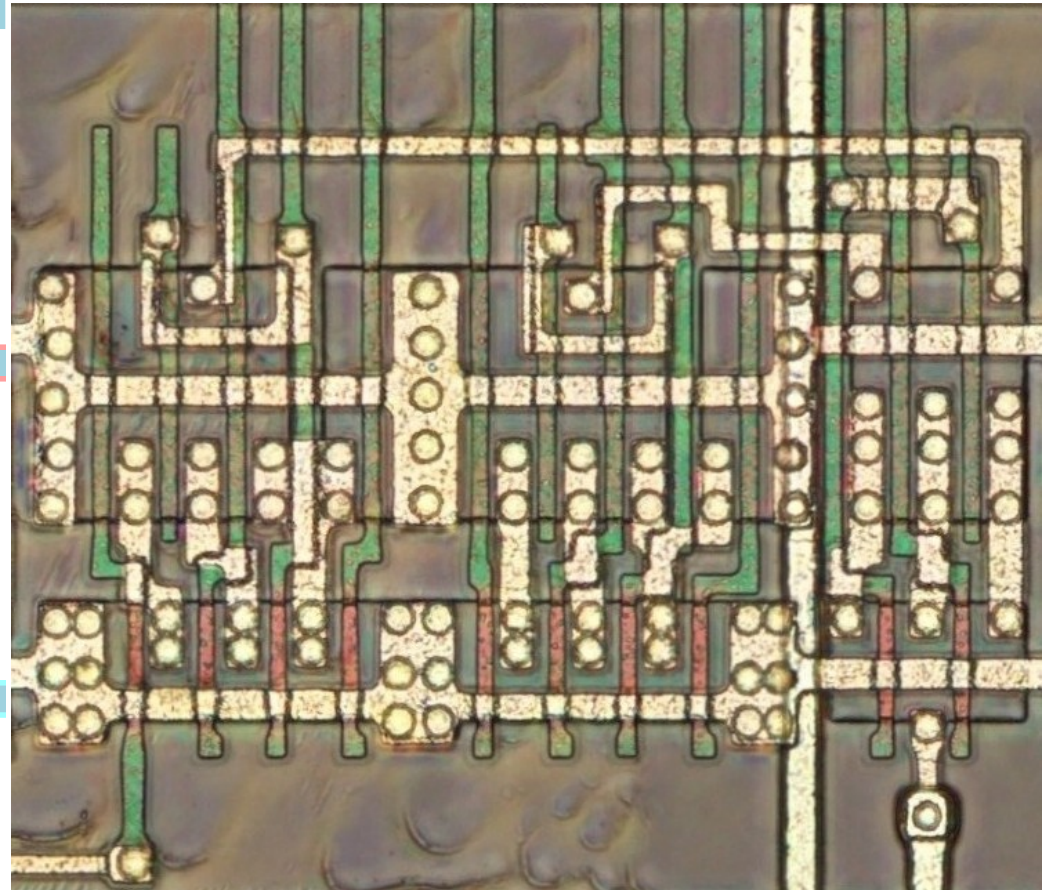
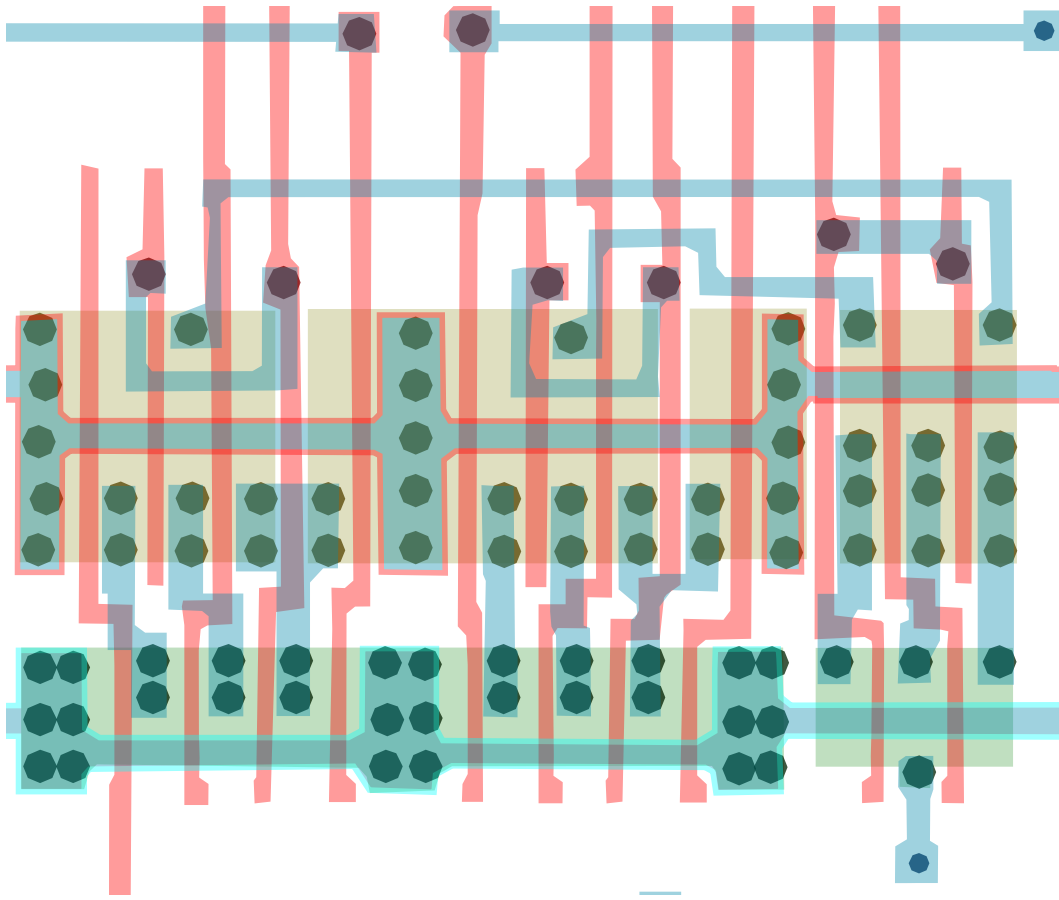
Maybe an SR latch?



Same cell on
LA32
(MB87136A)

#ffff00ff

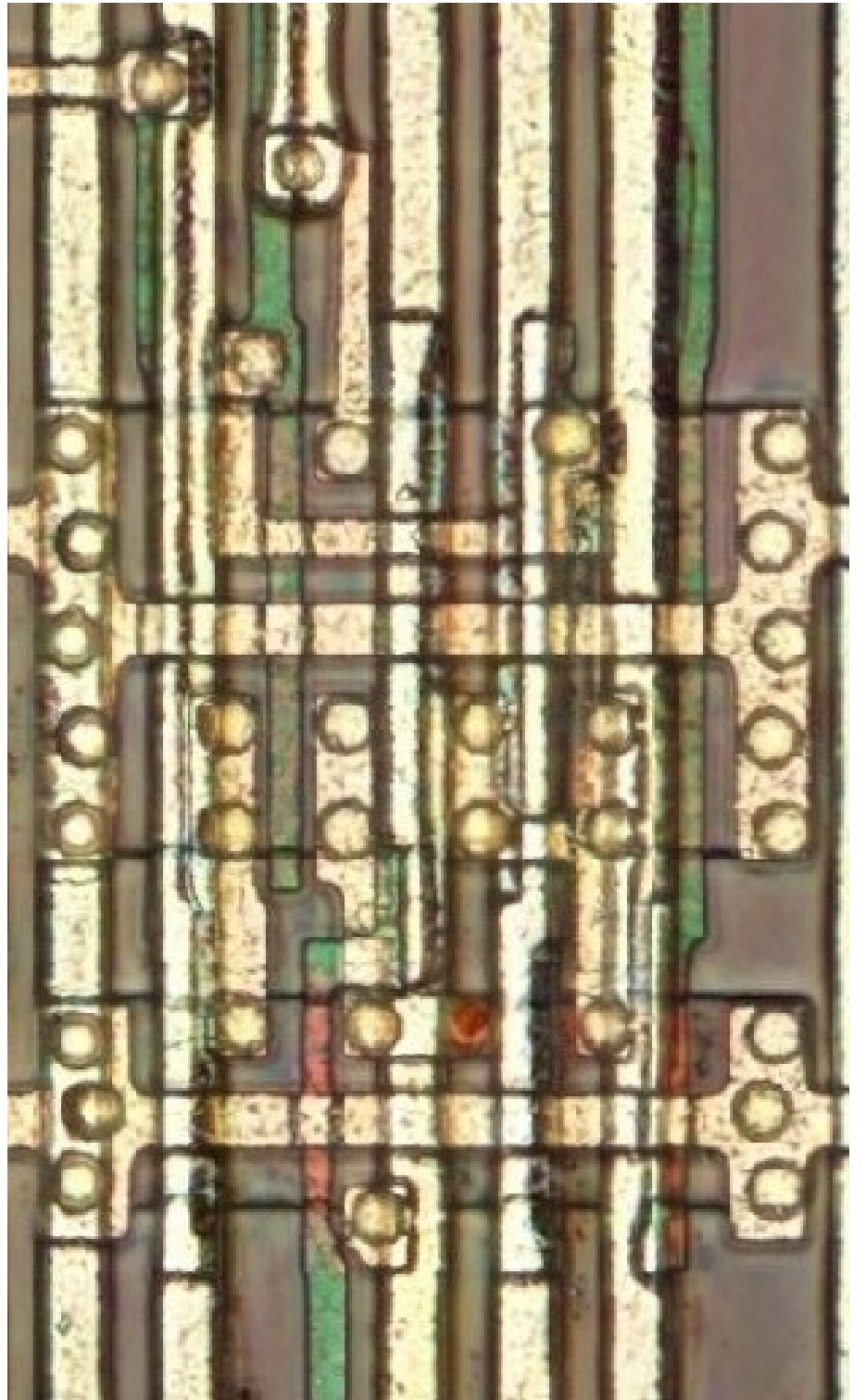
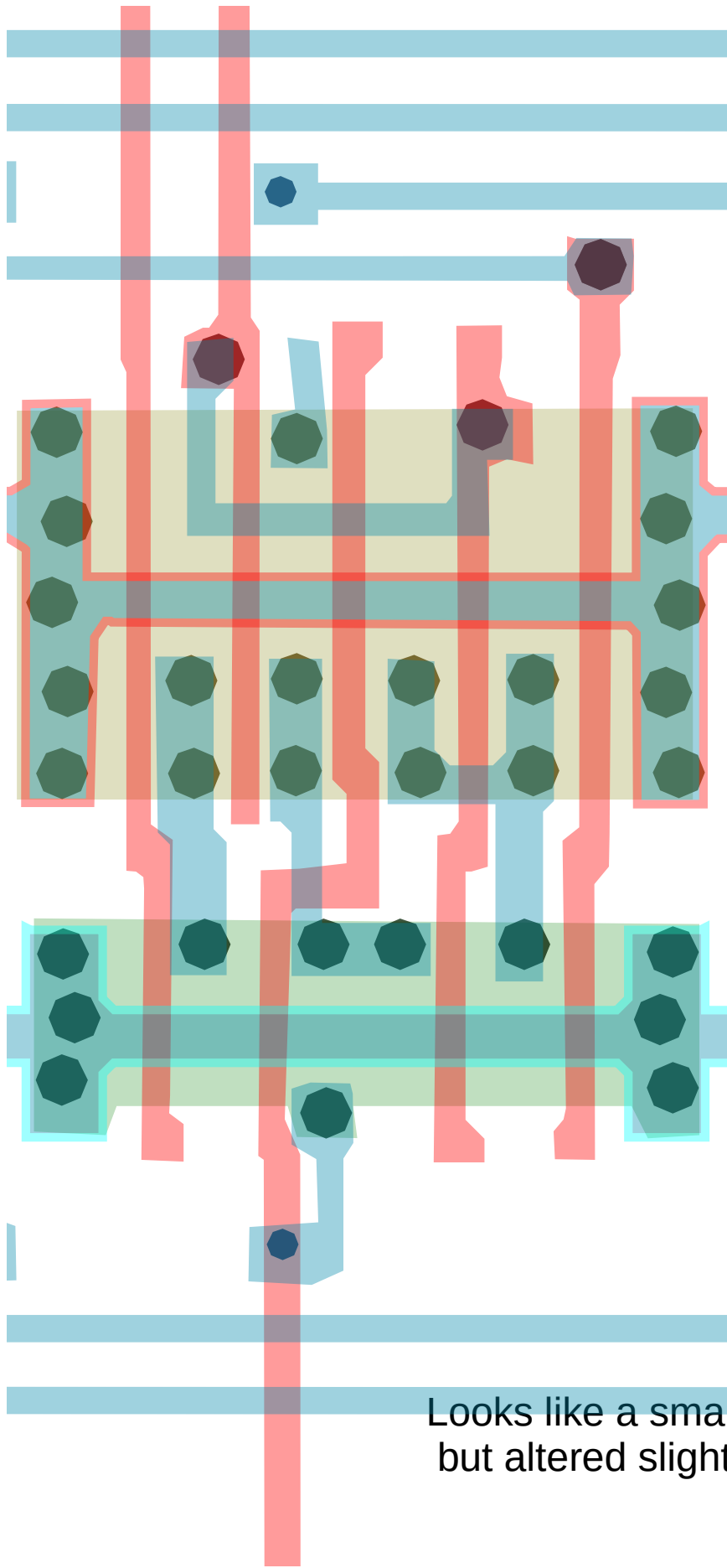
Cell type 3a



Unable to find on LA32

#b29e31ff

Cell type 3b

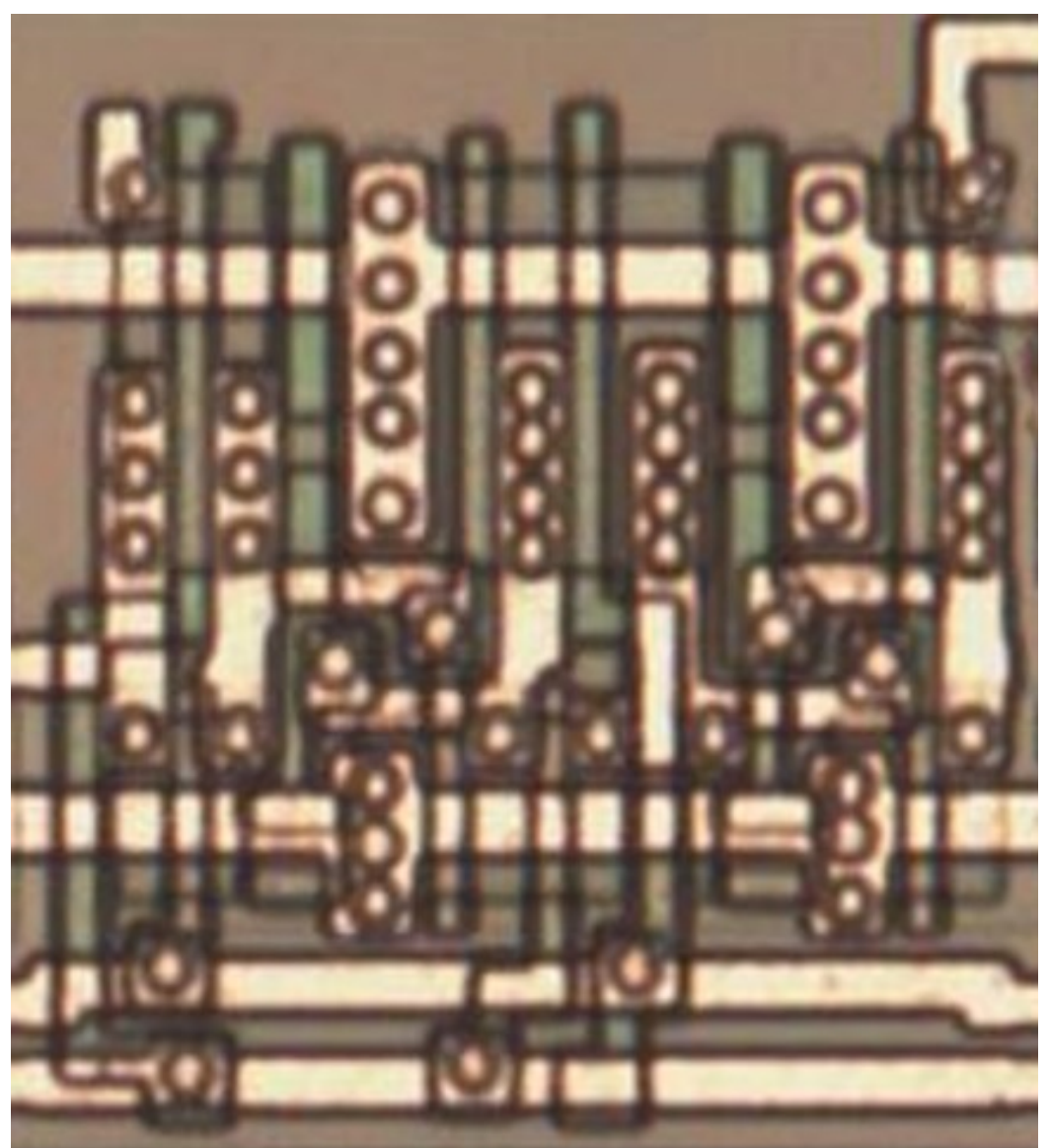
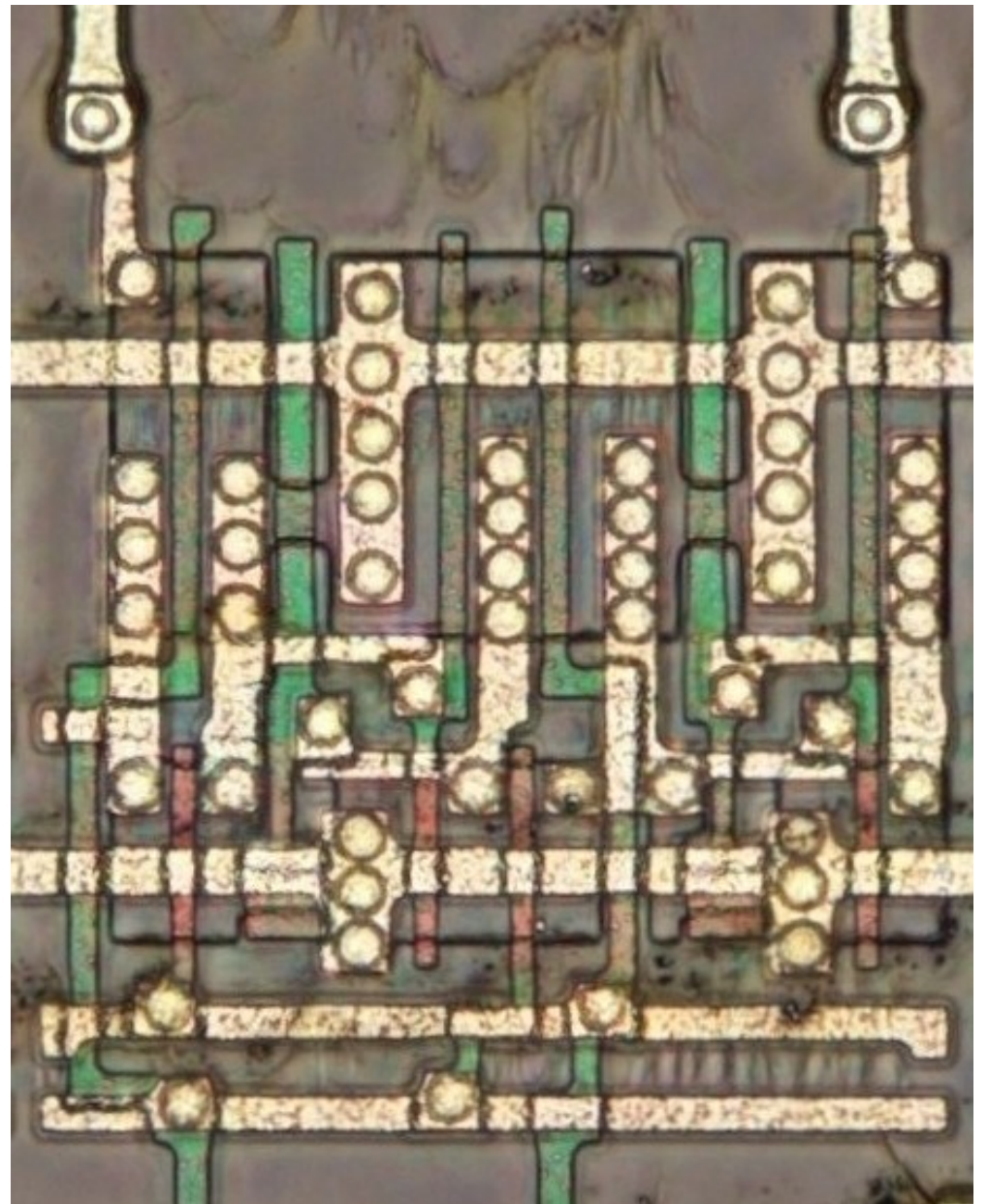
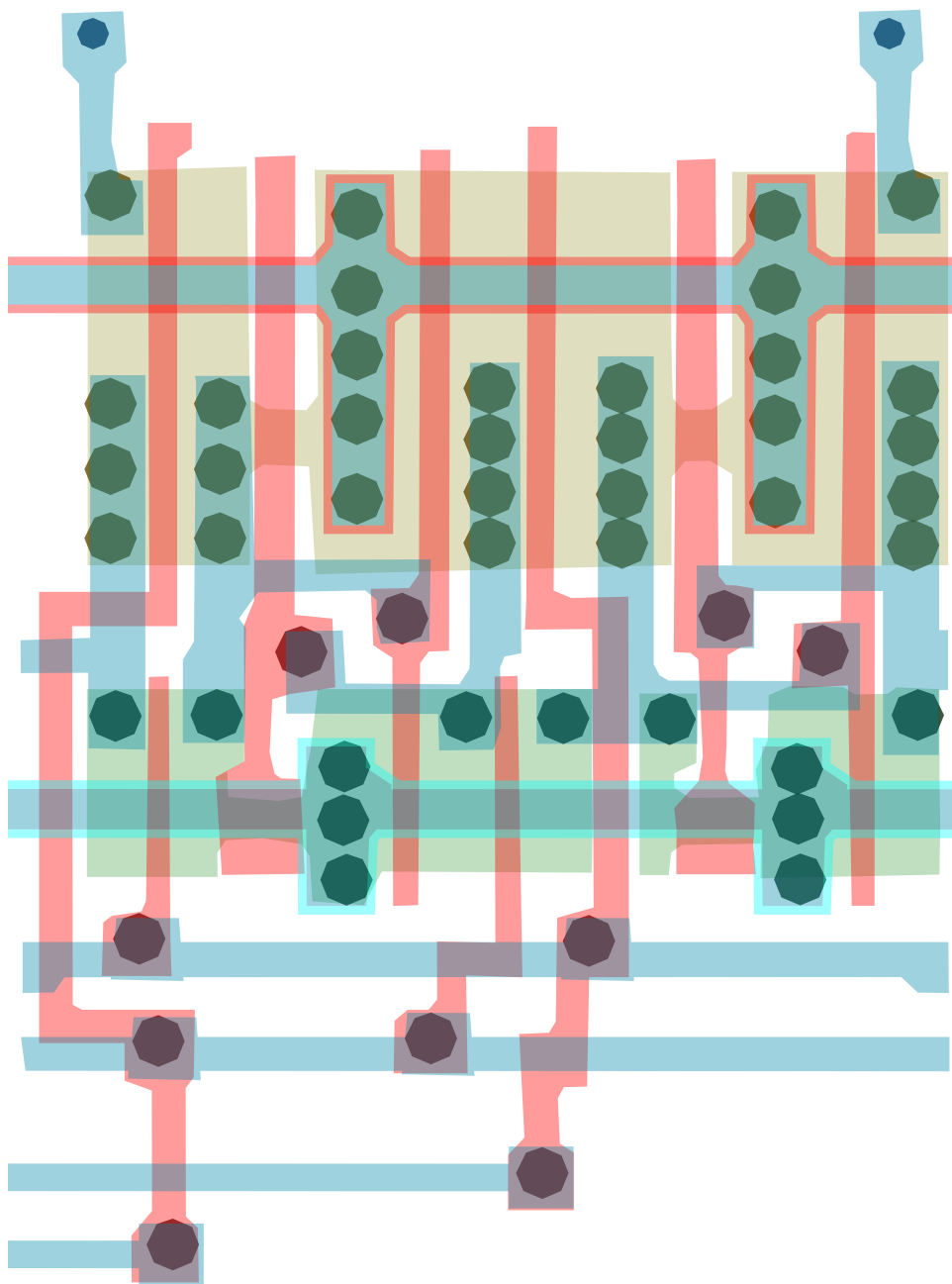


Looks like a small segment of 3a,
but altered slightly

#00ff00ff

Cell 4

Double or 'x2' drive
version of Cell 2

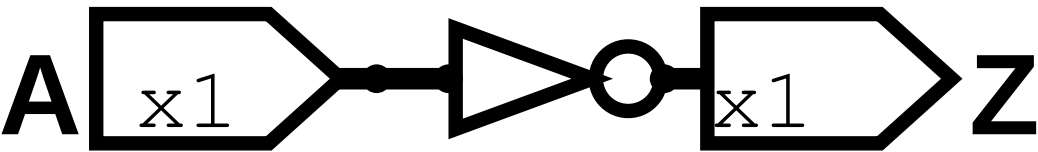


Same cell on
LA32
(MB87136A)

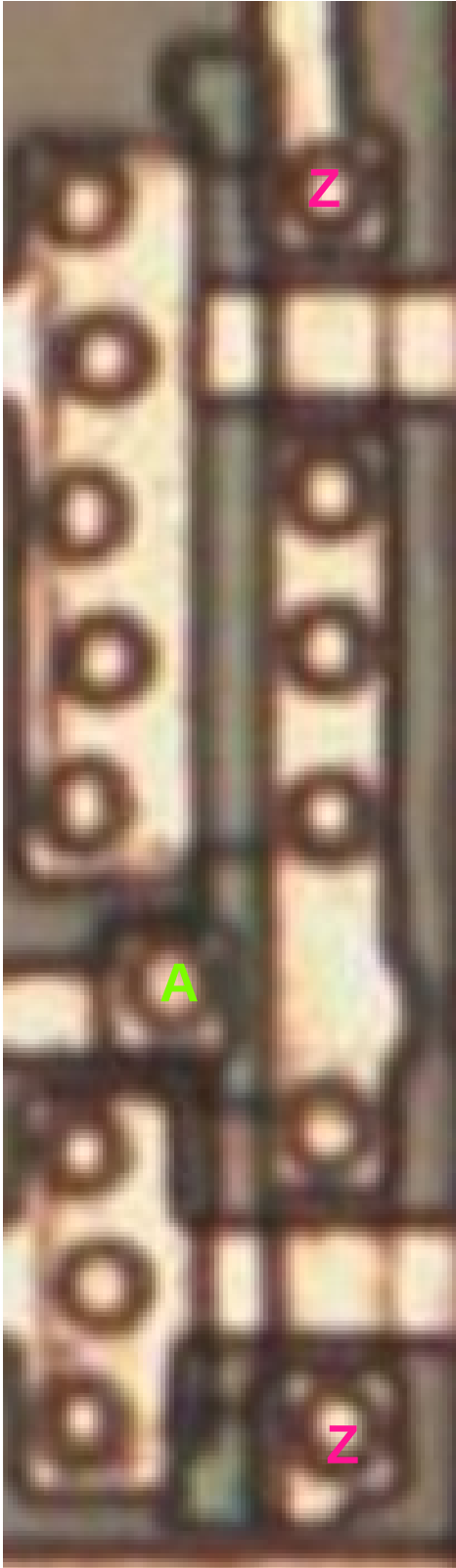
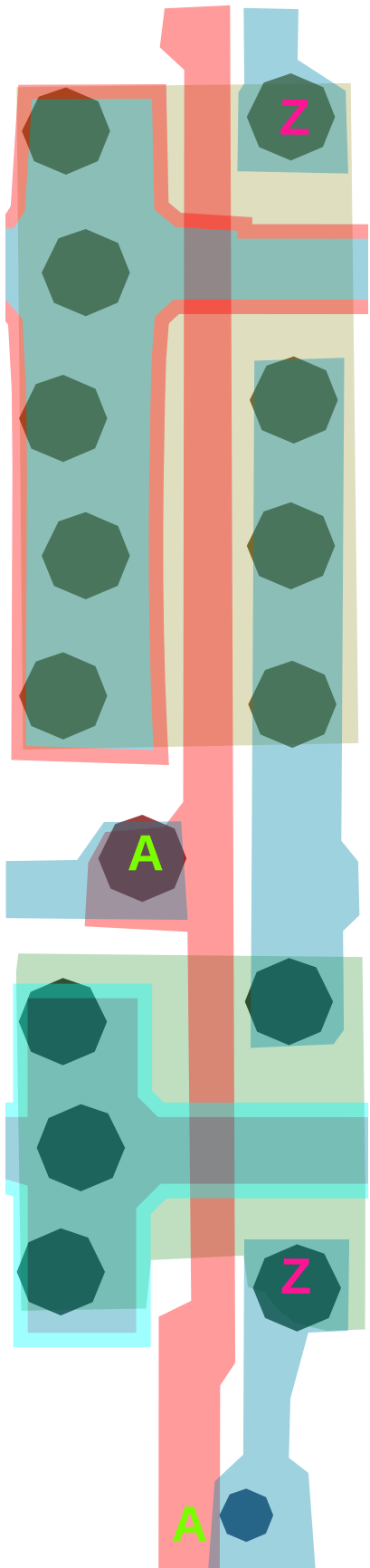
#51215aff

Inverter/NOT x1

Cell 5



A	Z
0	1
1	0

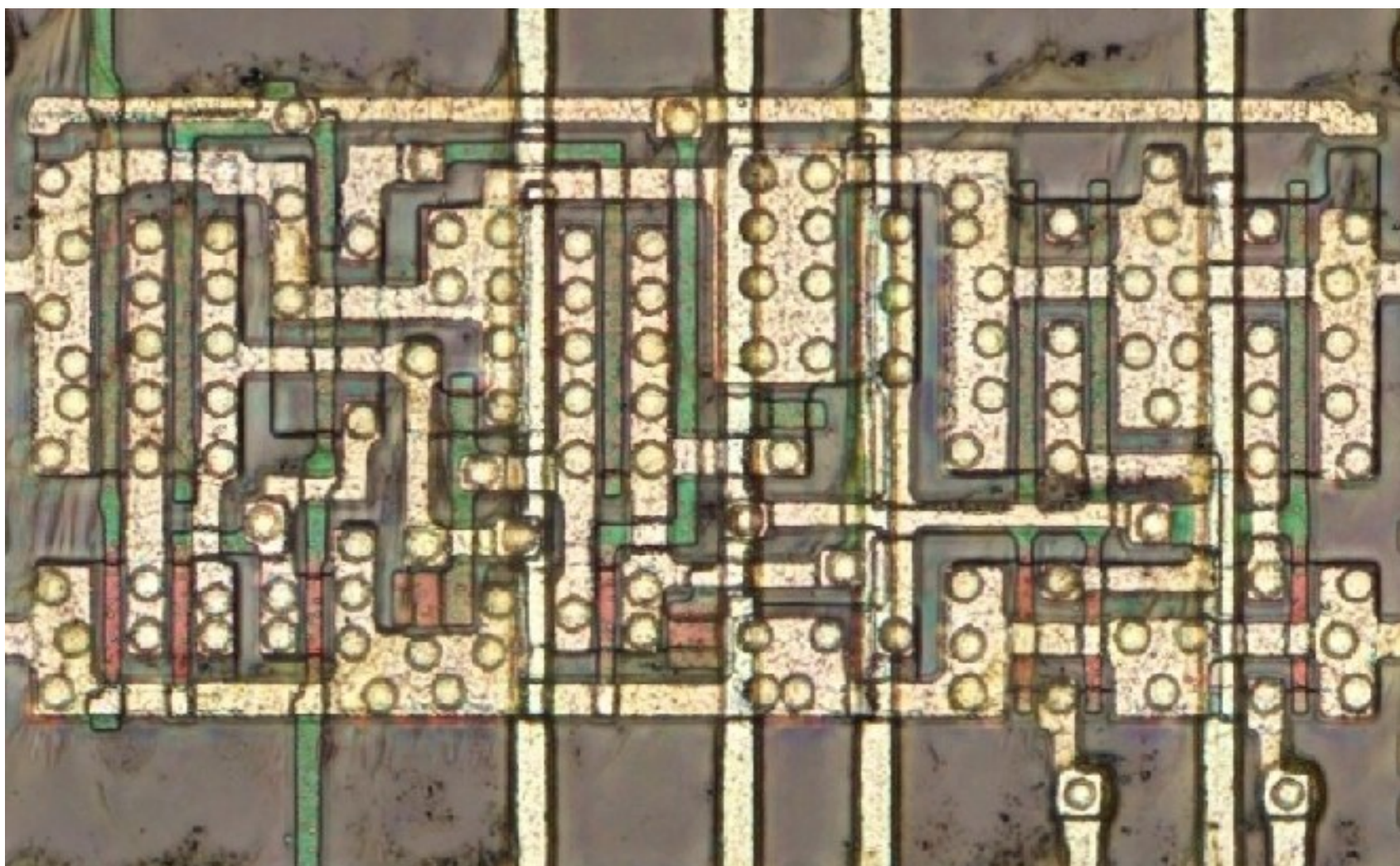
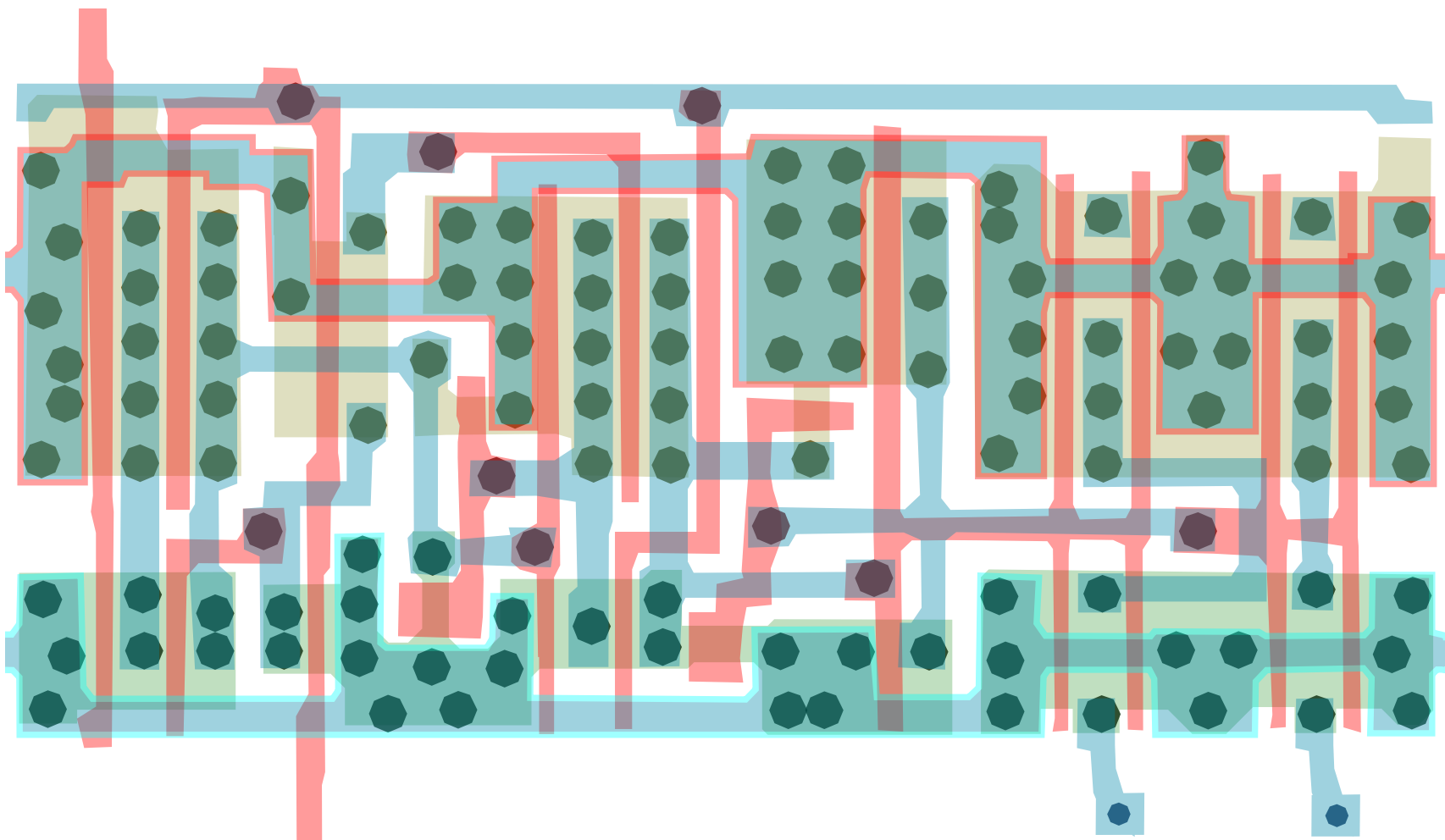


Same cell on
LA32
(MB87136A)



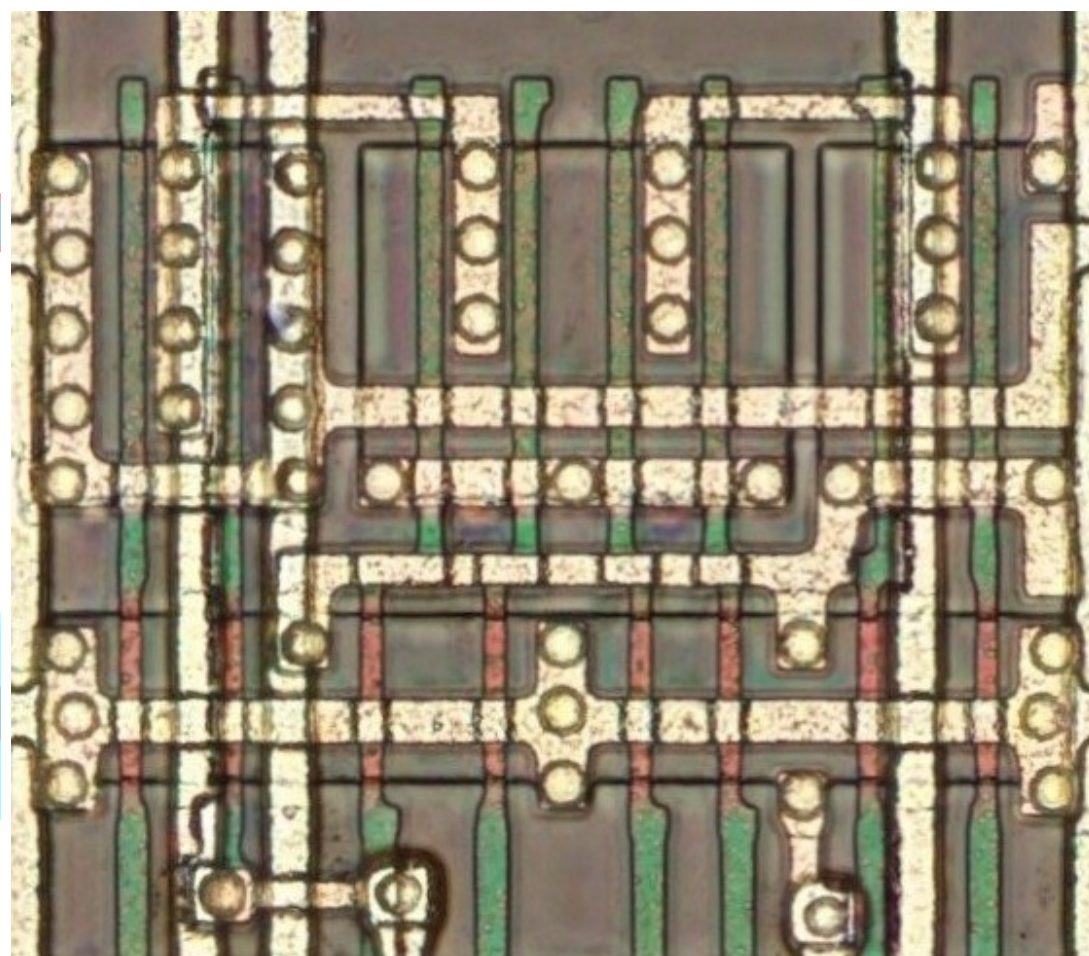
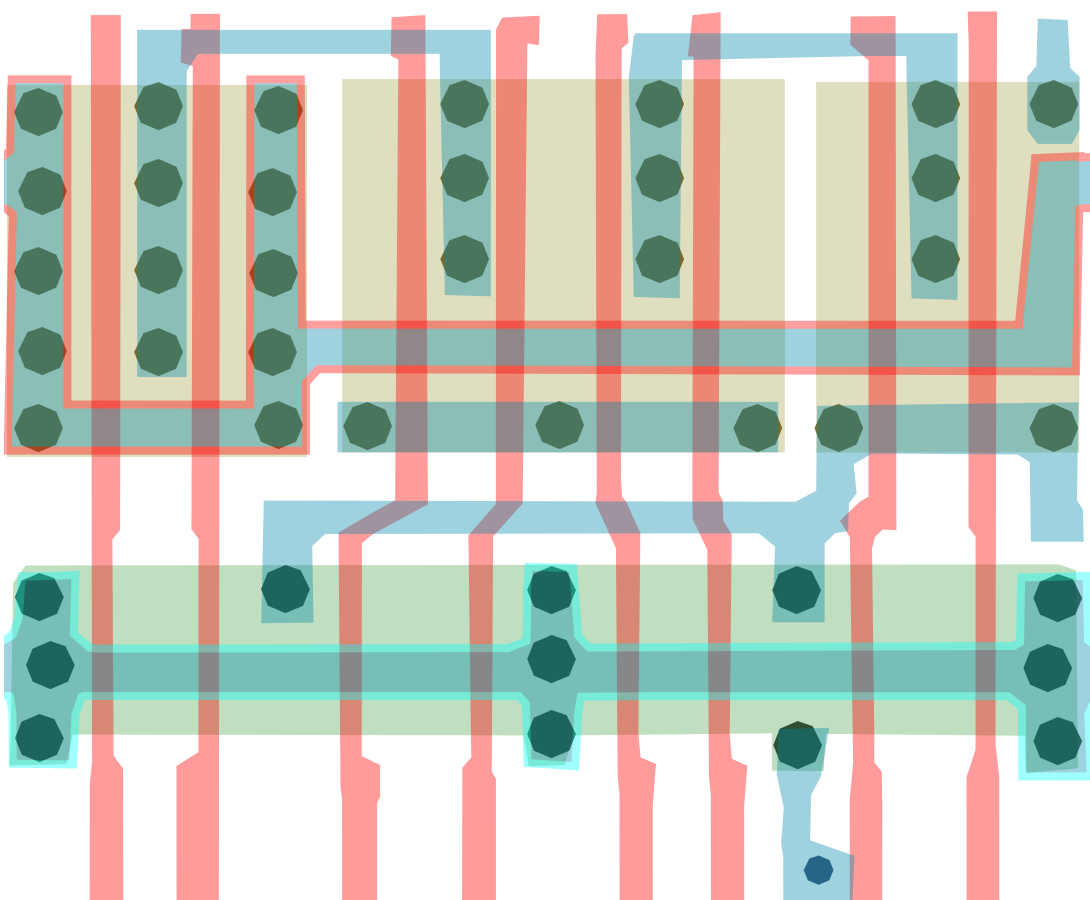
#955f00ff

Cell type 6

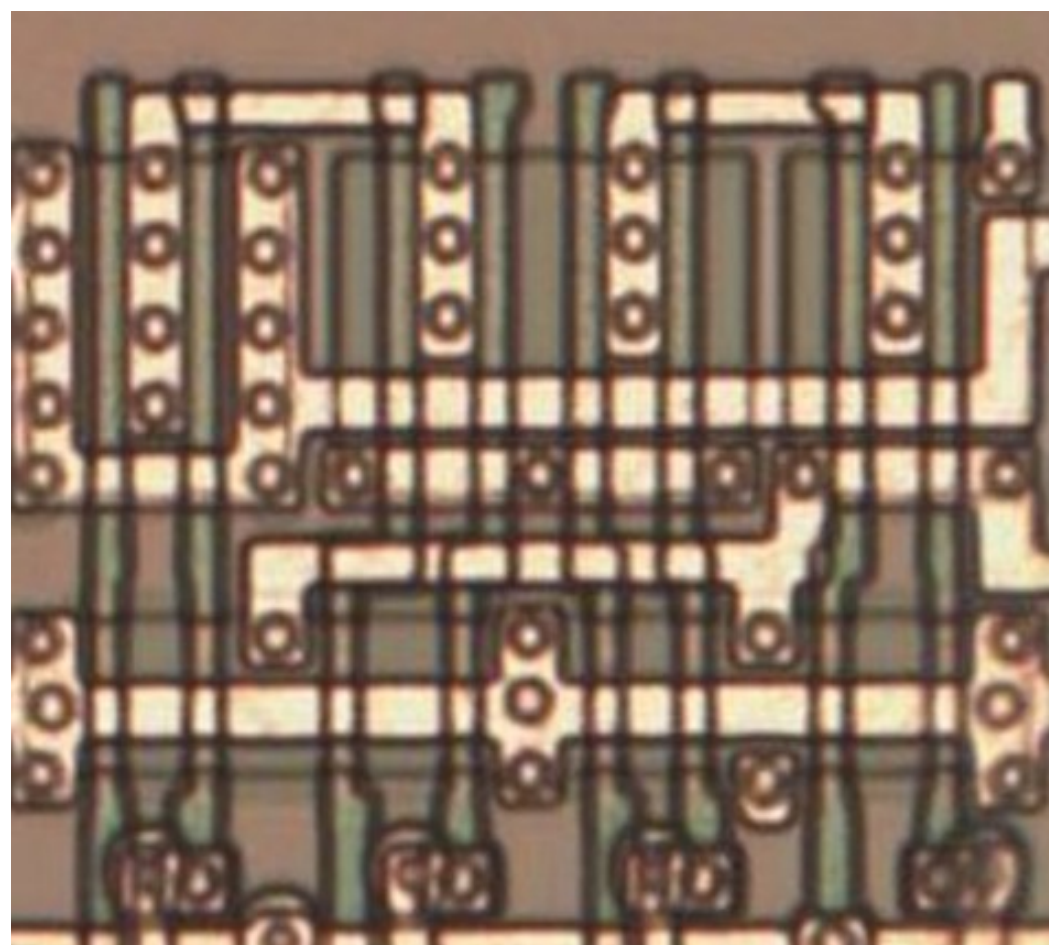


#13215aff

Cell type 7



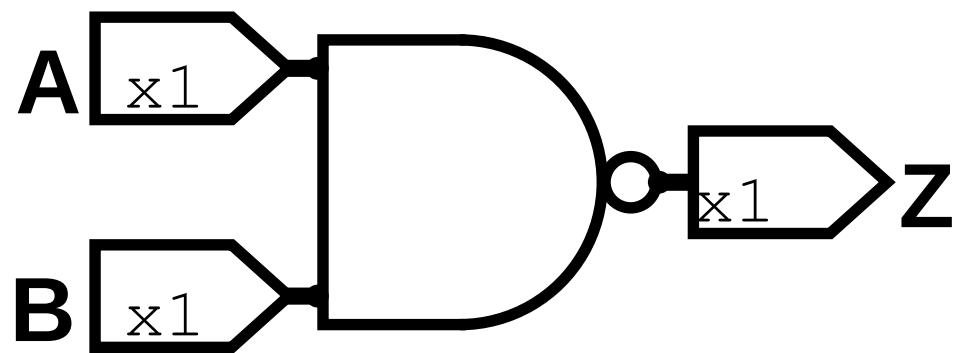
Same cell on
LA32
(MB87136A)



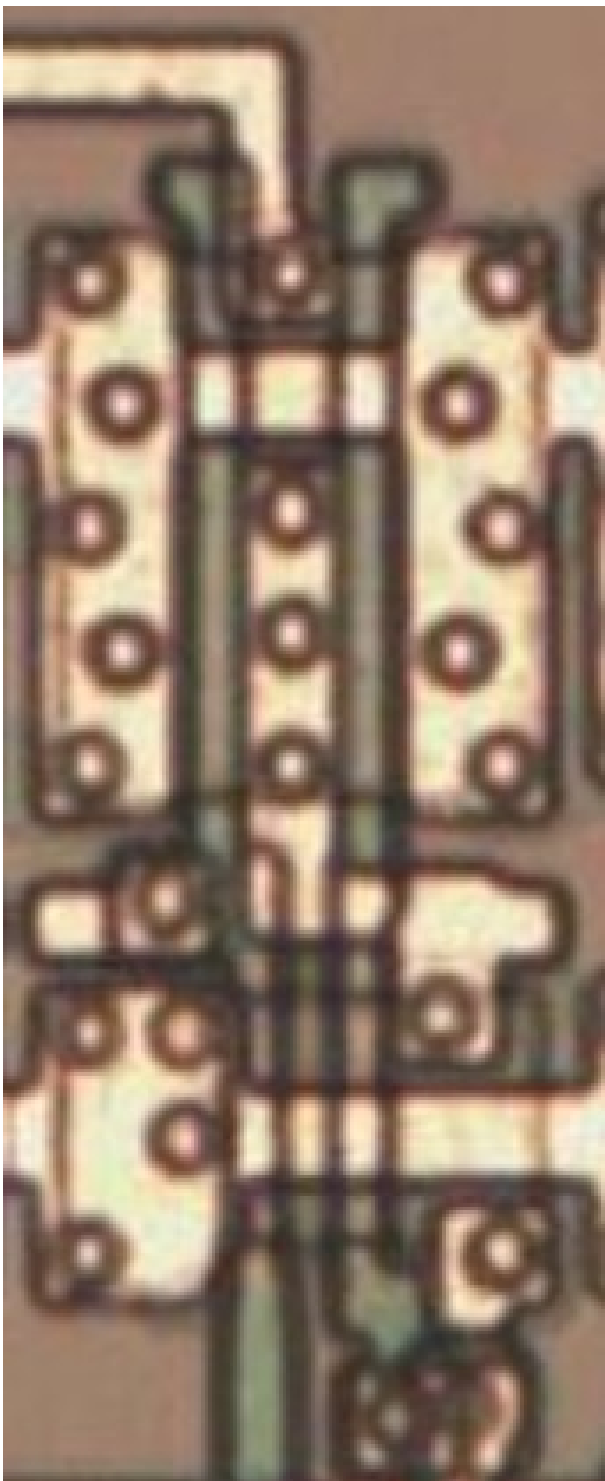
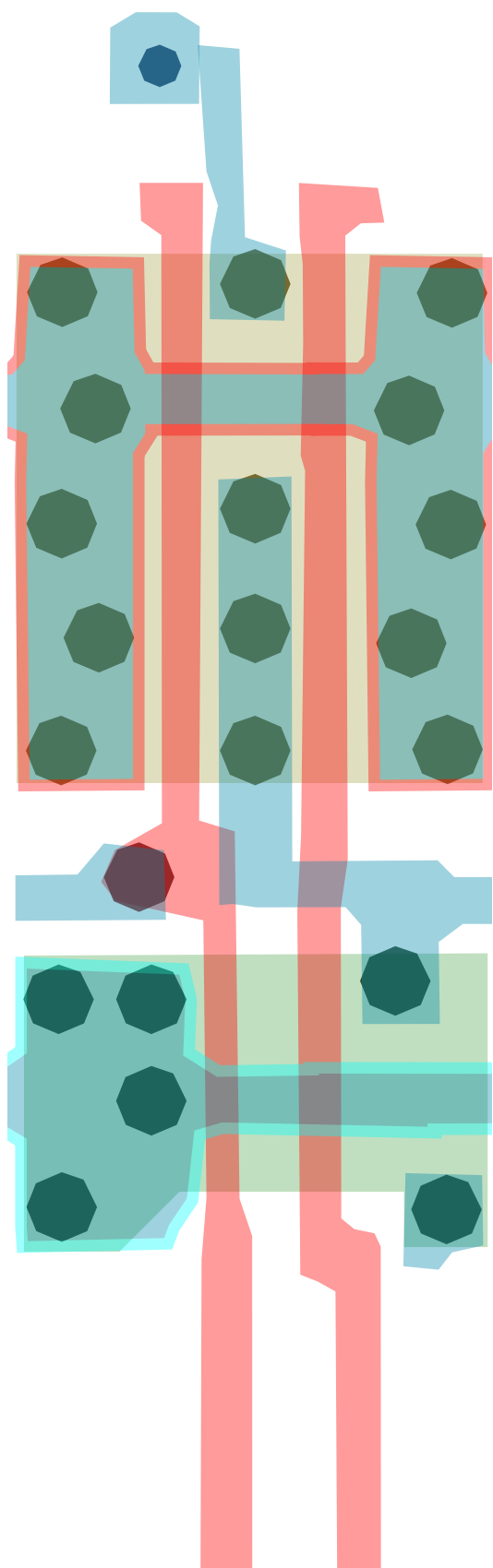
#89ff86ff

2-Input NAND

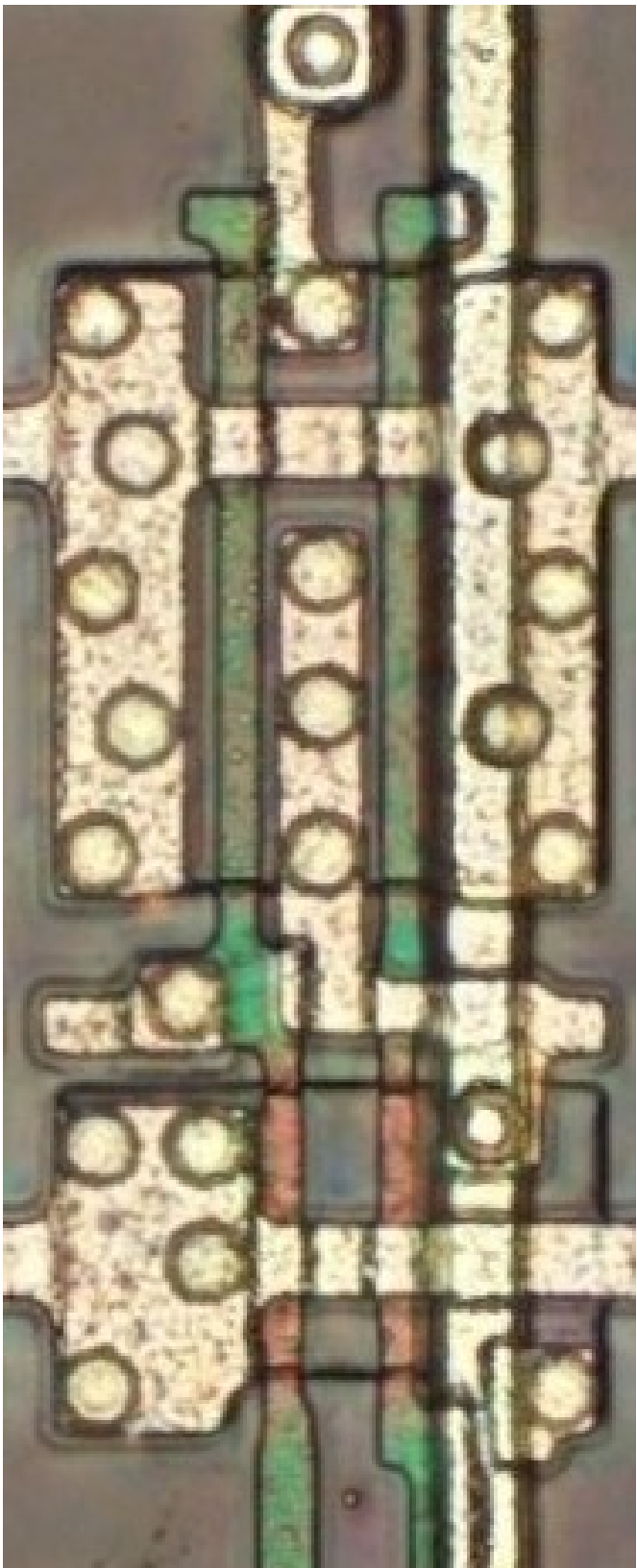
Cell 8



A	B	X
0	0	1
0	1	1
1	0	1
1	1	0



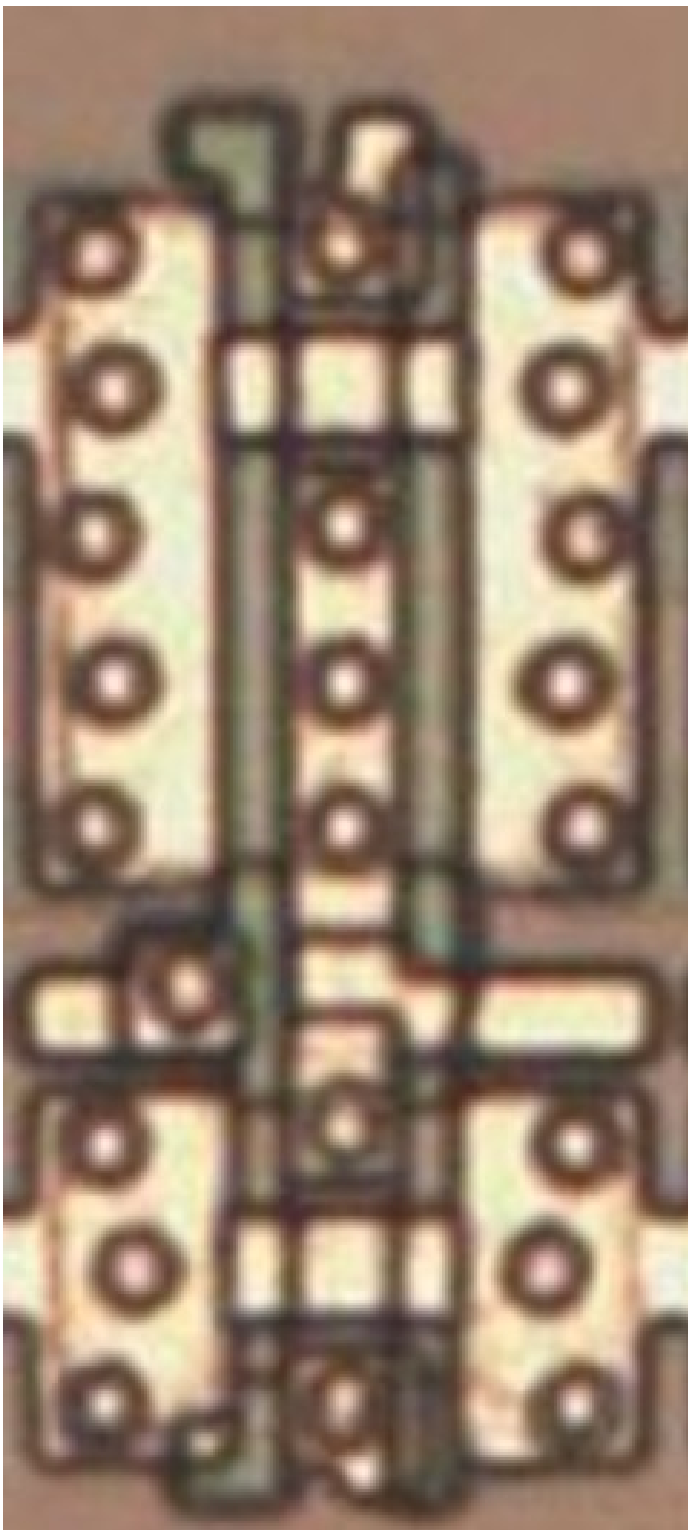
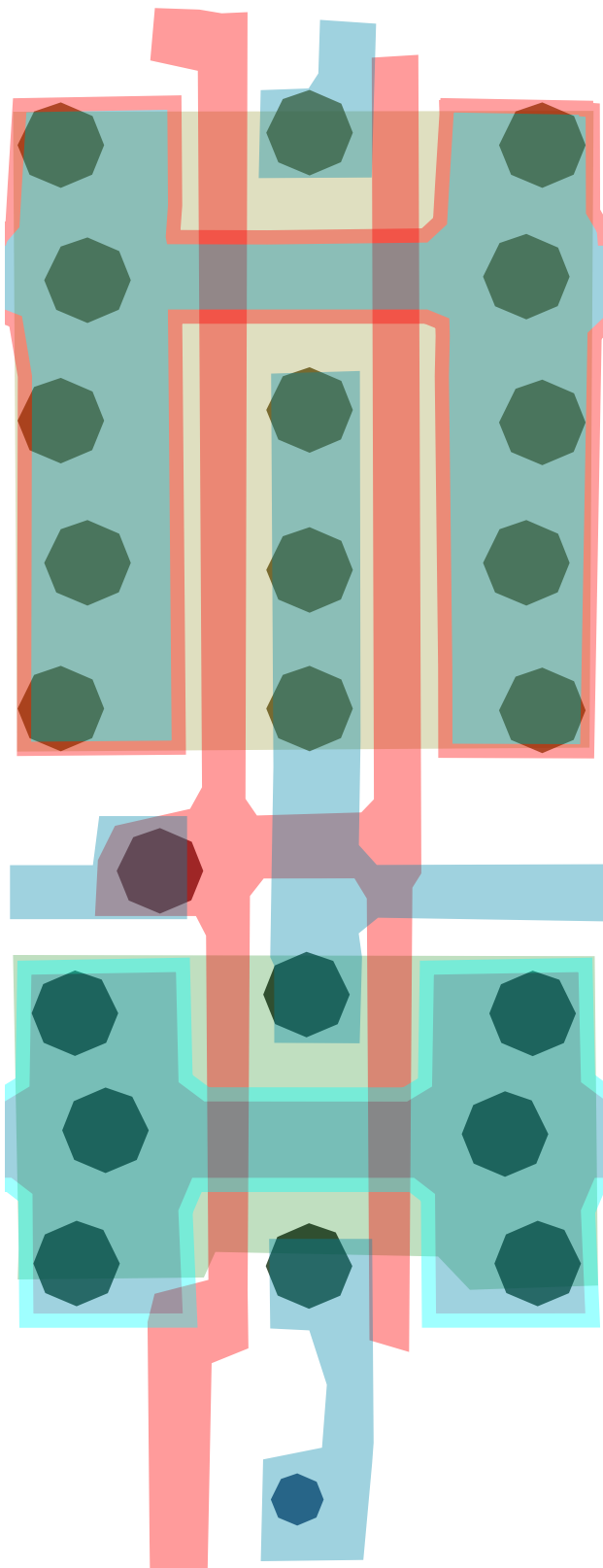
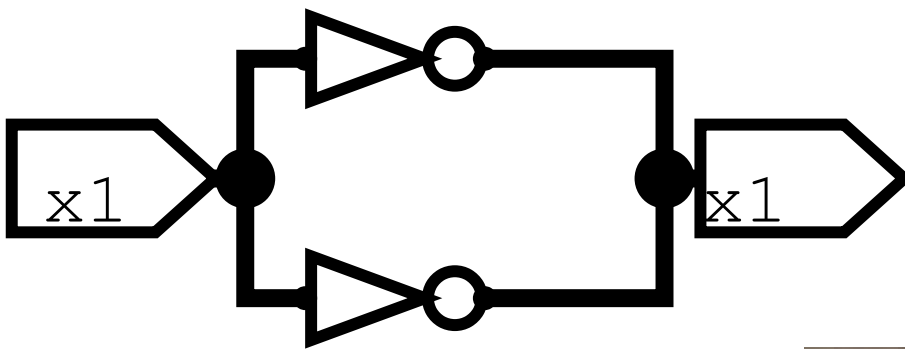
Same cell on
LA32
(MB87136A)



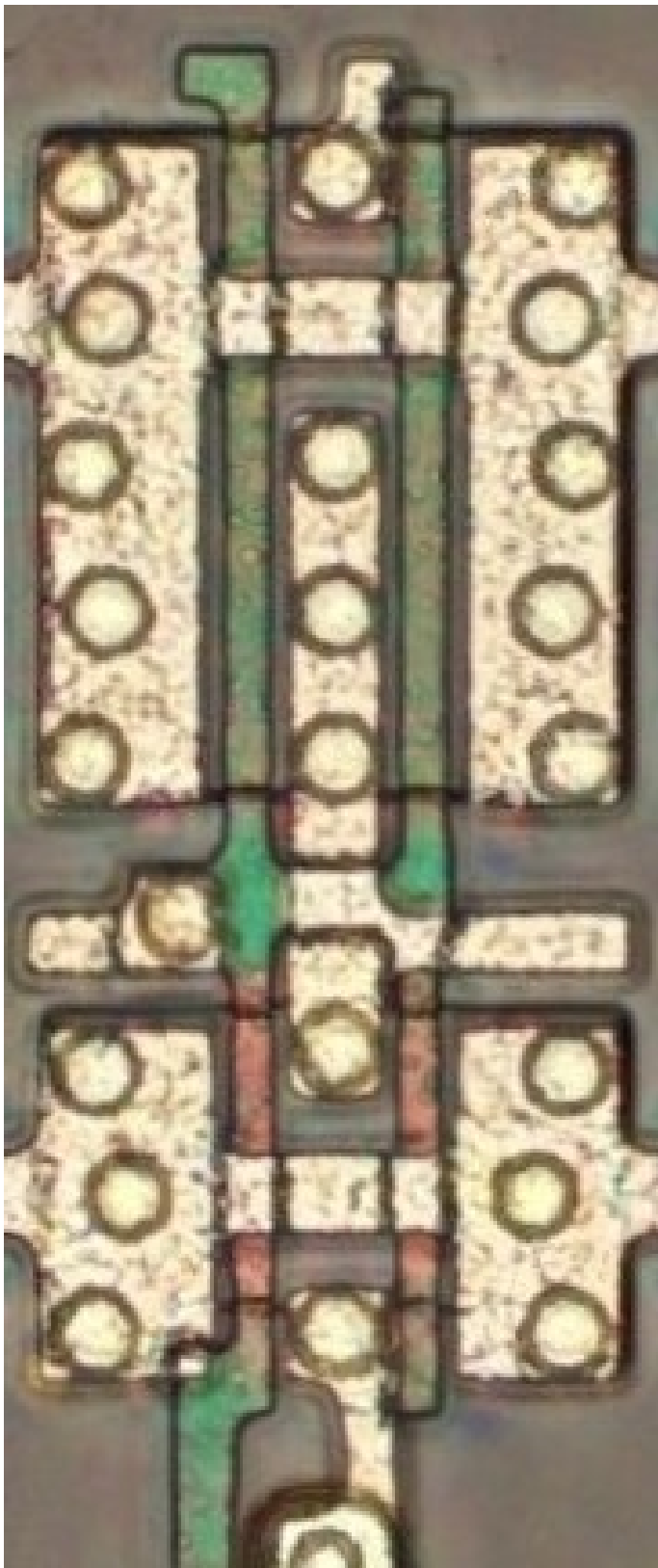
#393c3cff

Inverter/NOT X2

Cell 9

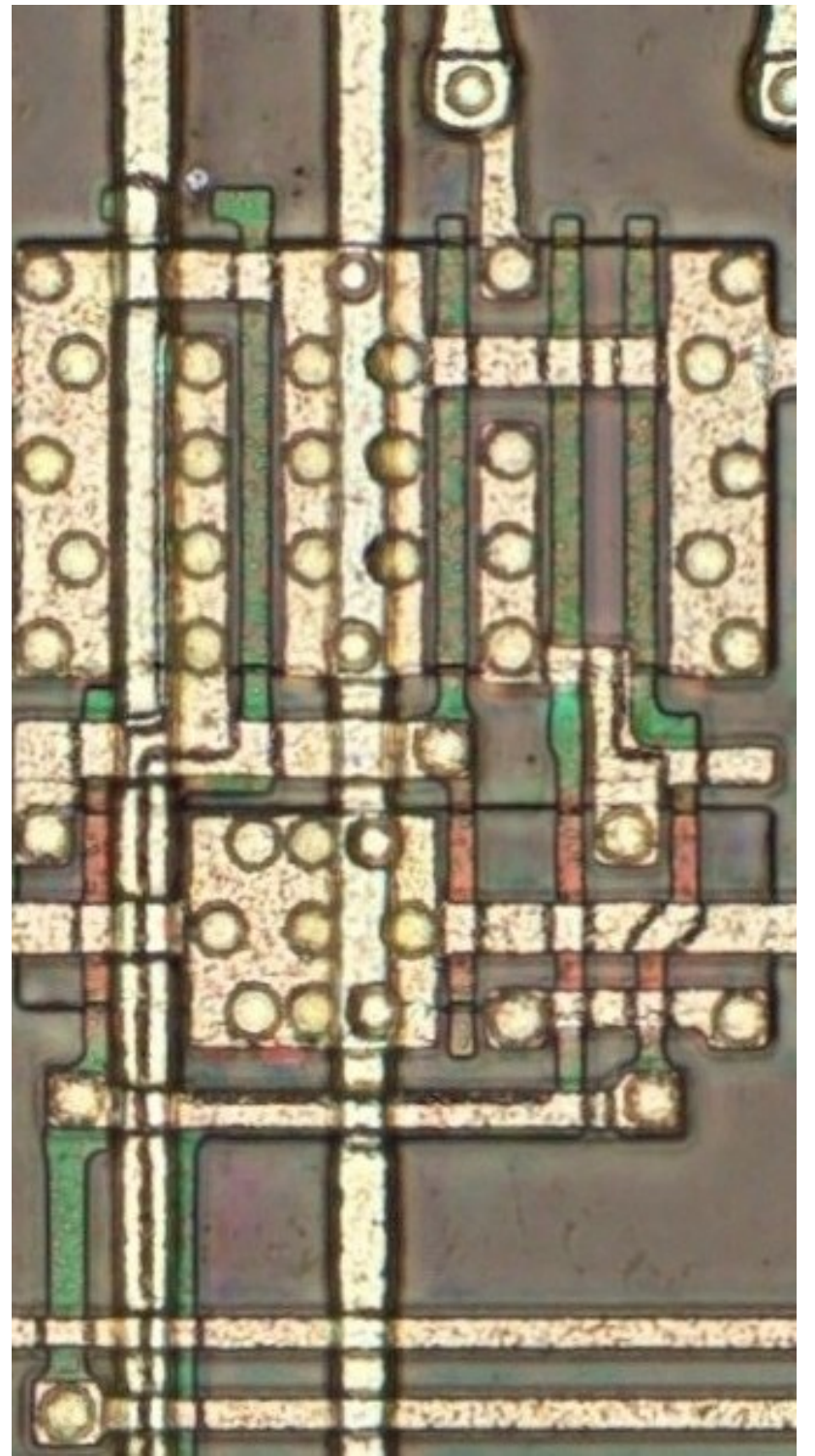
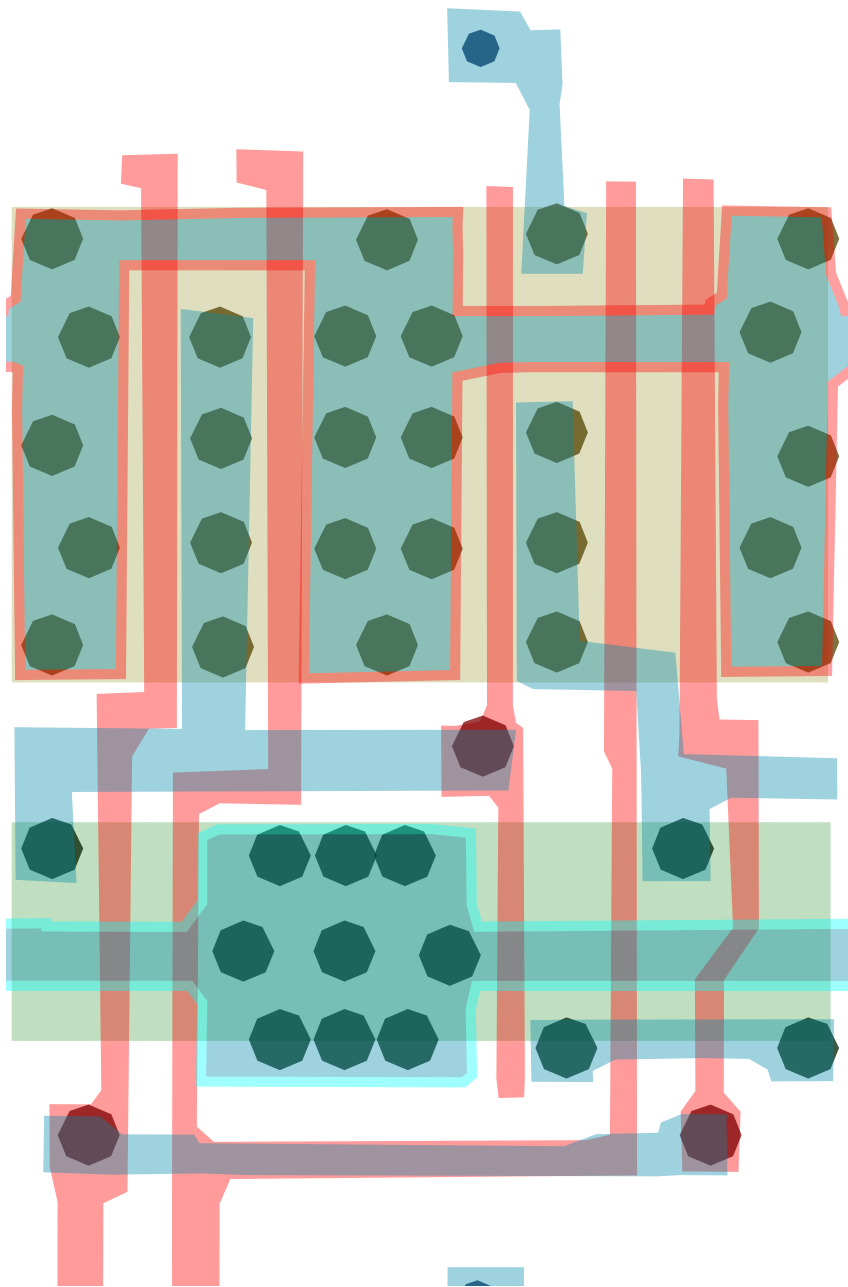


Same cell on
LA32
(MB87136A)

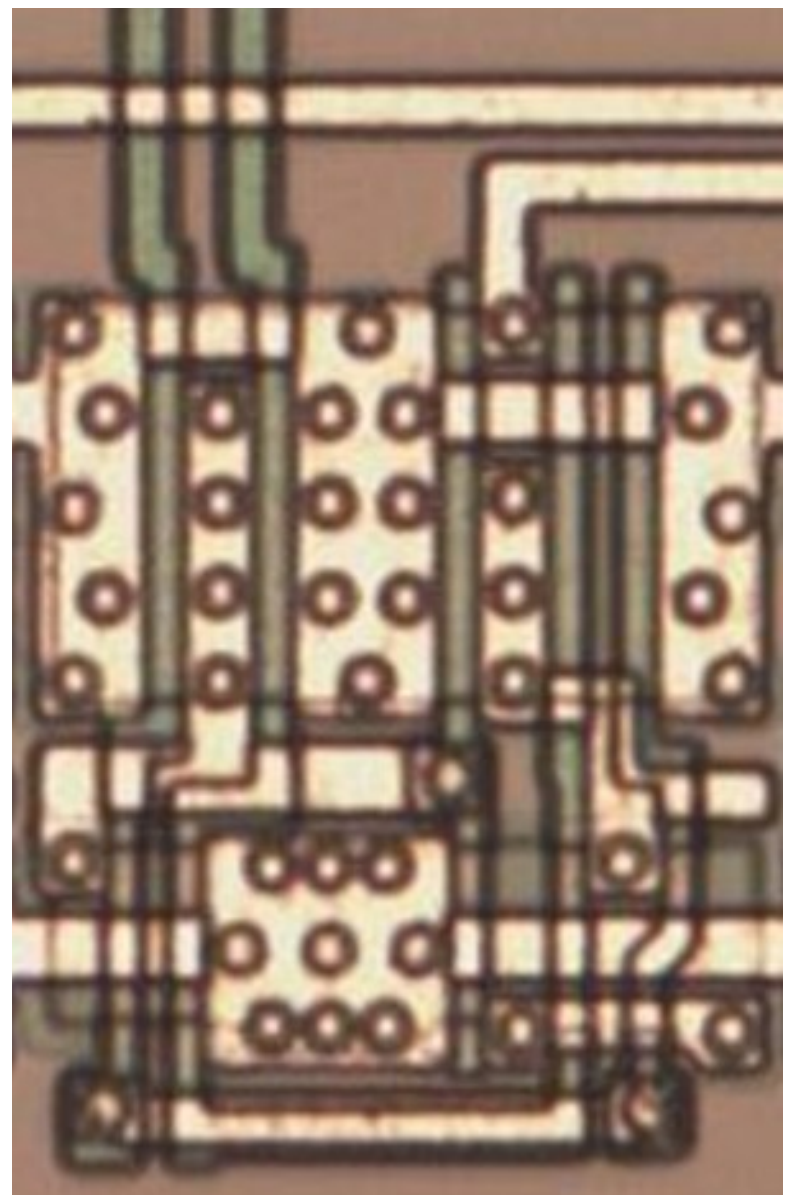


#000000ff

Cell type 10



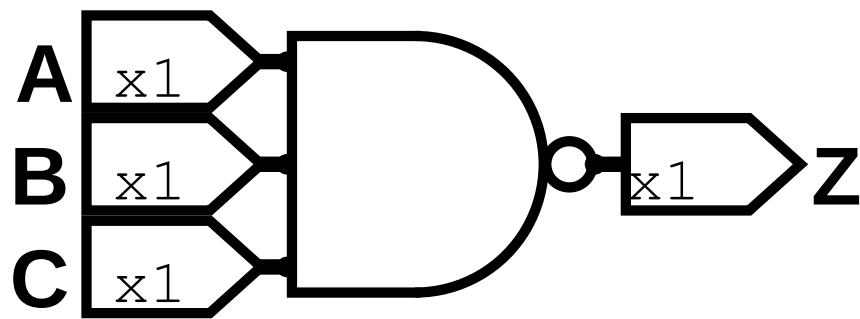
Same cell on
LA32
(MB87136A)



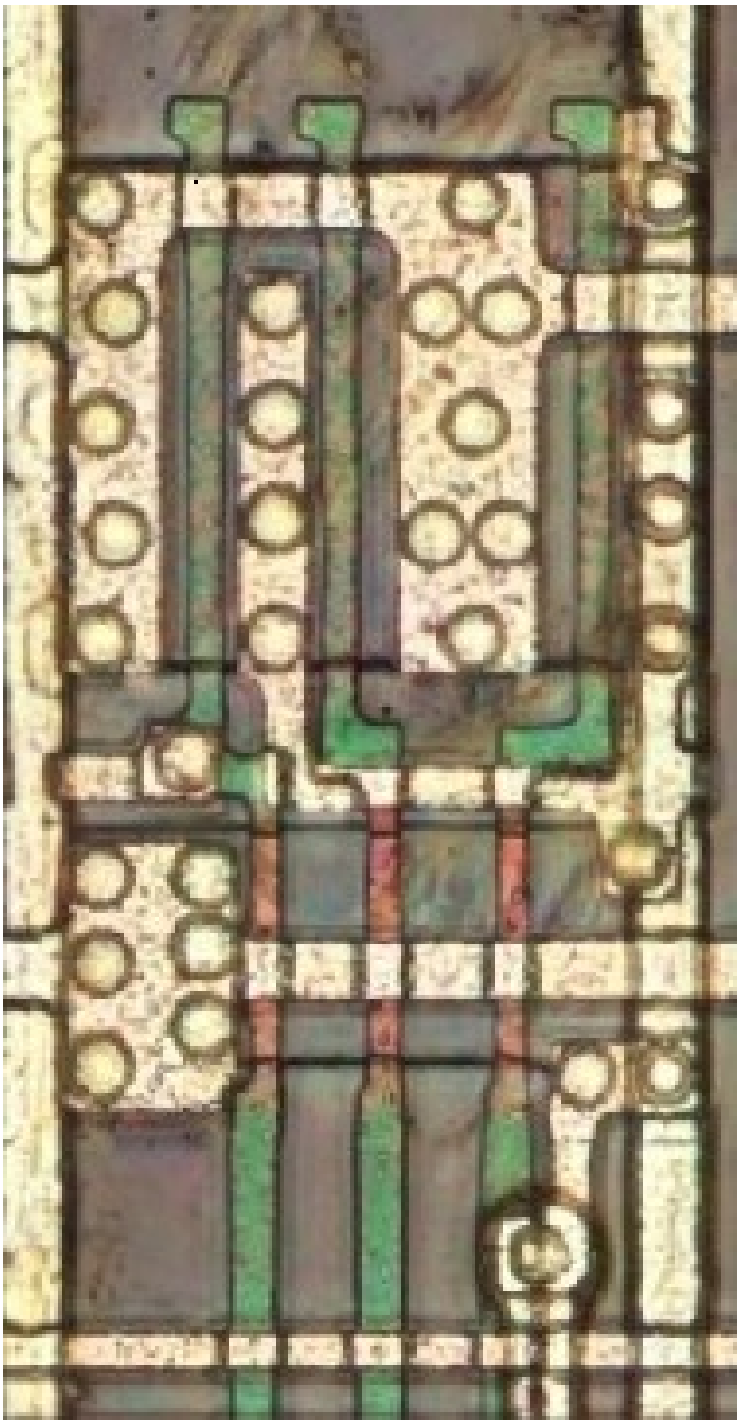
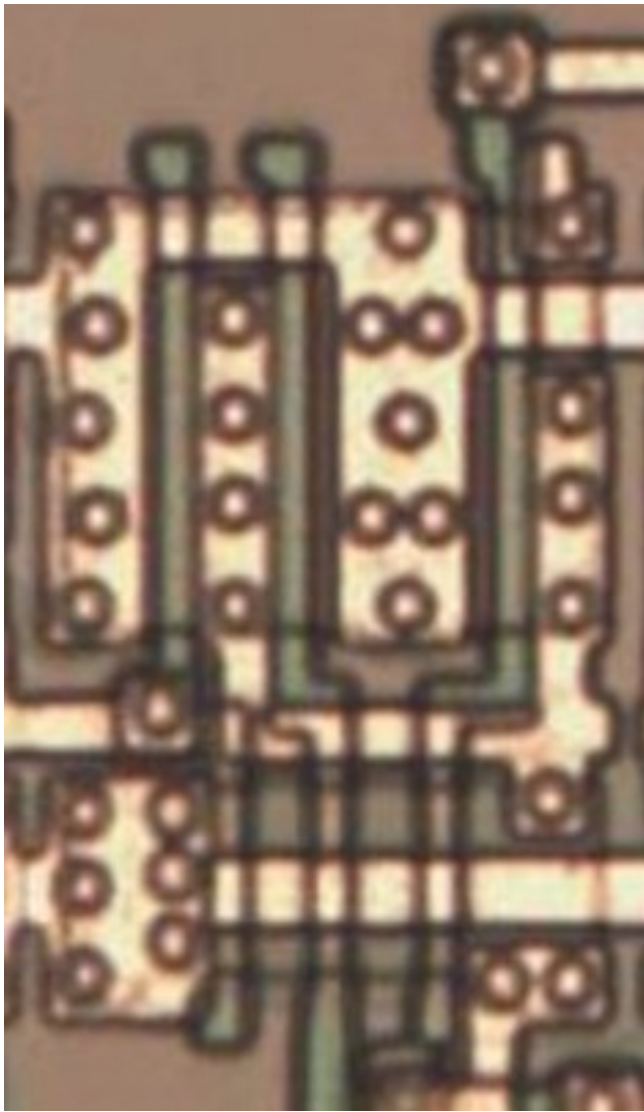
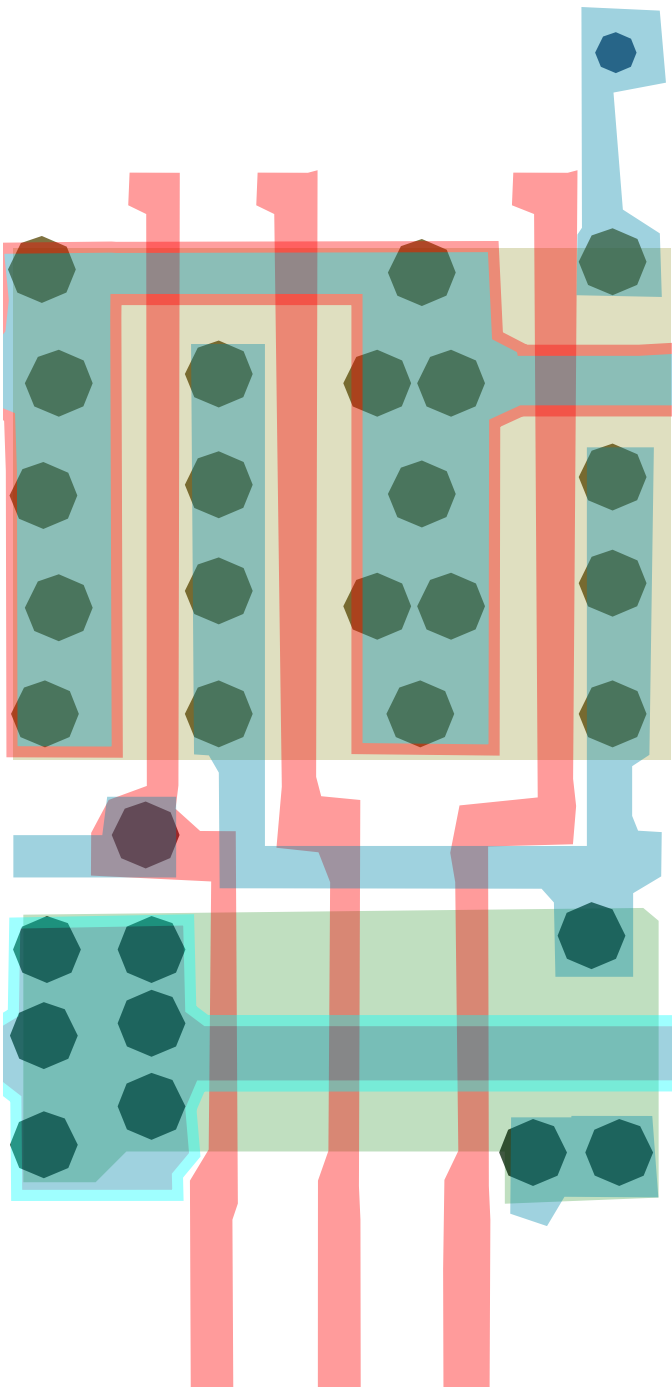
#f3ffffff

3-Input NAND

Cell 11



A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

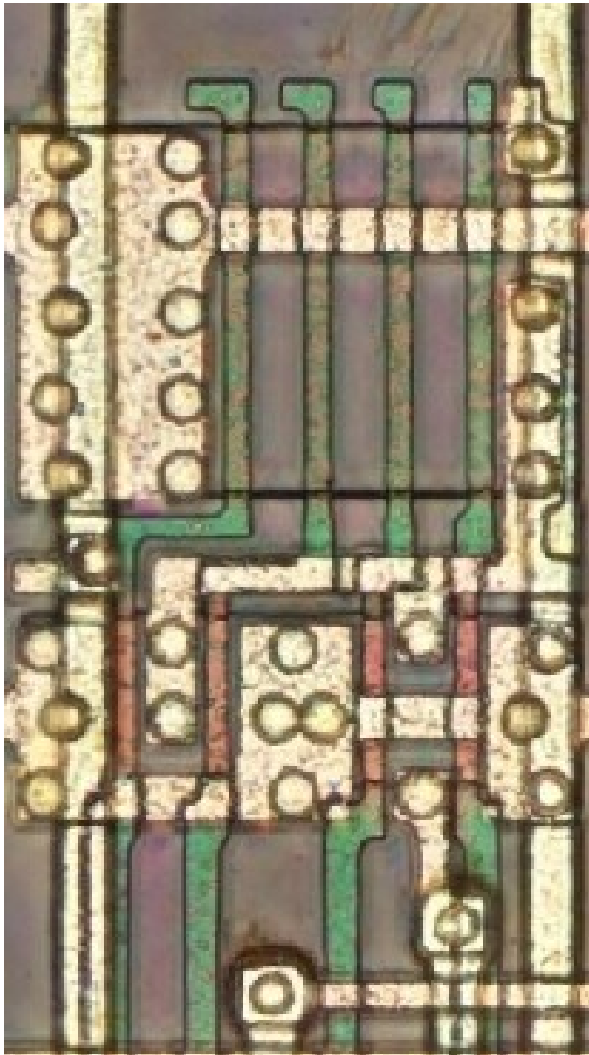
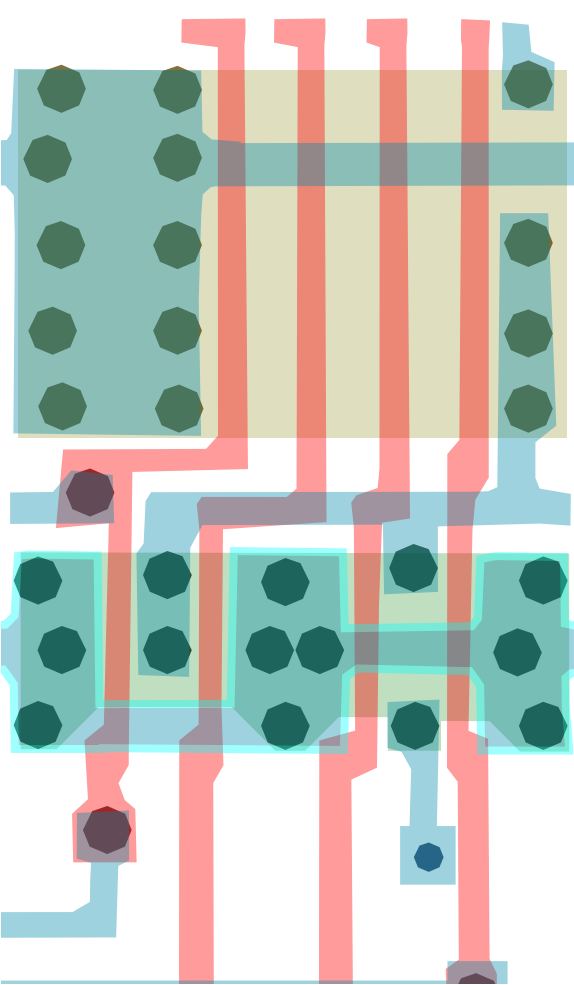
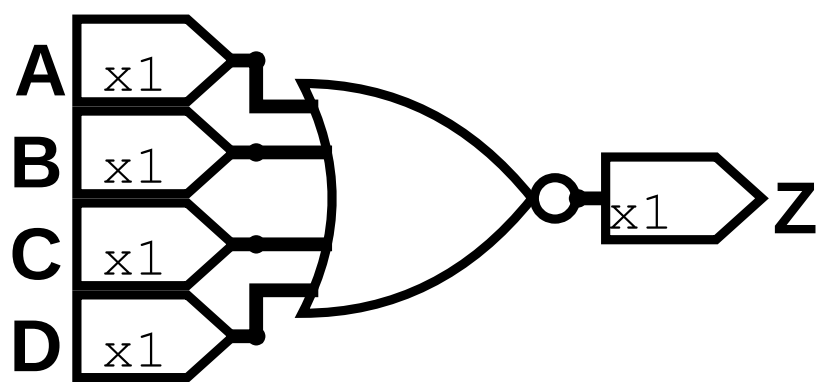


Same cell on
LA32
(MB87136A)

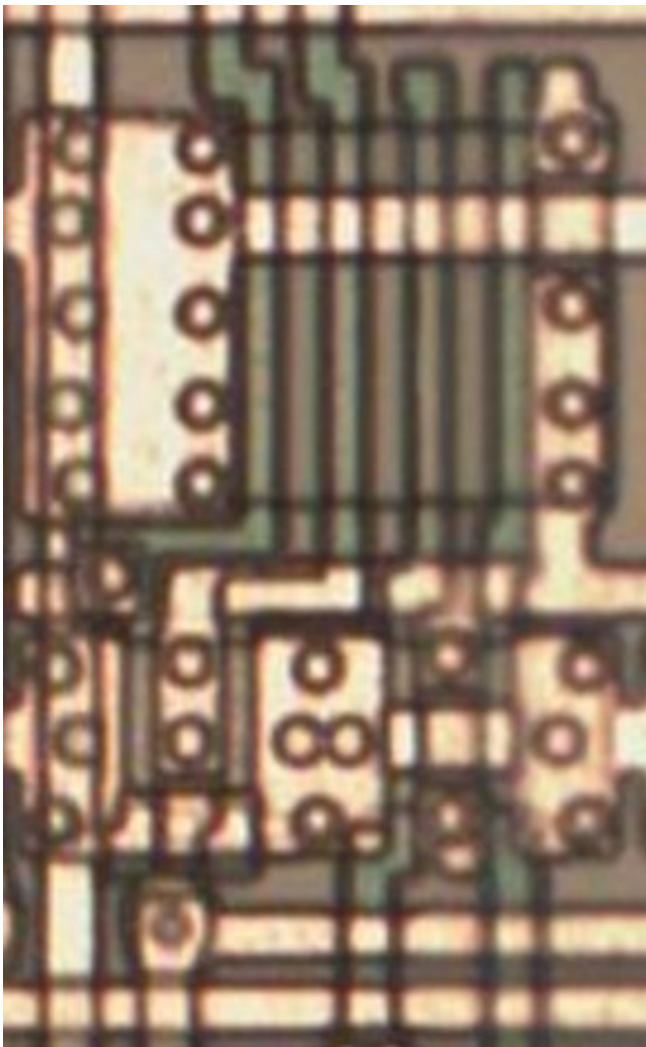
#007affff

4-Input NOR

Cell 12



A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

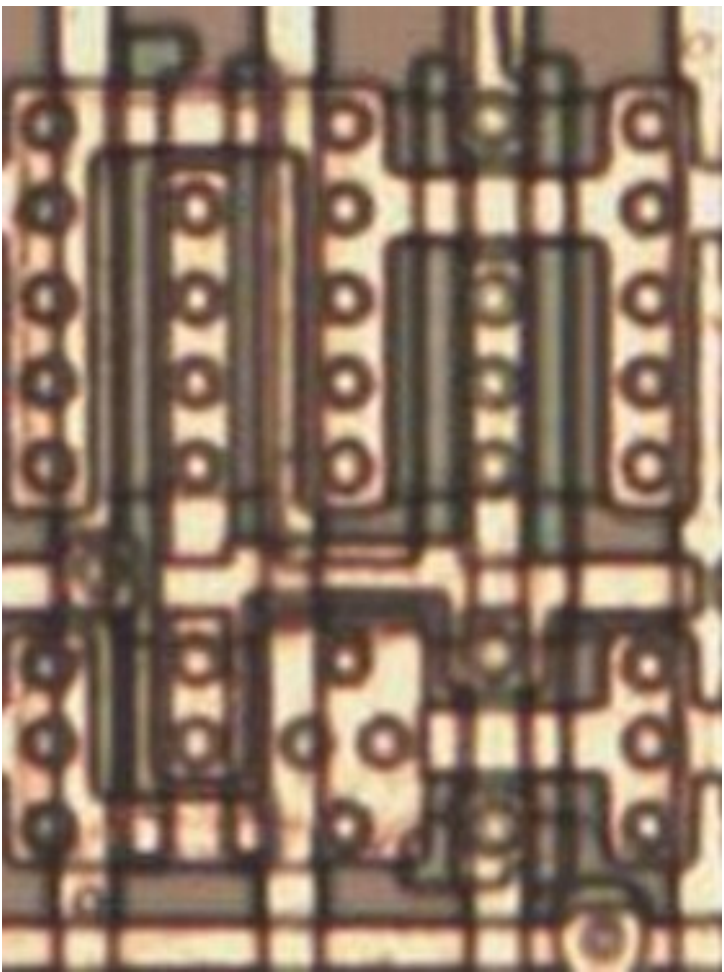
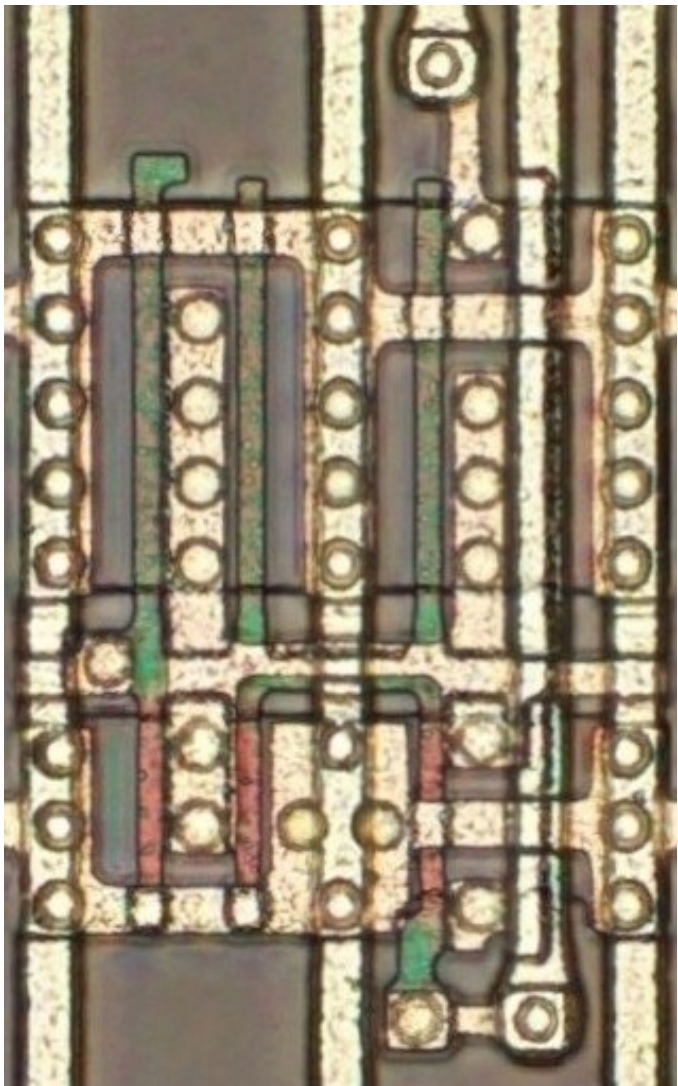
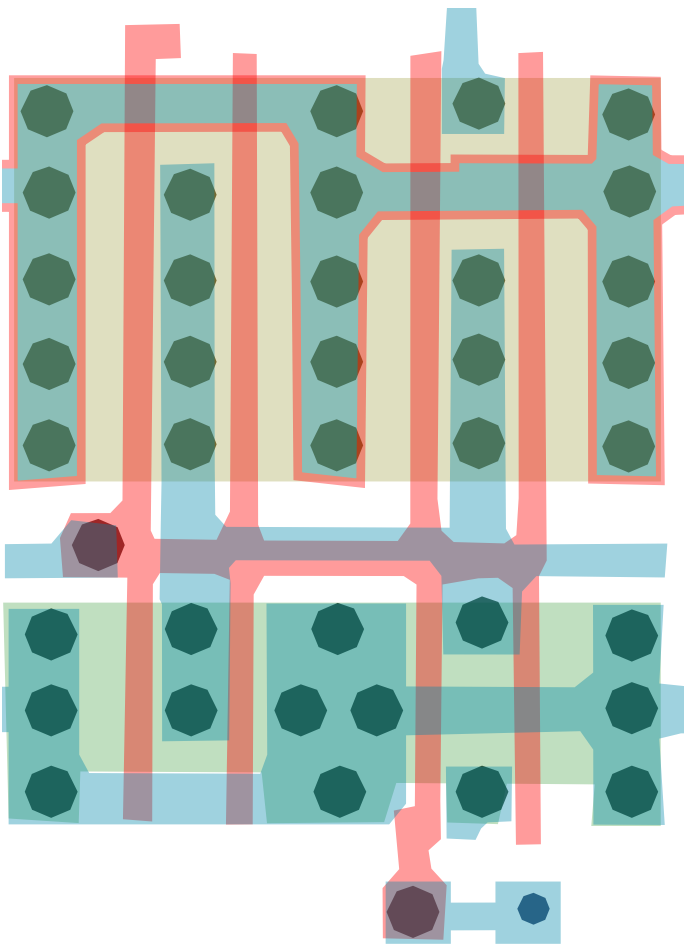
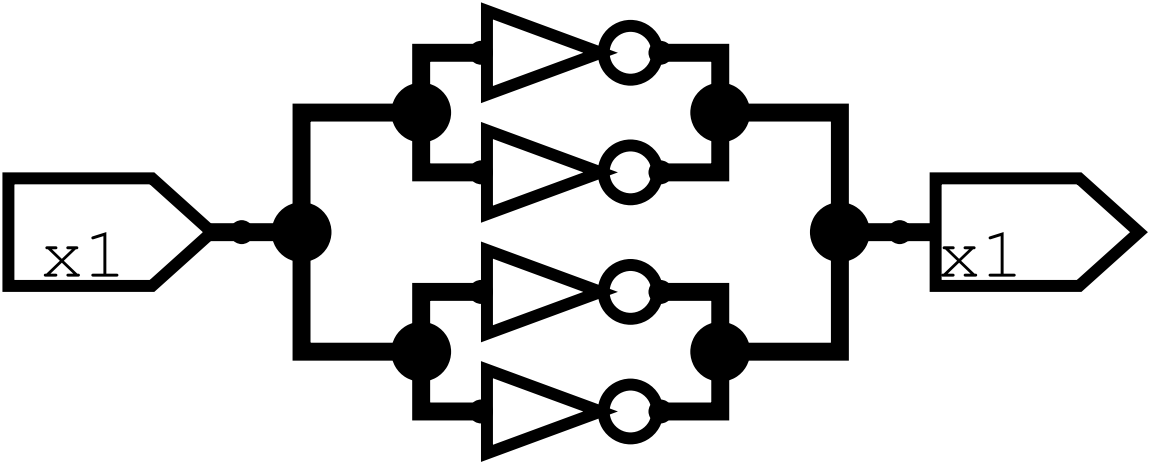


Same cell on
LA32
(MB87136A)

#b9ff43ff

Inverter/NOT x4

Cell 13a

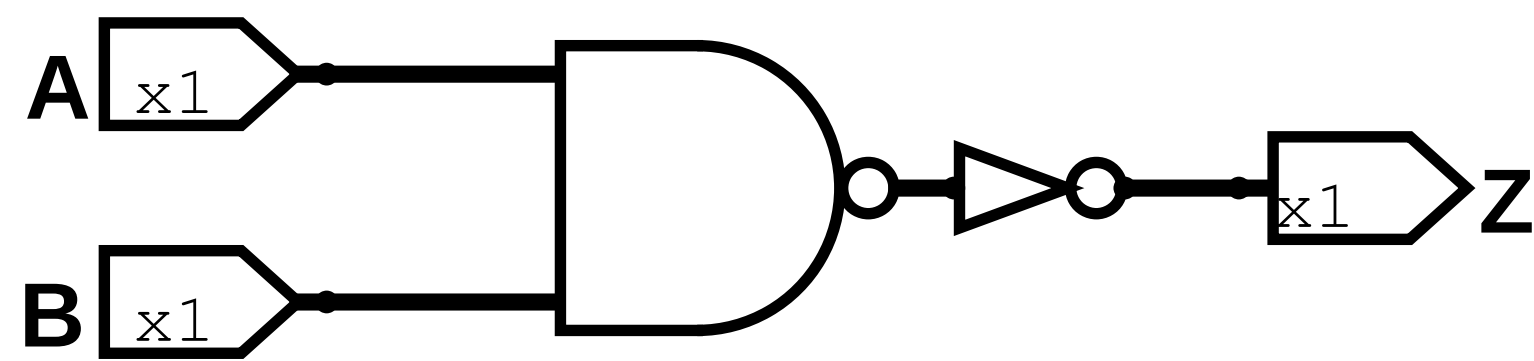


Same cell on
LA32
(MB87136A)

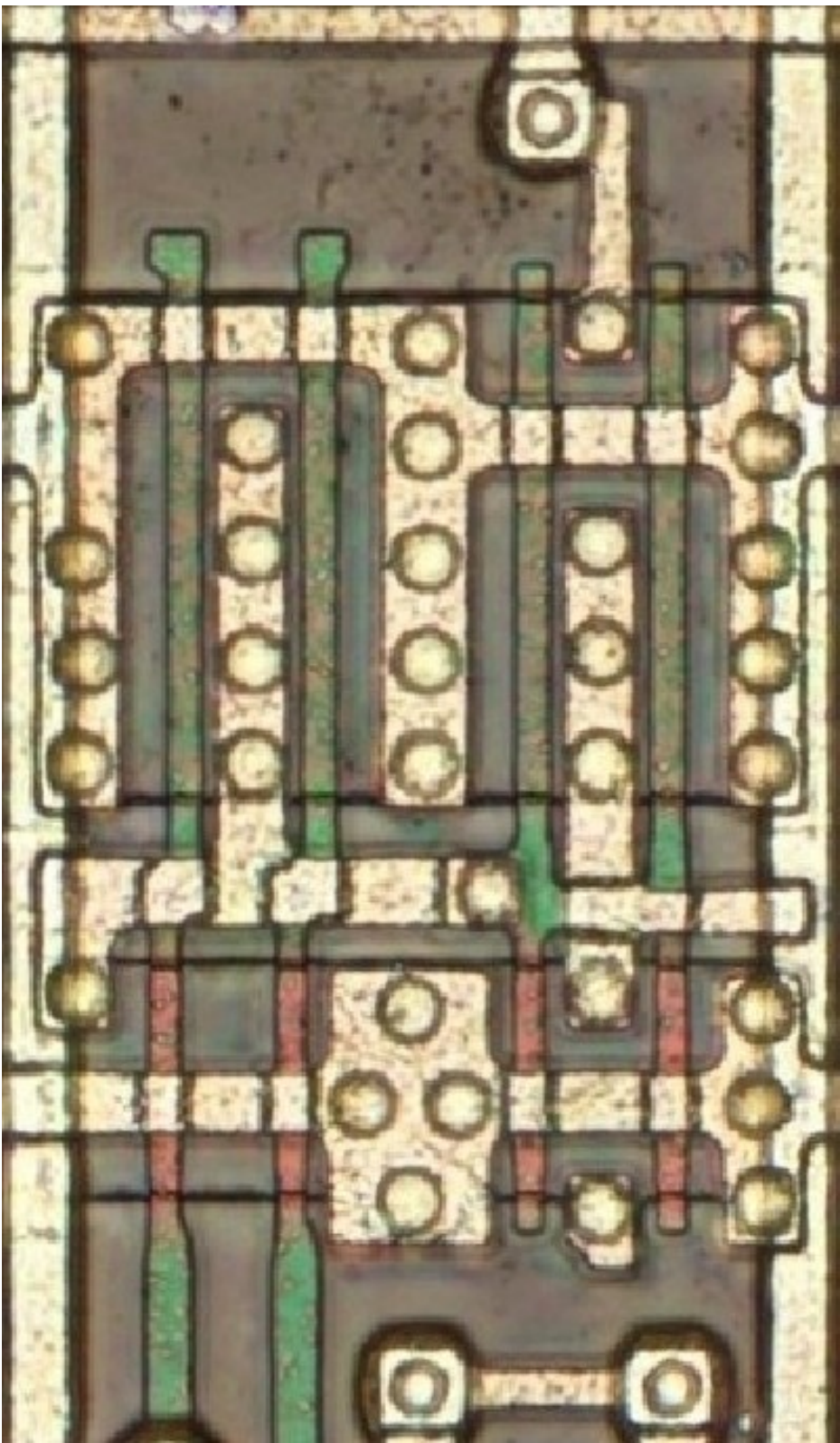
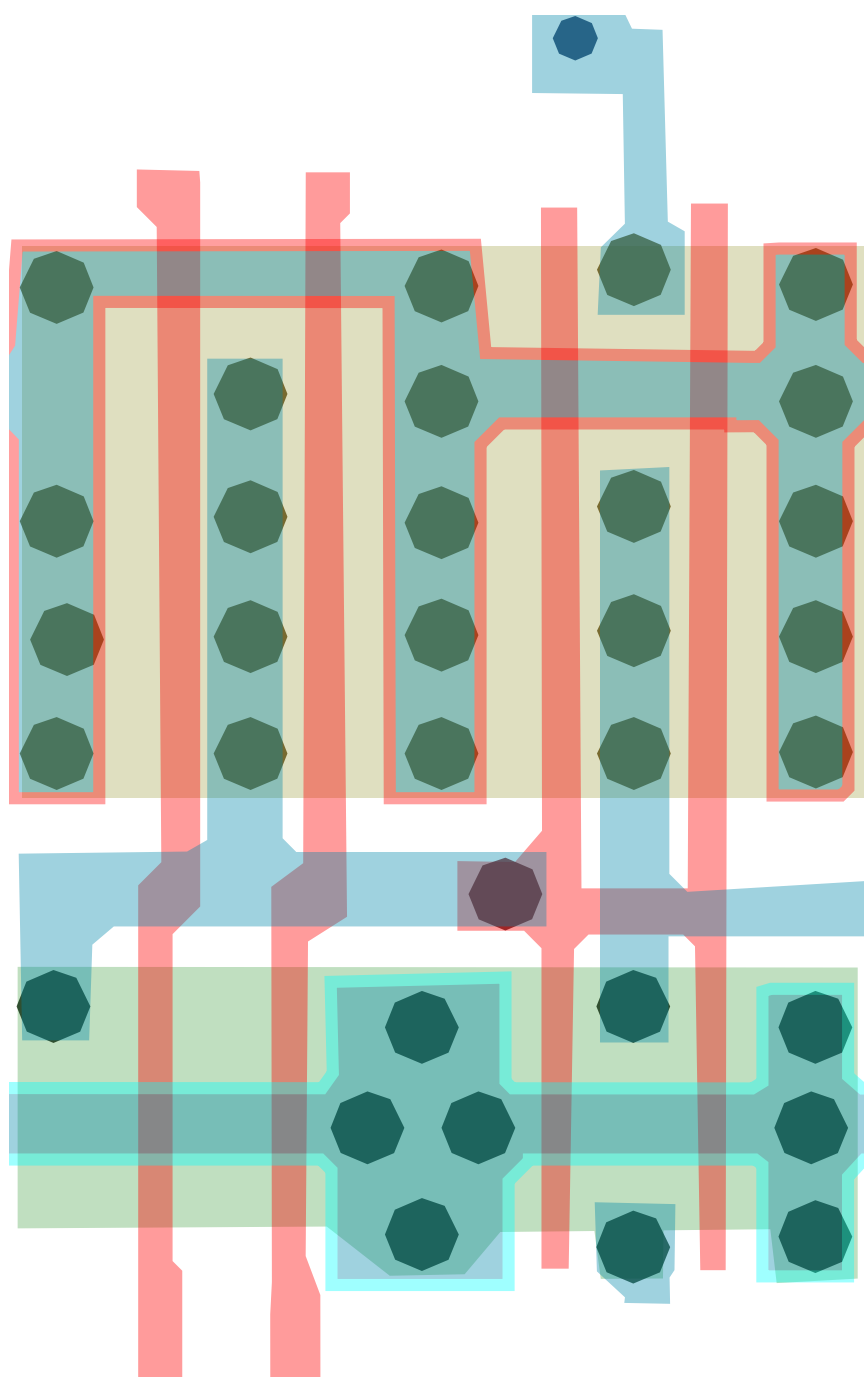
#bfa867ff

2-Input AND

Cell 13b



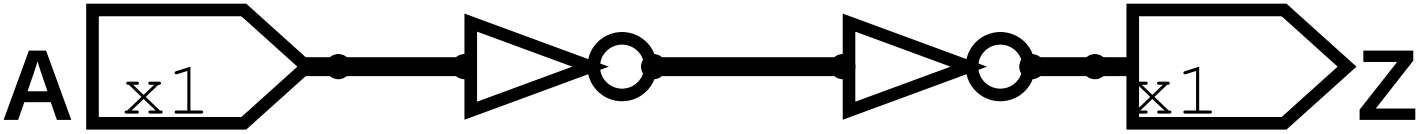
A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1



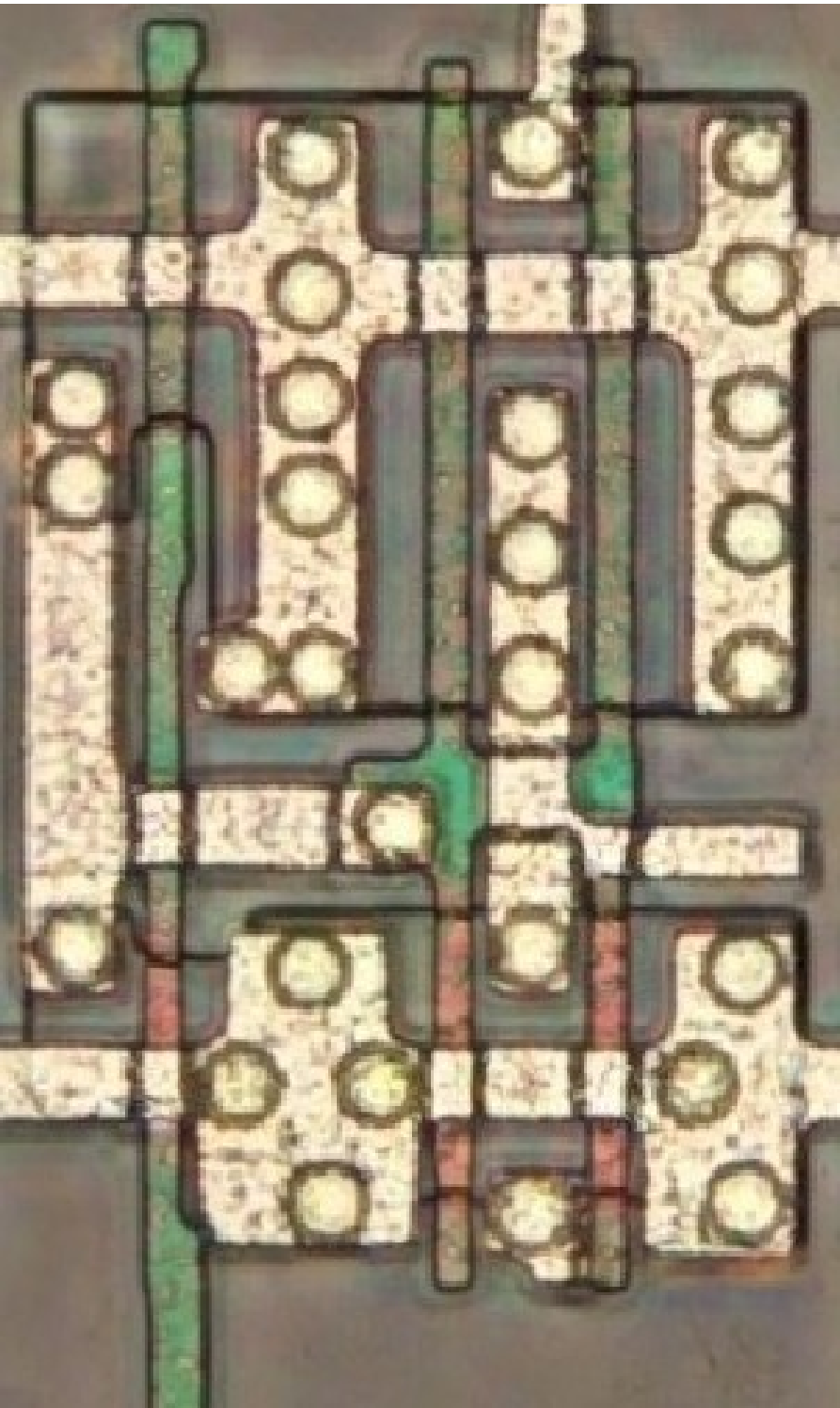
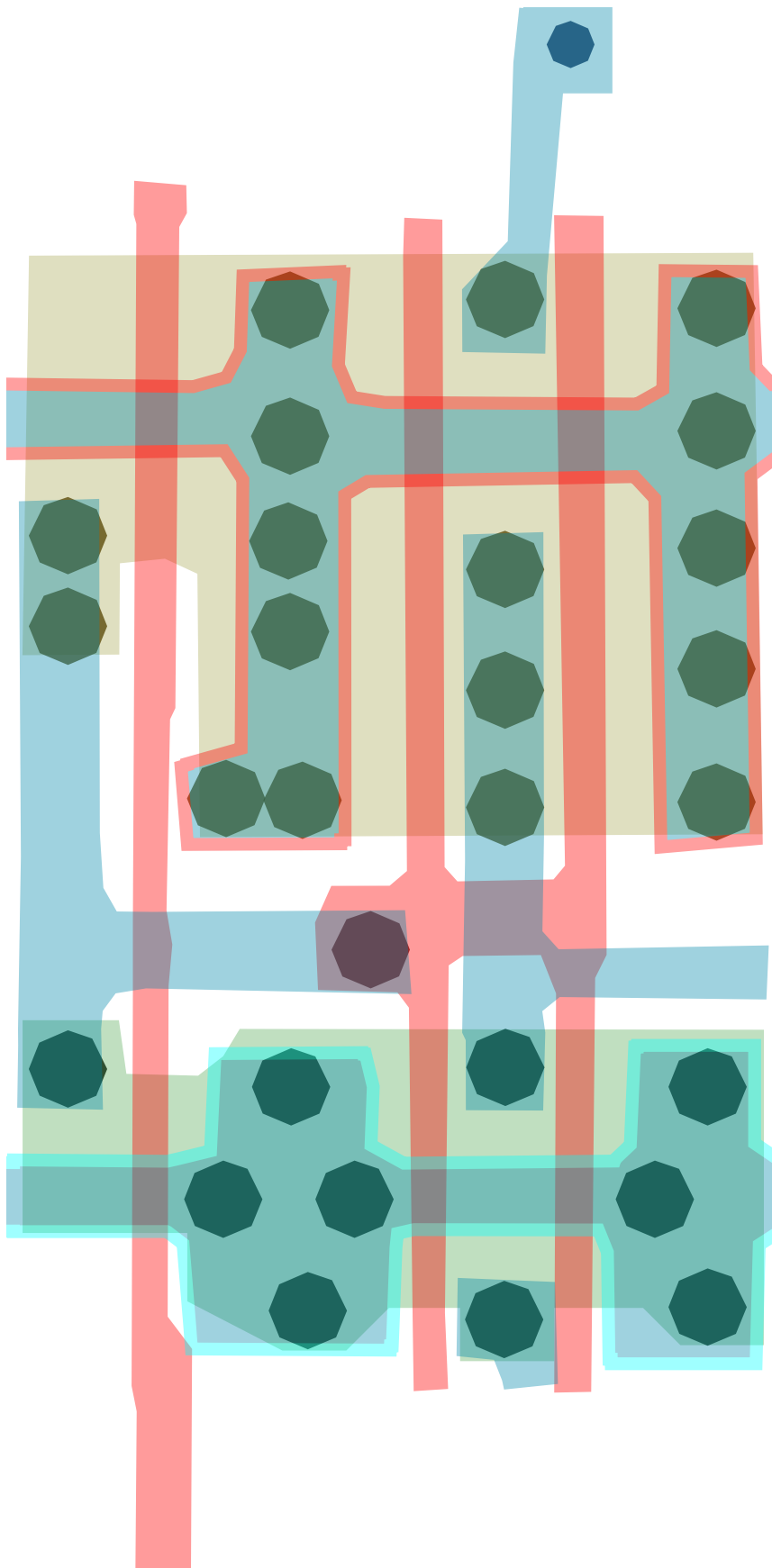
#a03ac0ff

Buffer

Cell 13c



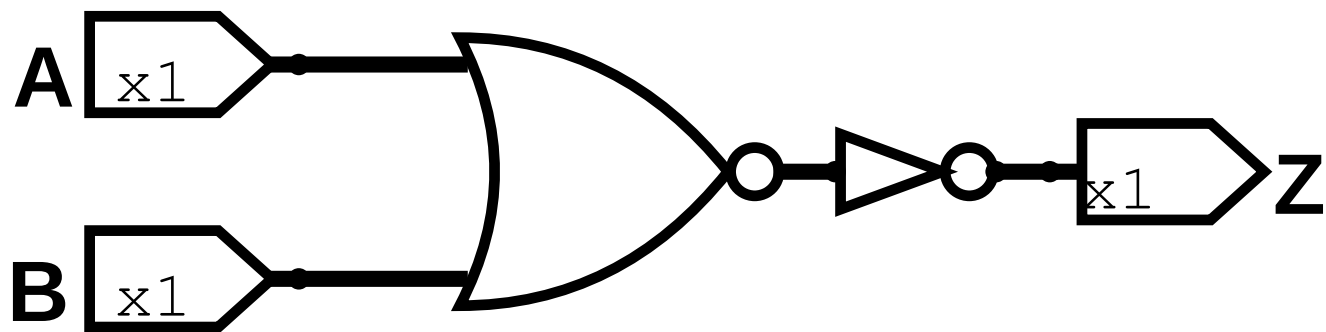
A	Z
0	0
1	1



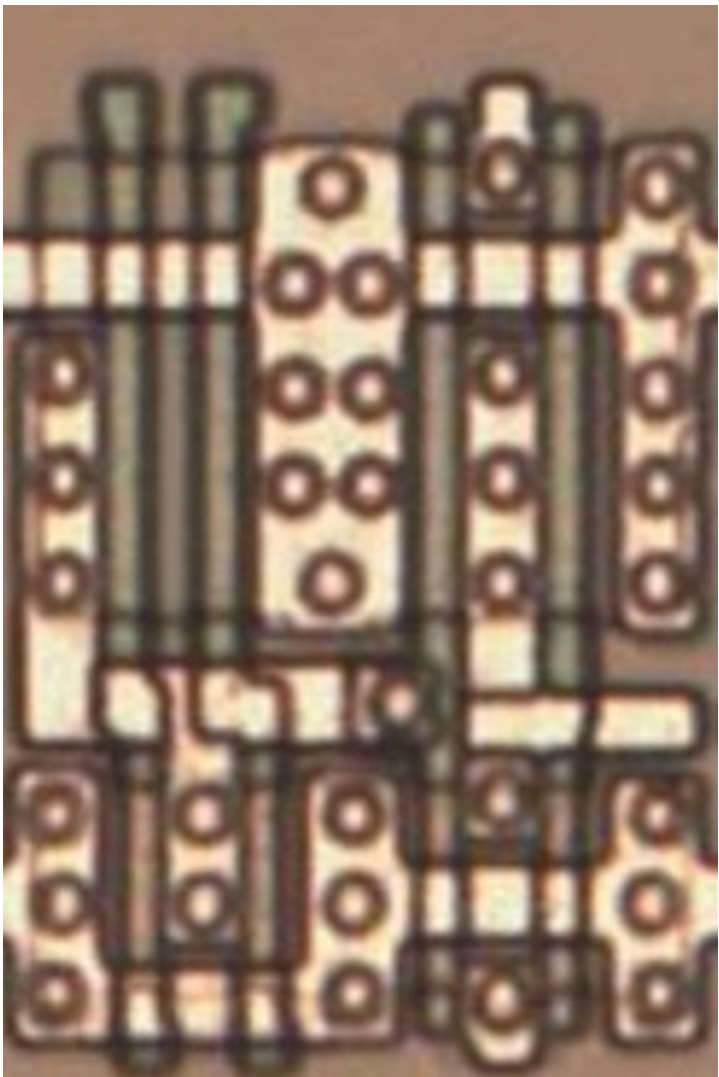
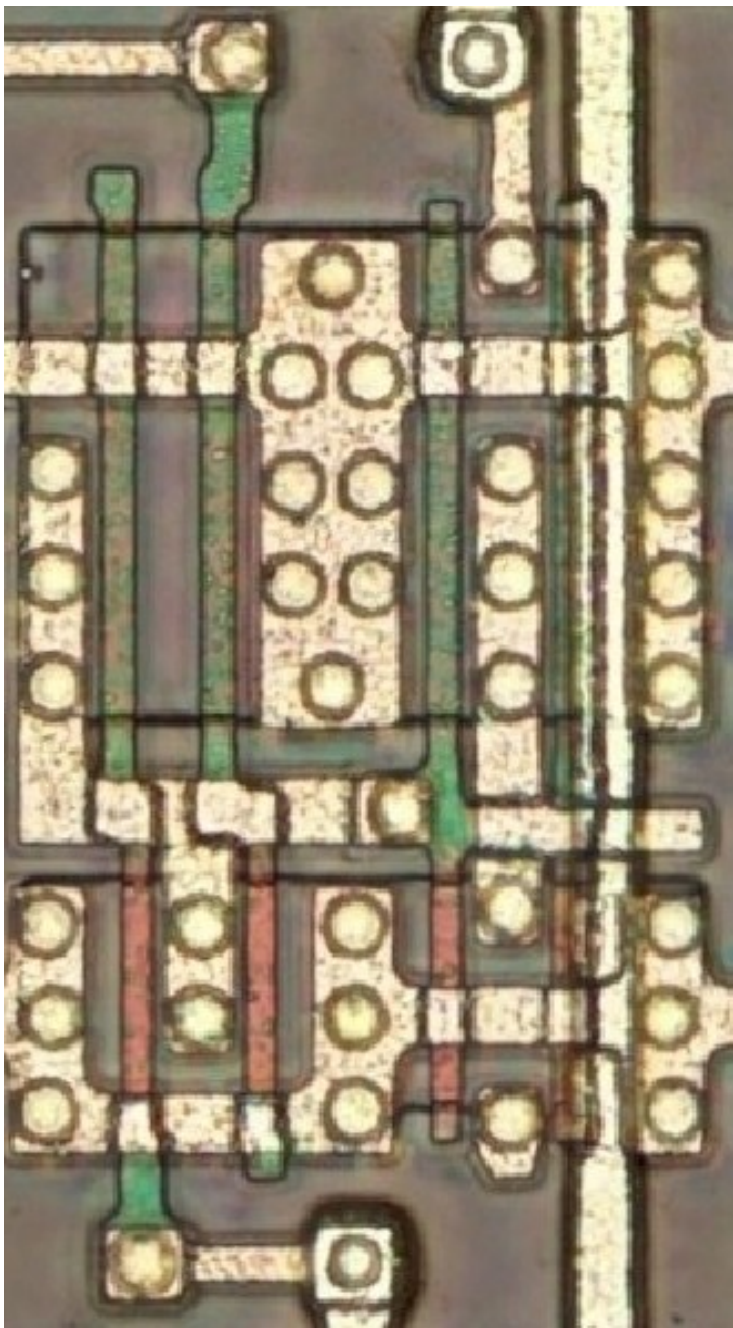
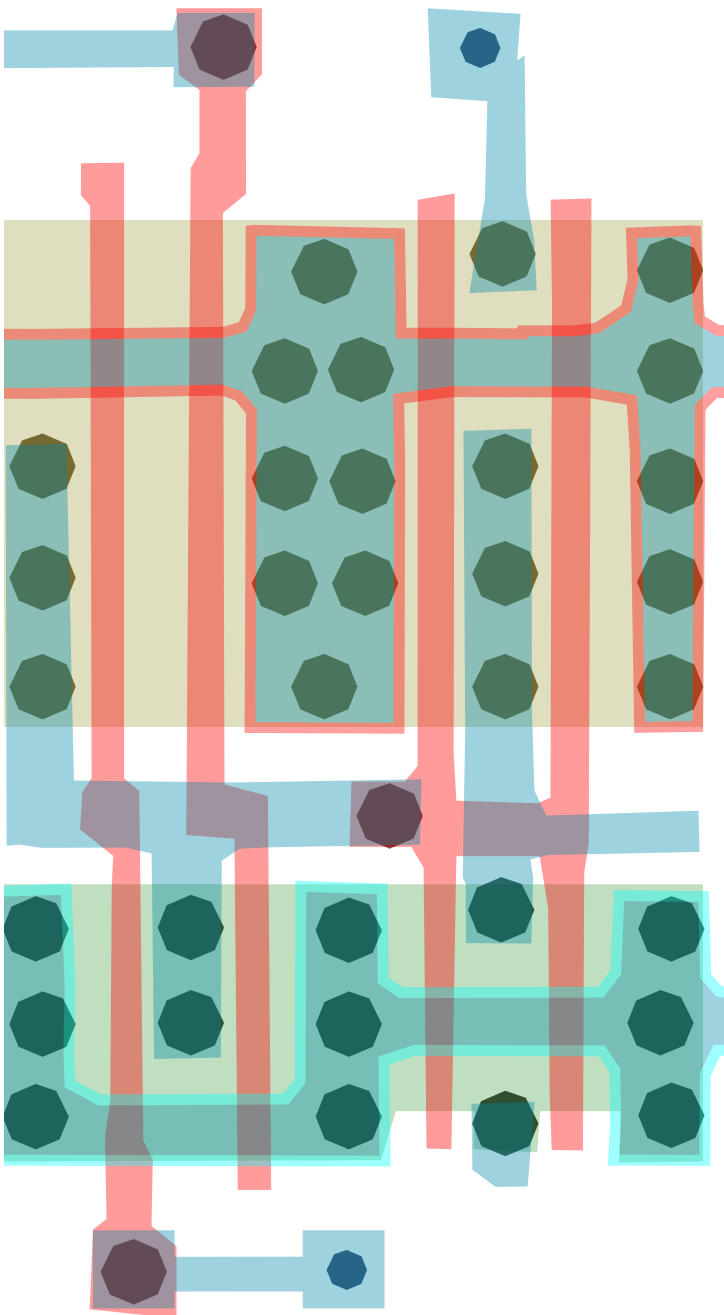
#ffbcacff

2-Input OR x1

Cell 13d



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

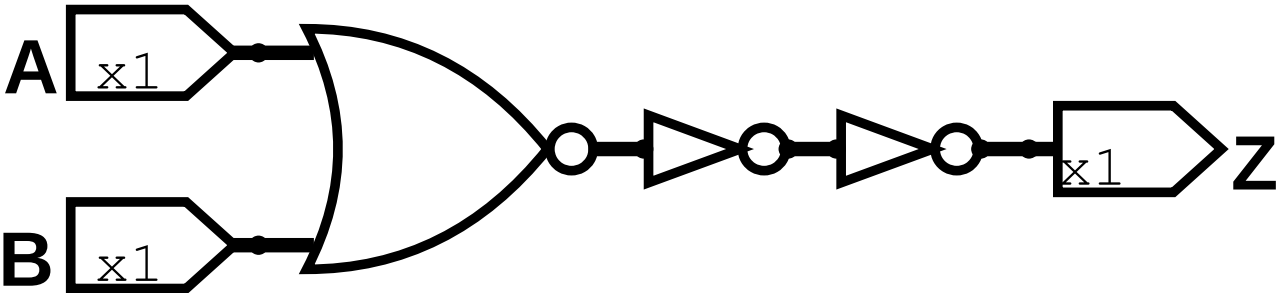


Same cell on
LA32
(MB87136A)

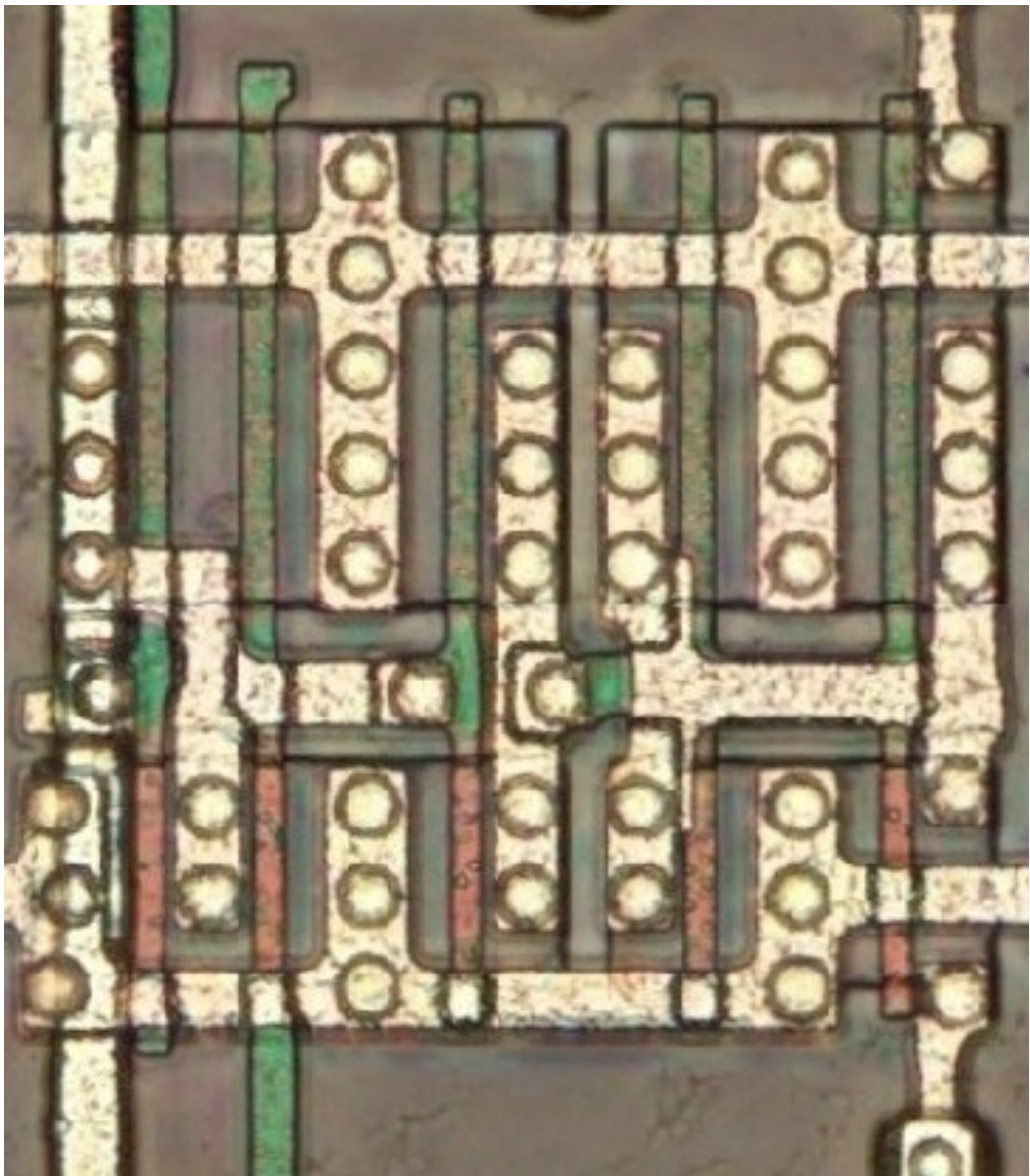
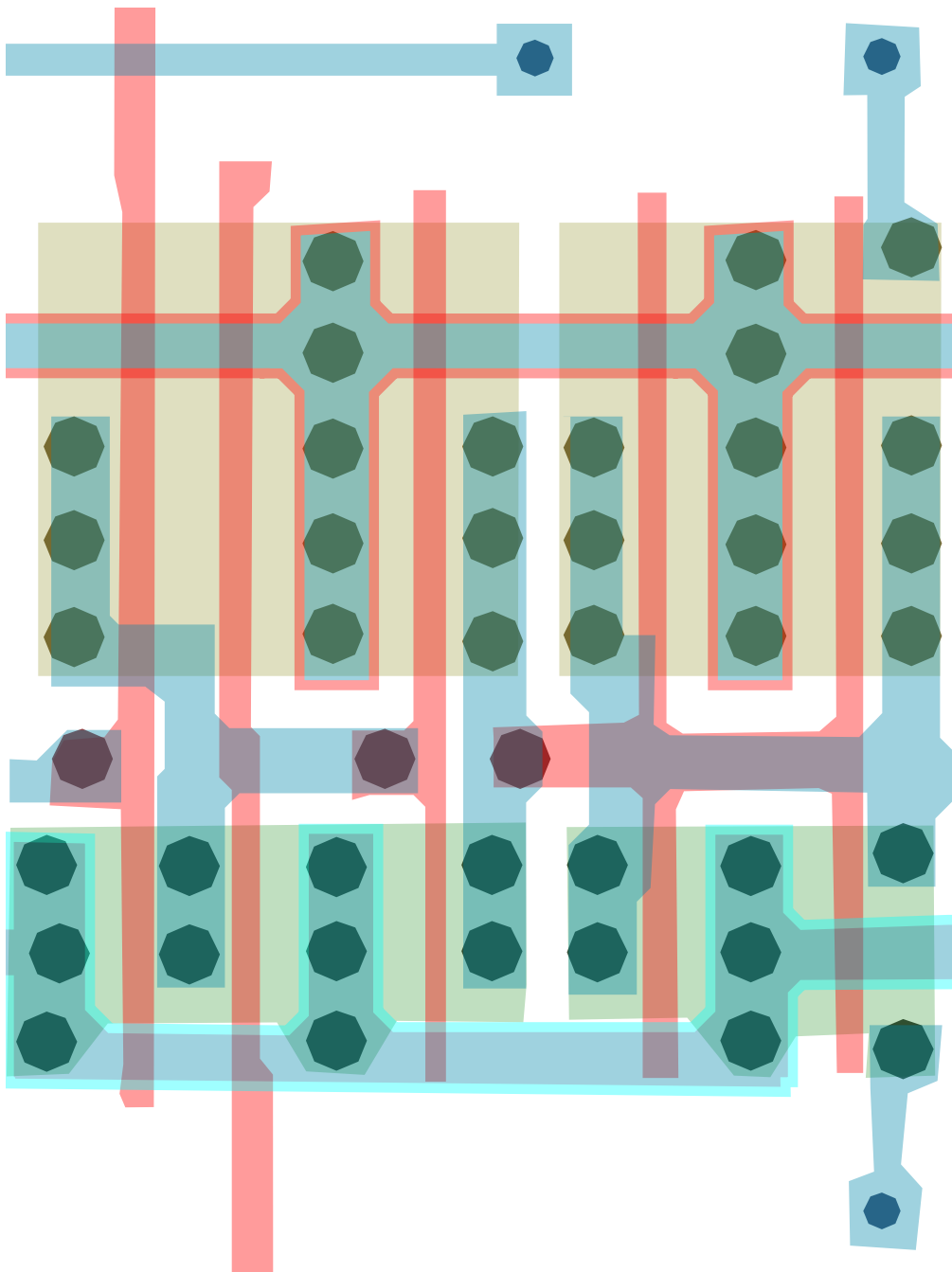
#ff7b94ff

2-Input NOR x2

Cell 13e



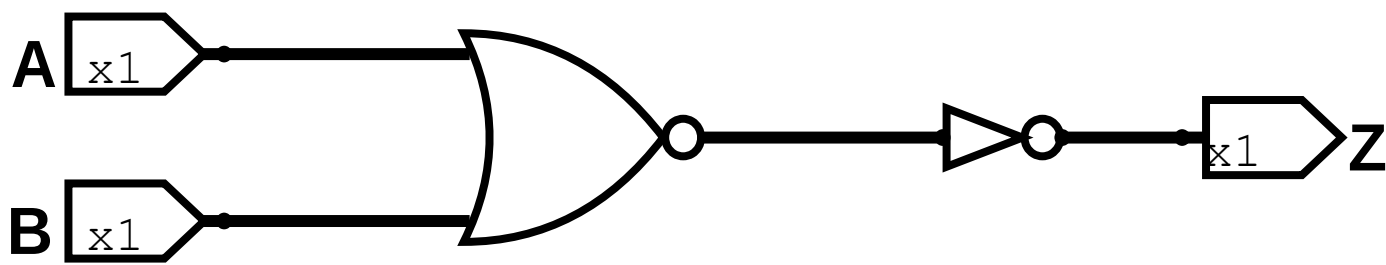
A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0



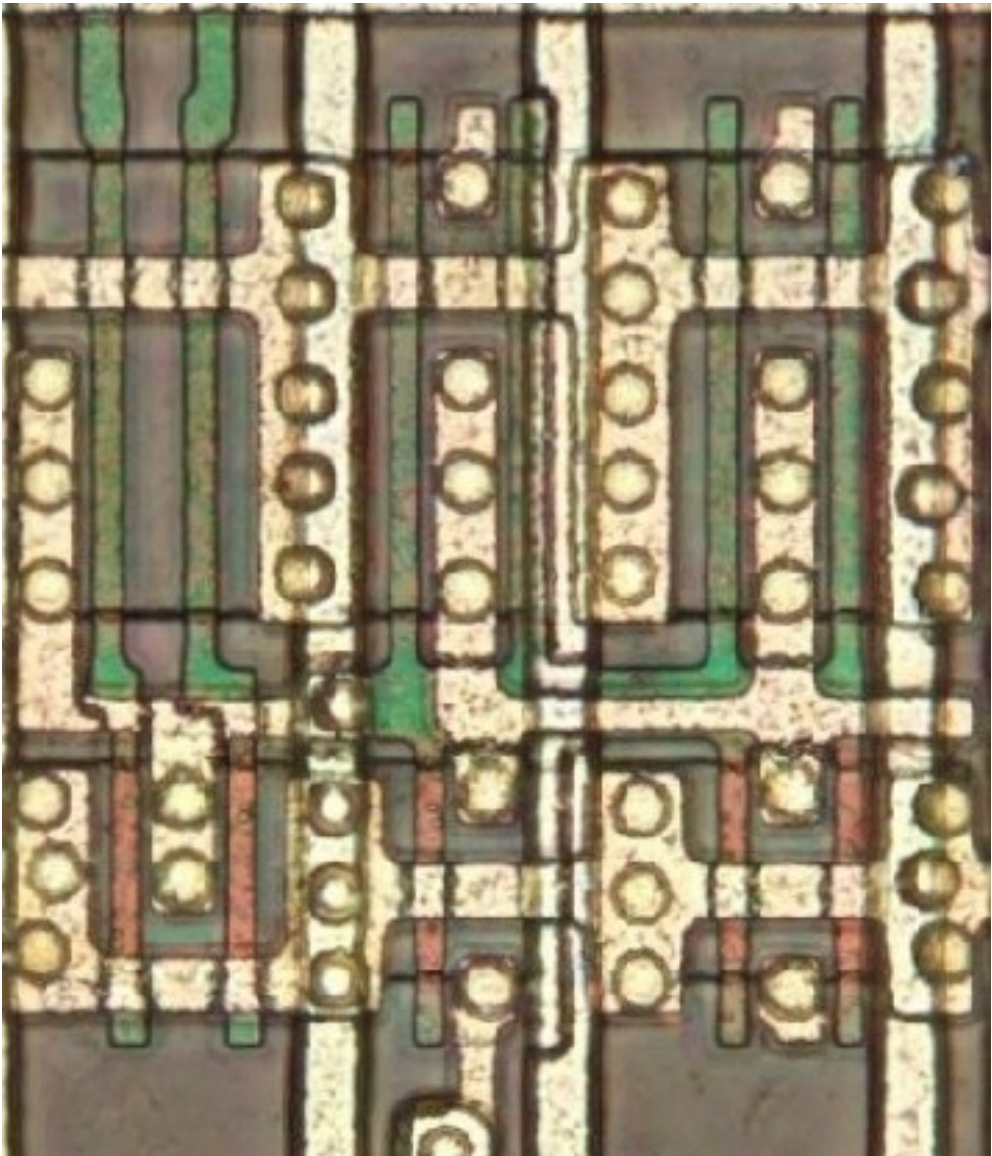
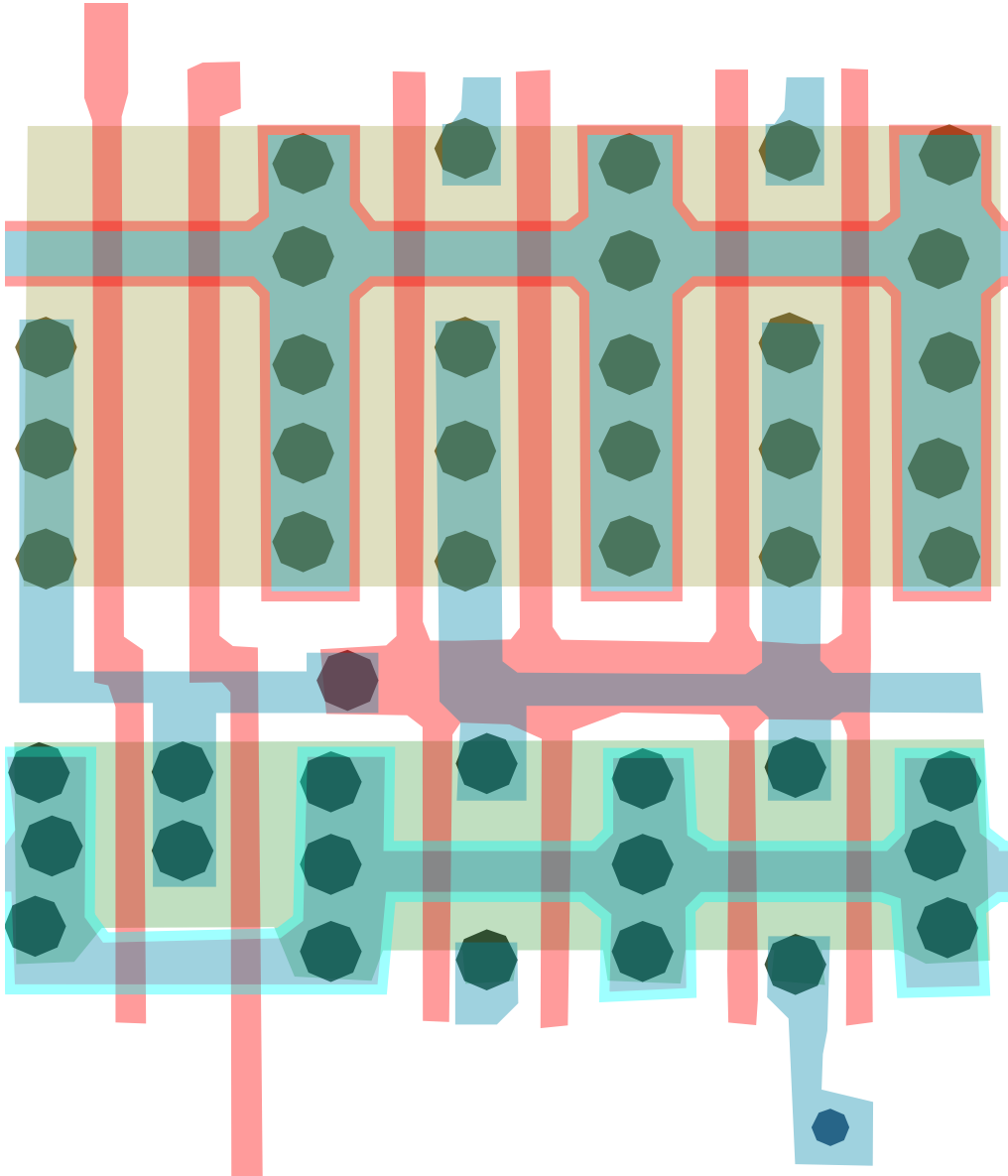
#fba968ff

2-Input OR x2

Cell 13f

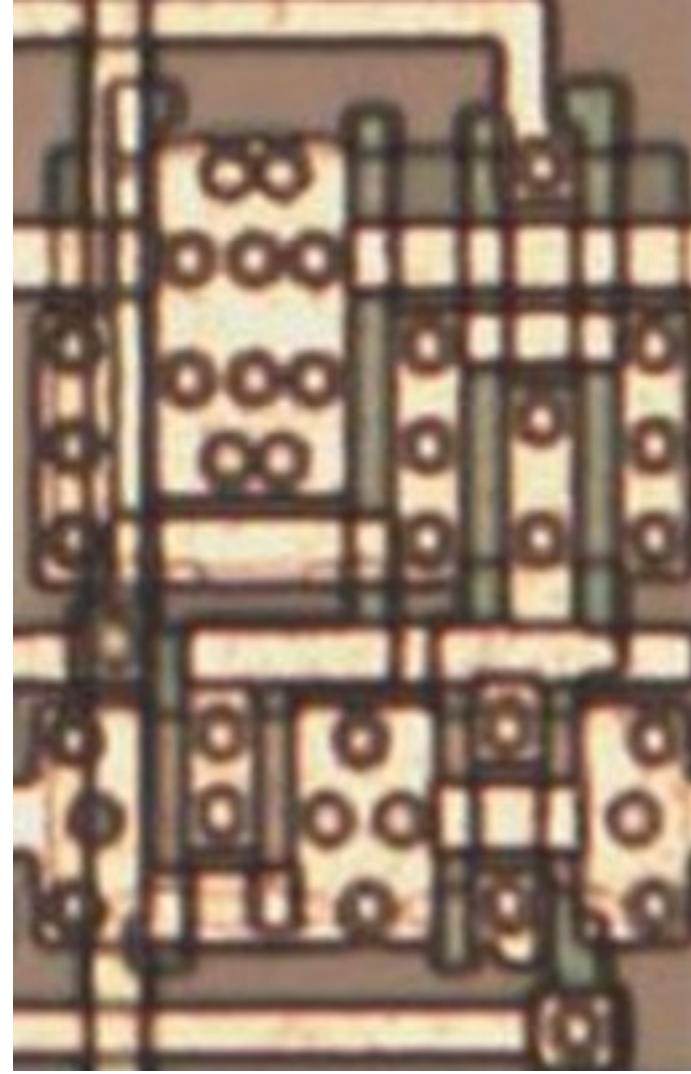
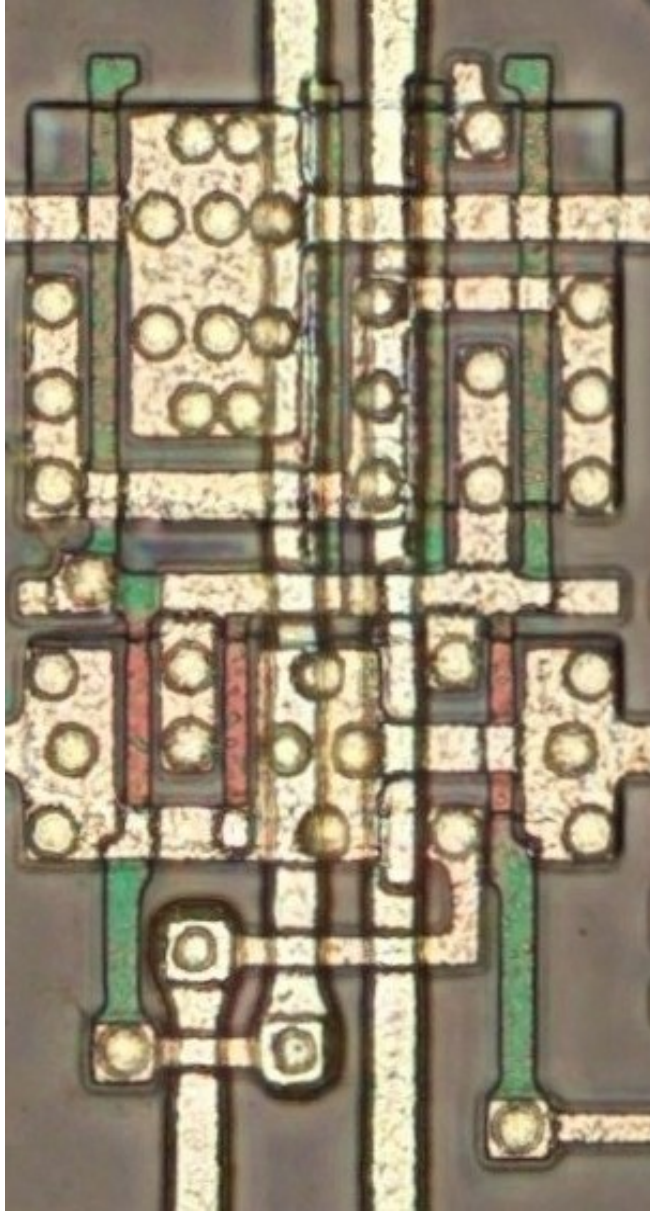
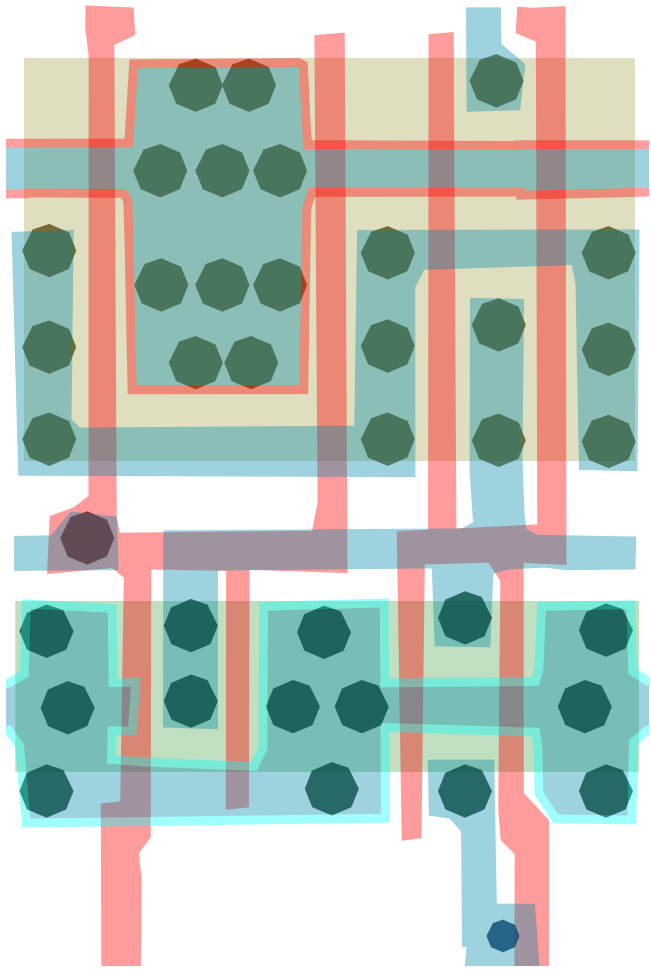


A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1



#f3ffa7ff

Cell type 14

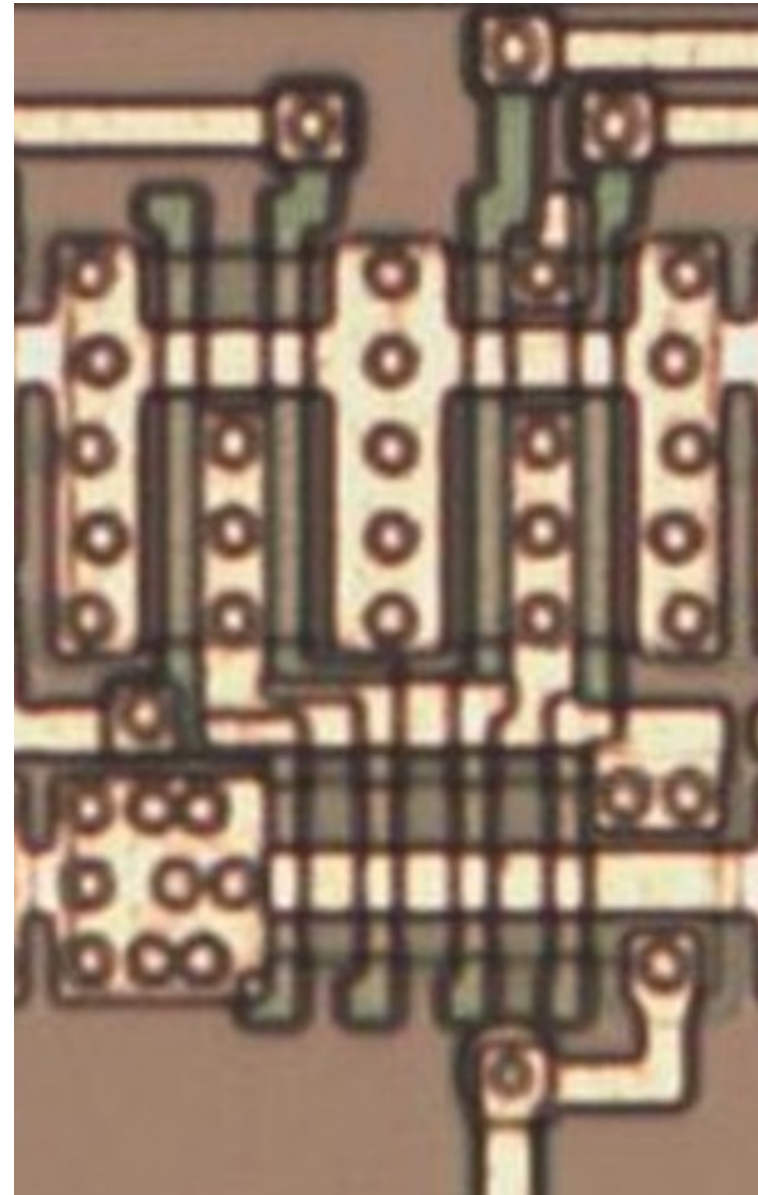
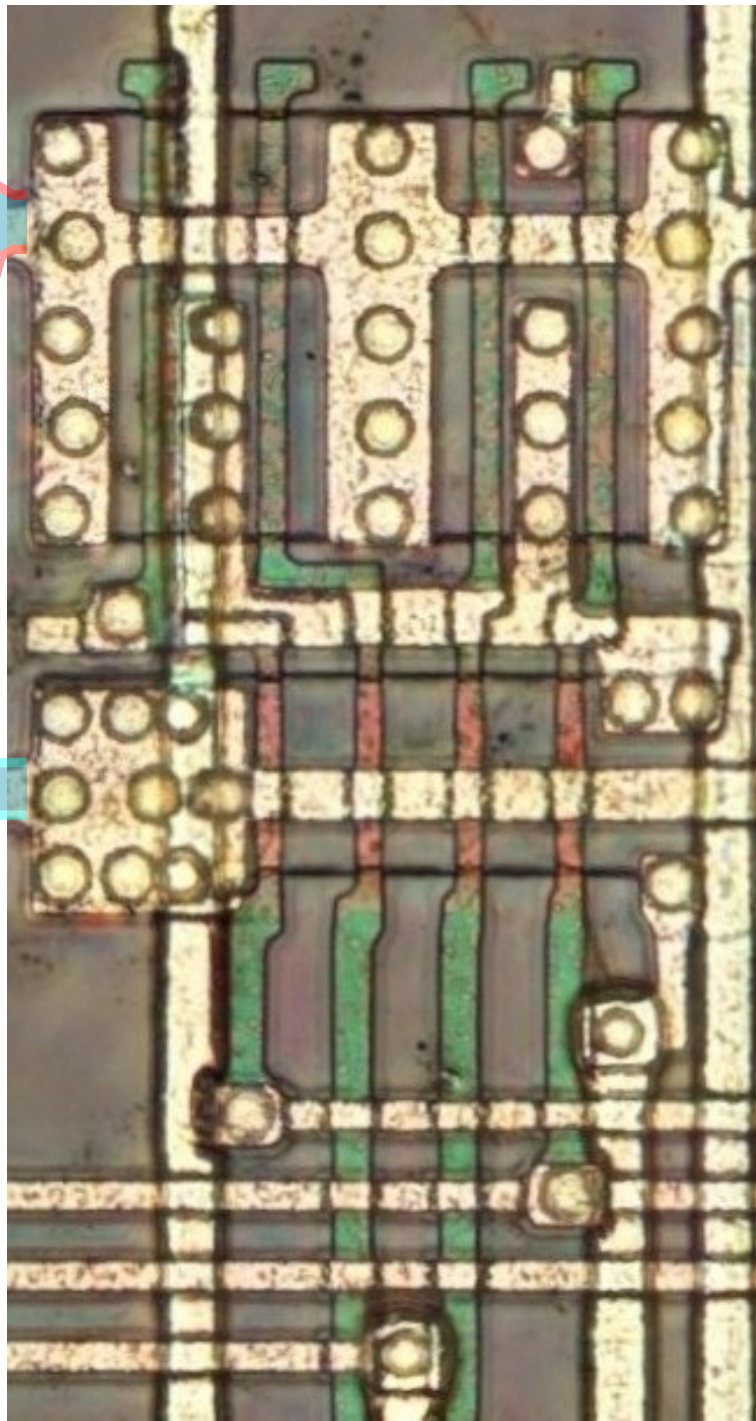
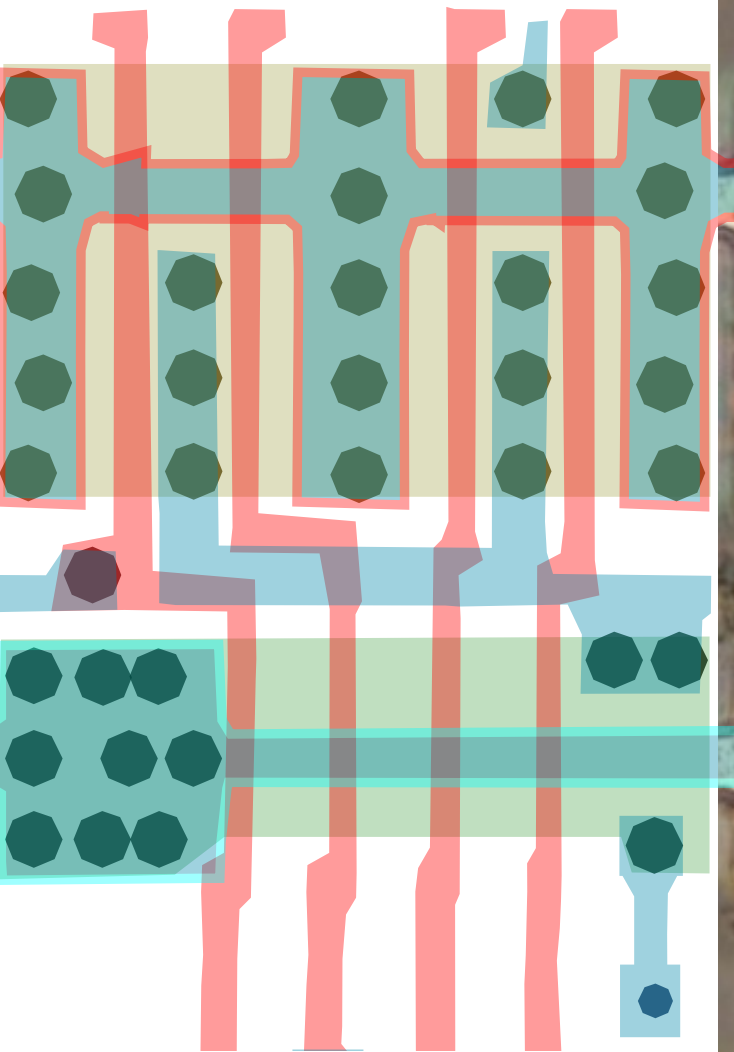


Same cell on
LA32
(MB87136A)

#00ffffff

4-Input NAND

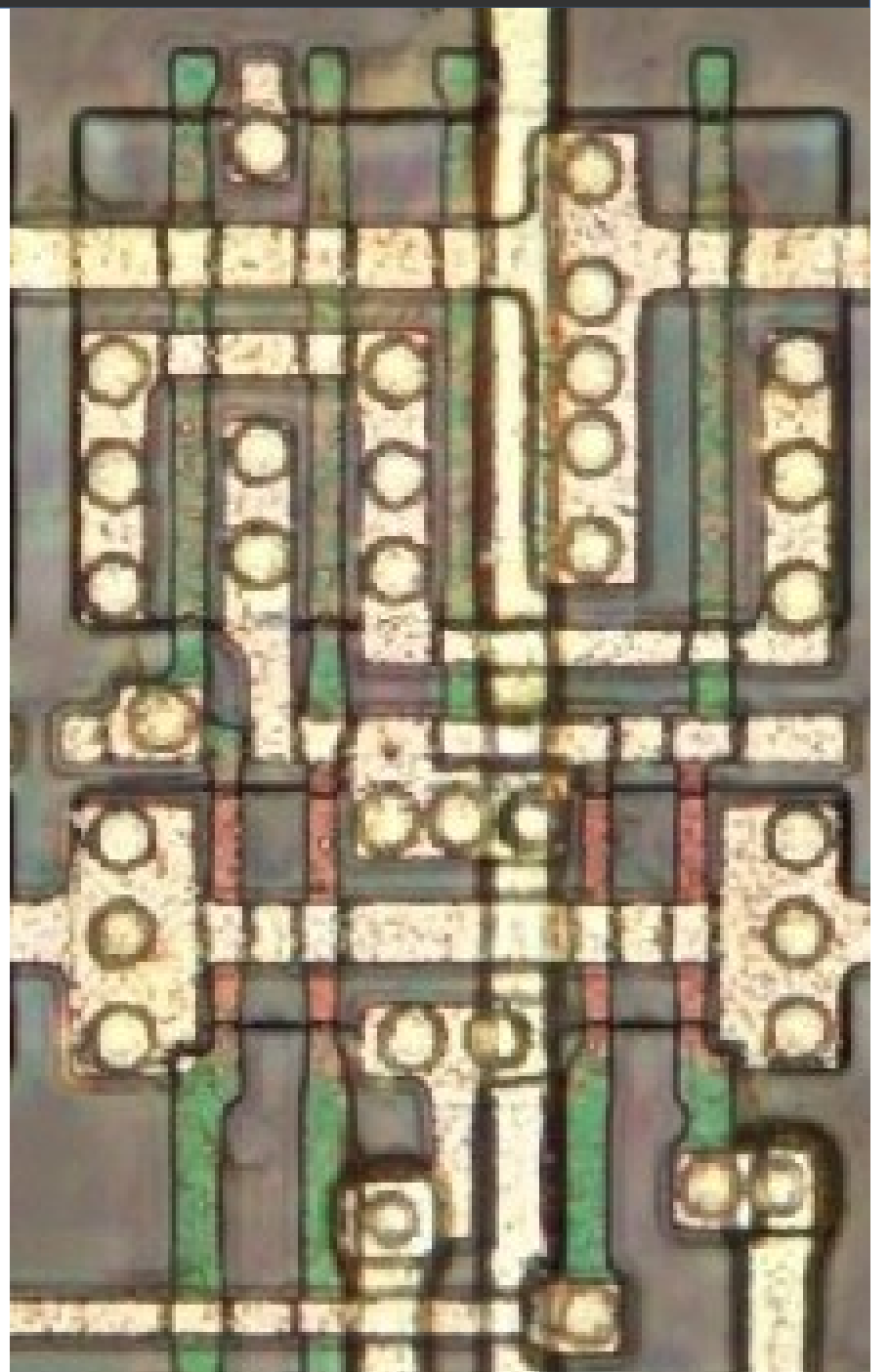
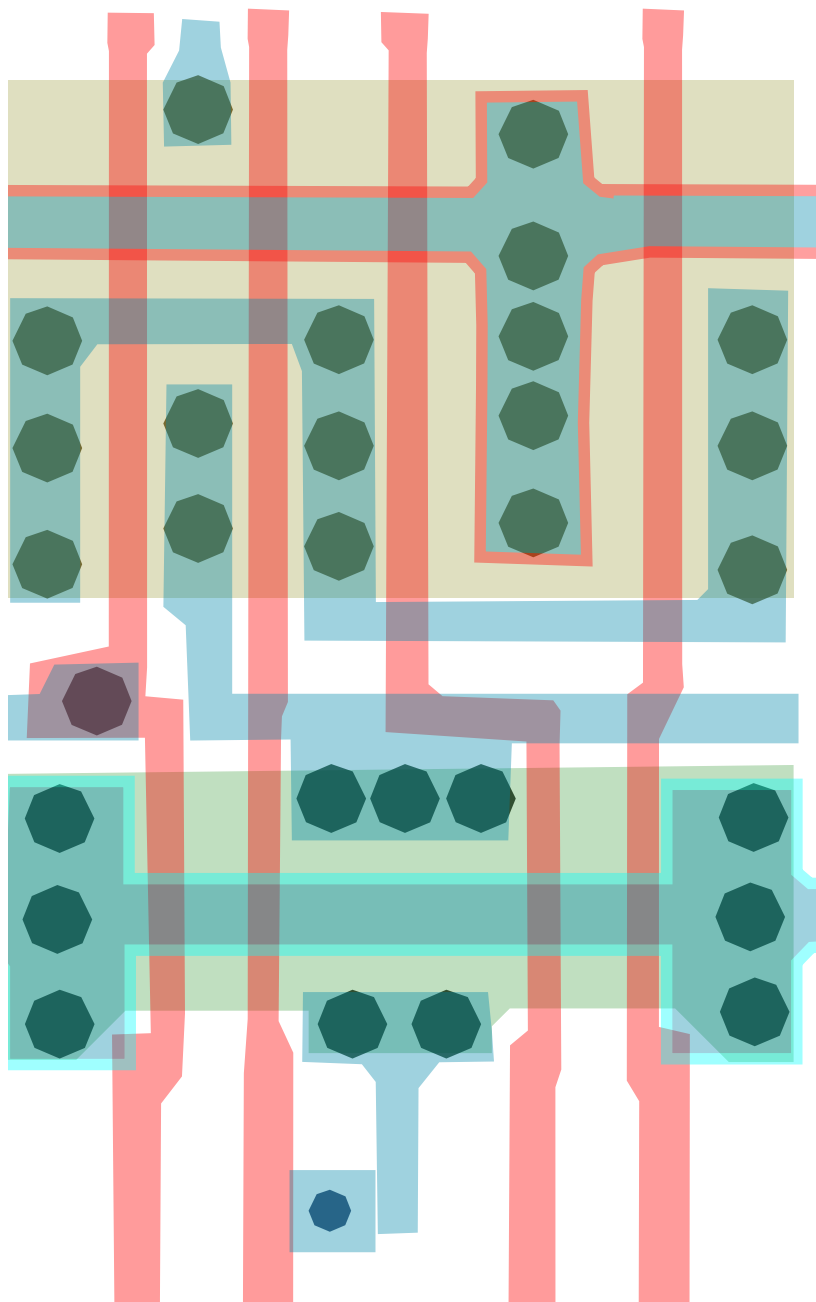
Cell 15



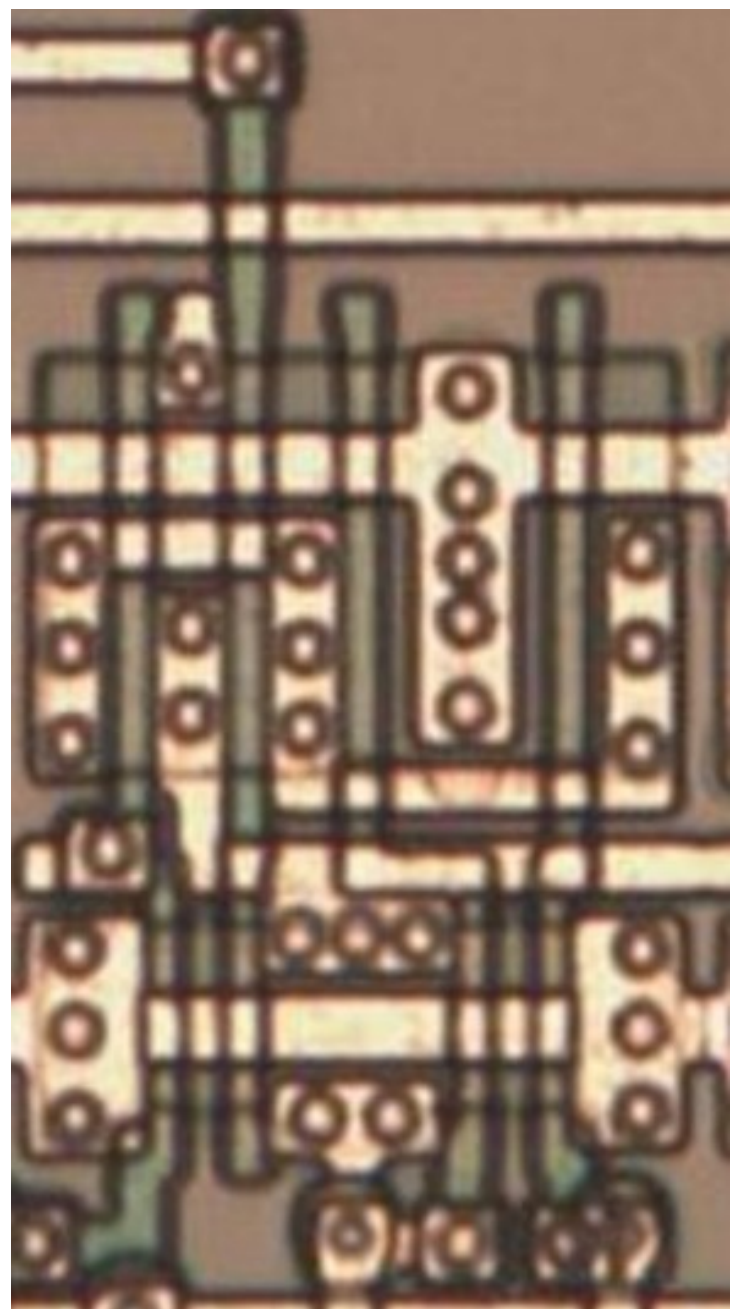
Same cell on
LA32
(MB87136A)

#ffb0ffff

Cell type 16

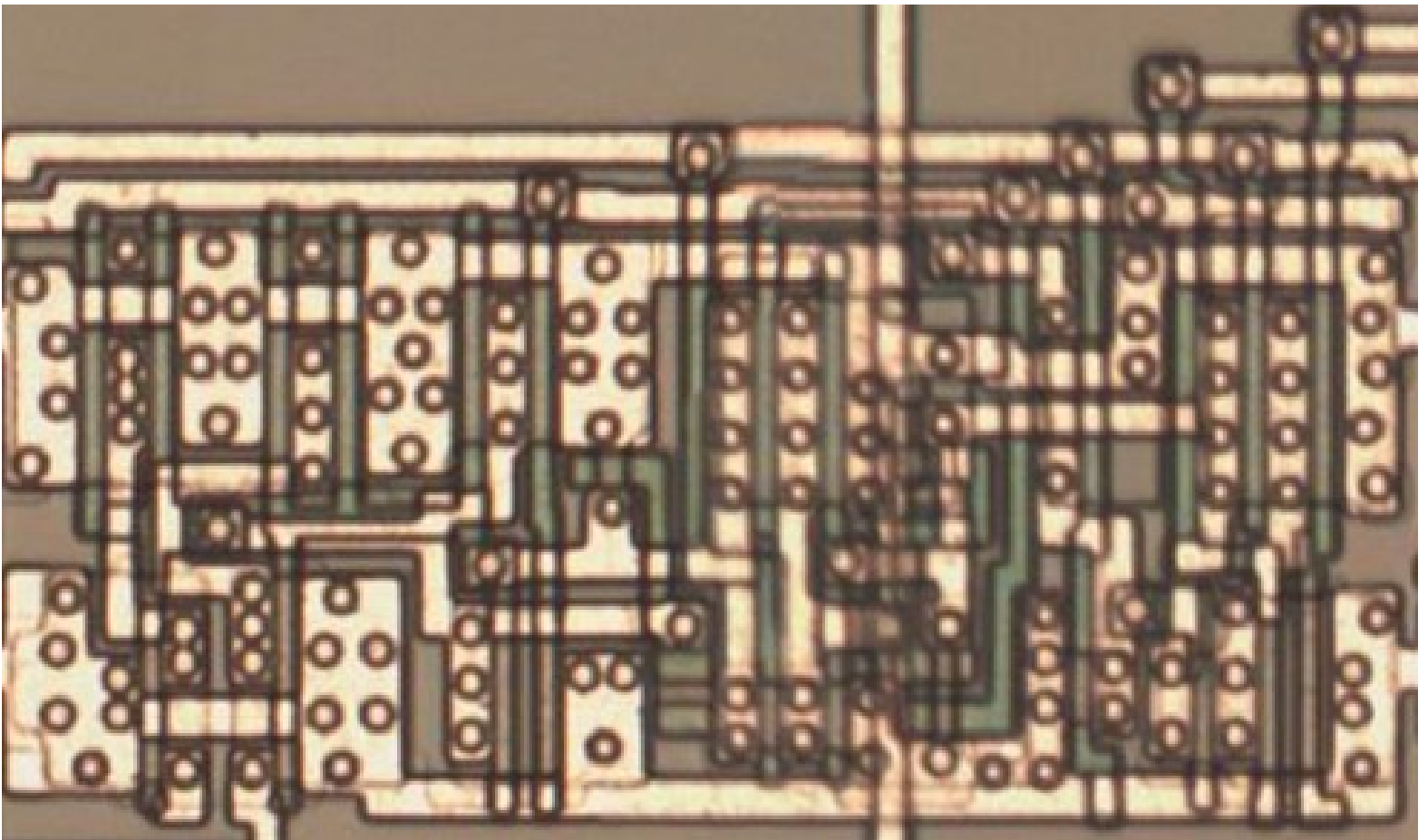
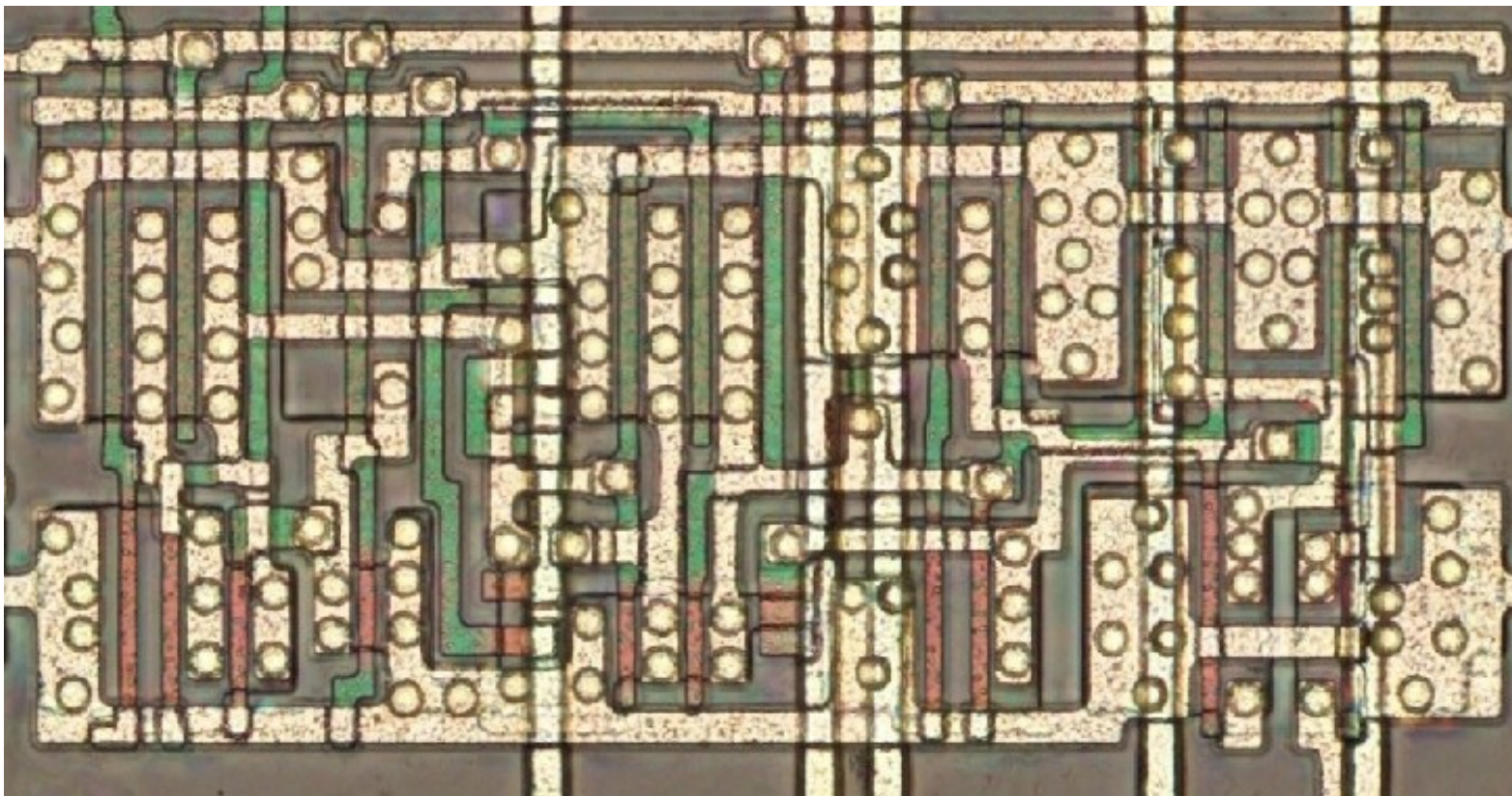
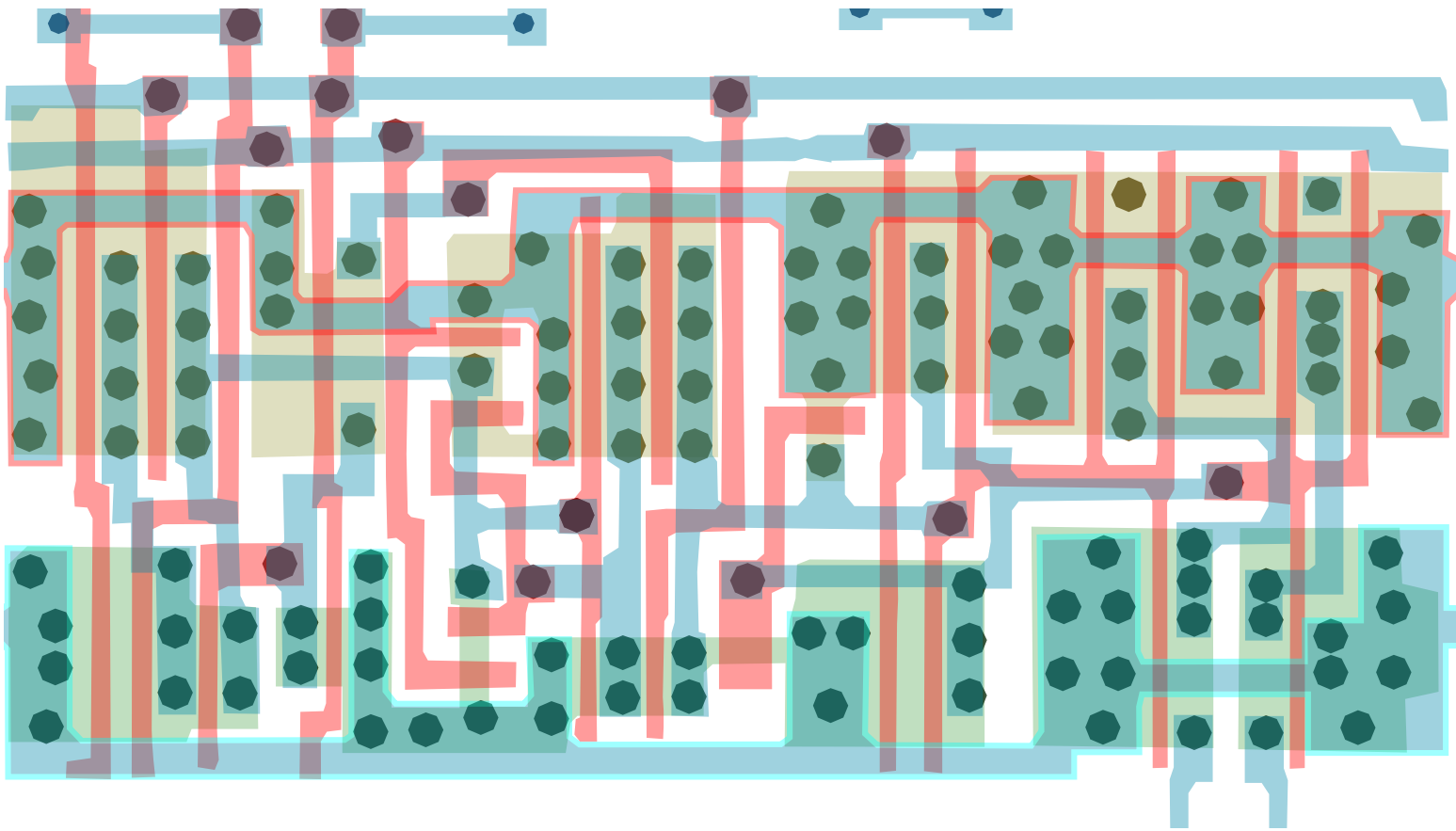


Same cell on
LA32
(MB87136A)



#0000ffff

Cell type 17

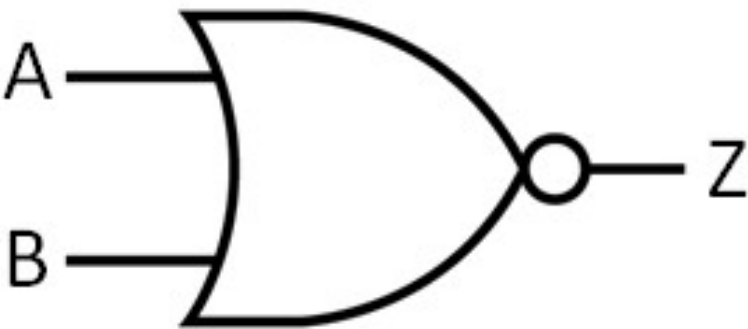


Same cell on
LA32
(mirrored)
(MB87136A)

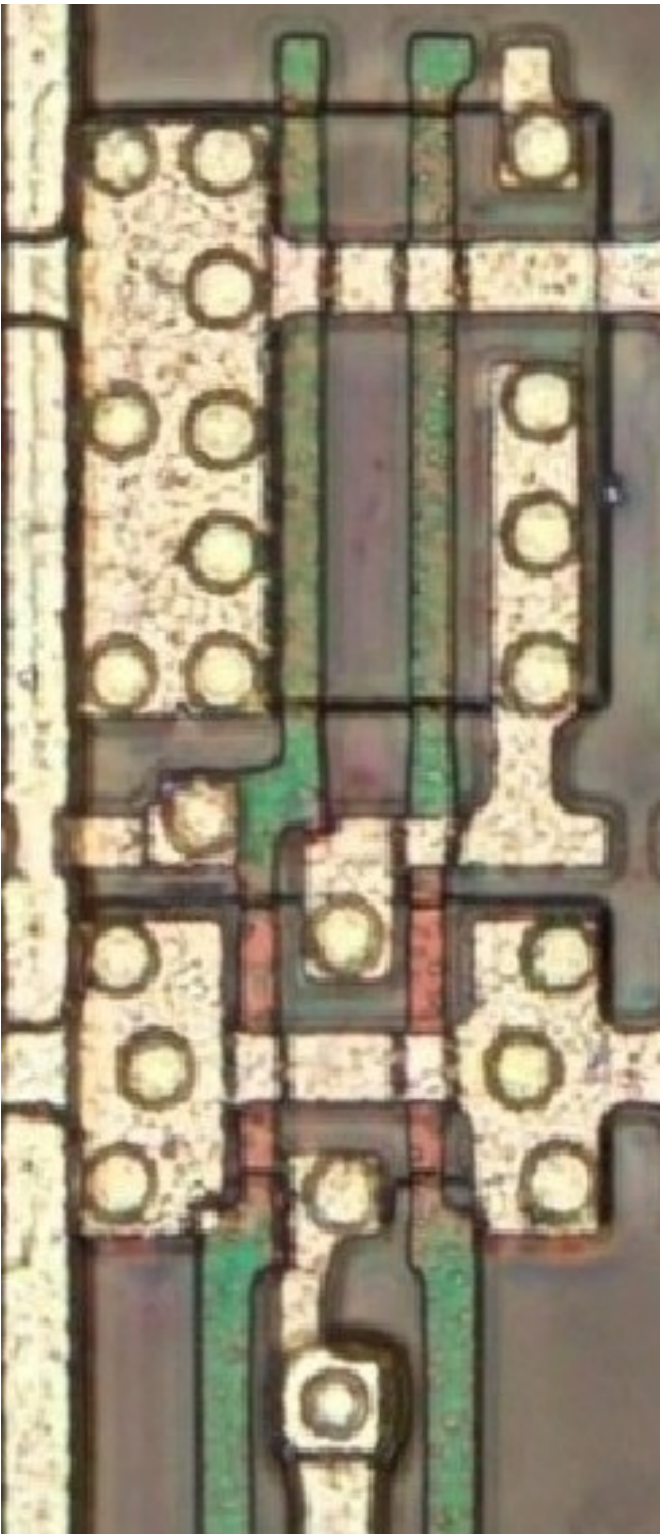
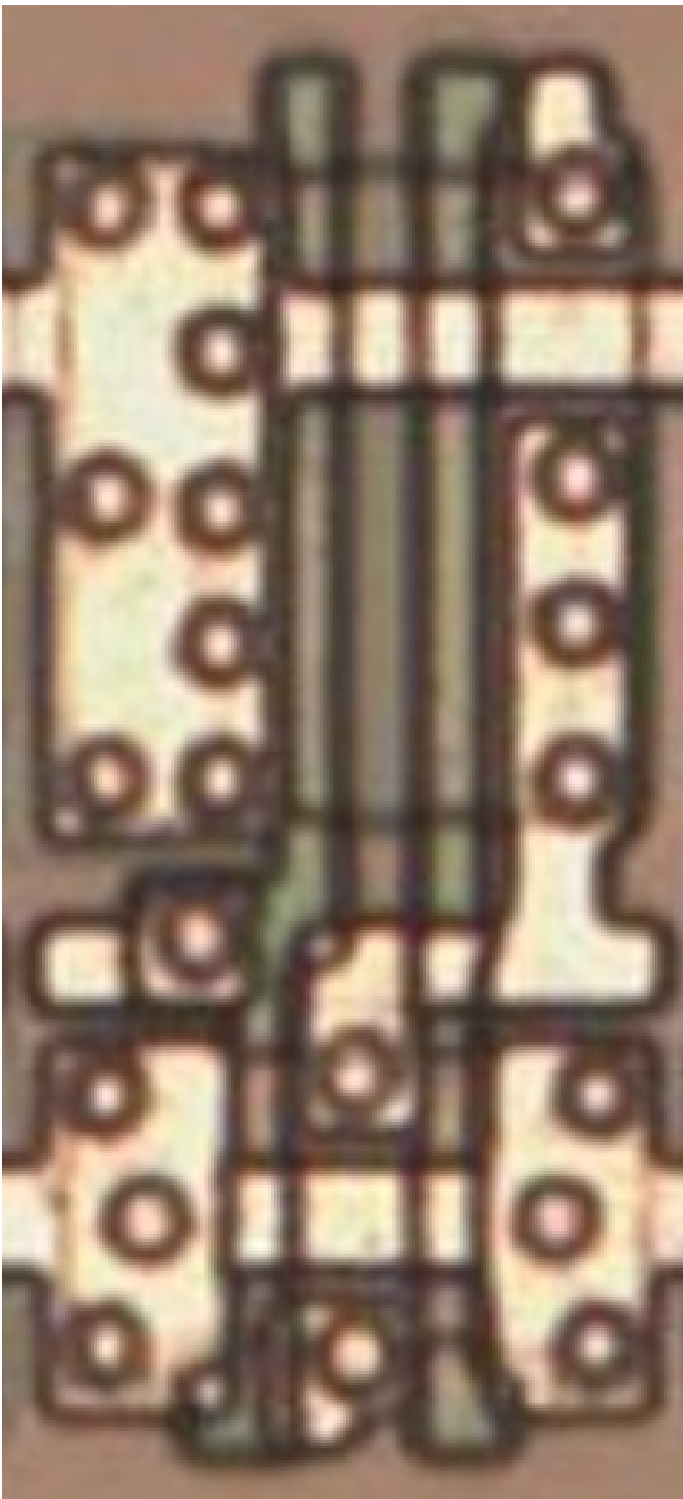
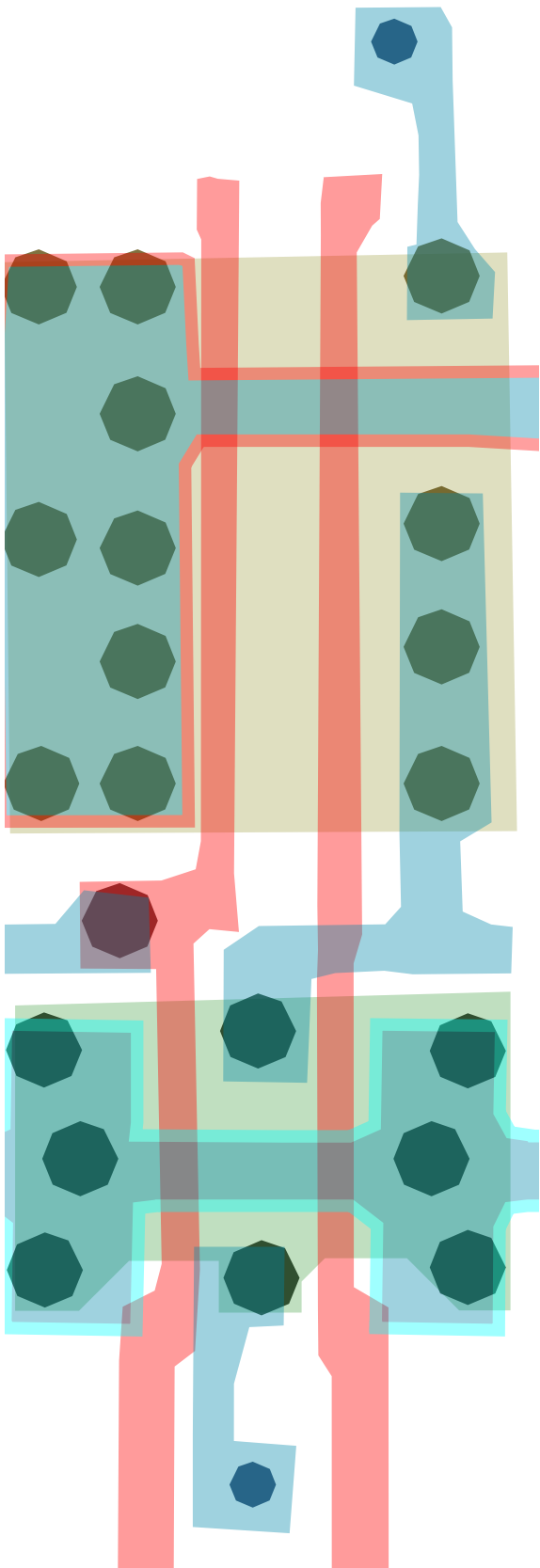
#005600ff

2-Input NOR

Cell 18



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

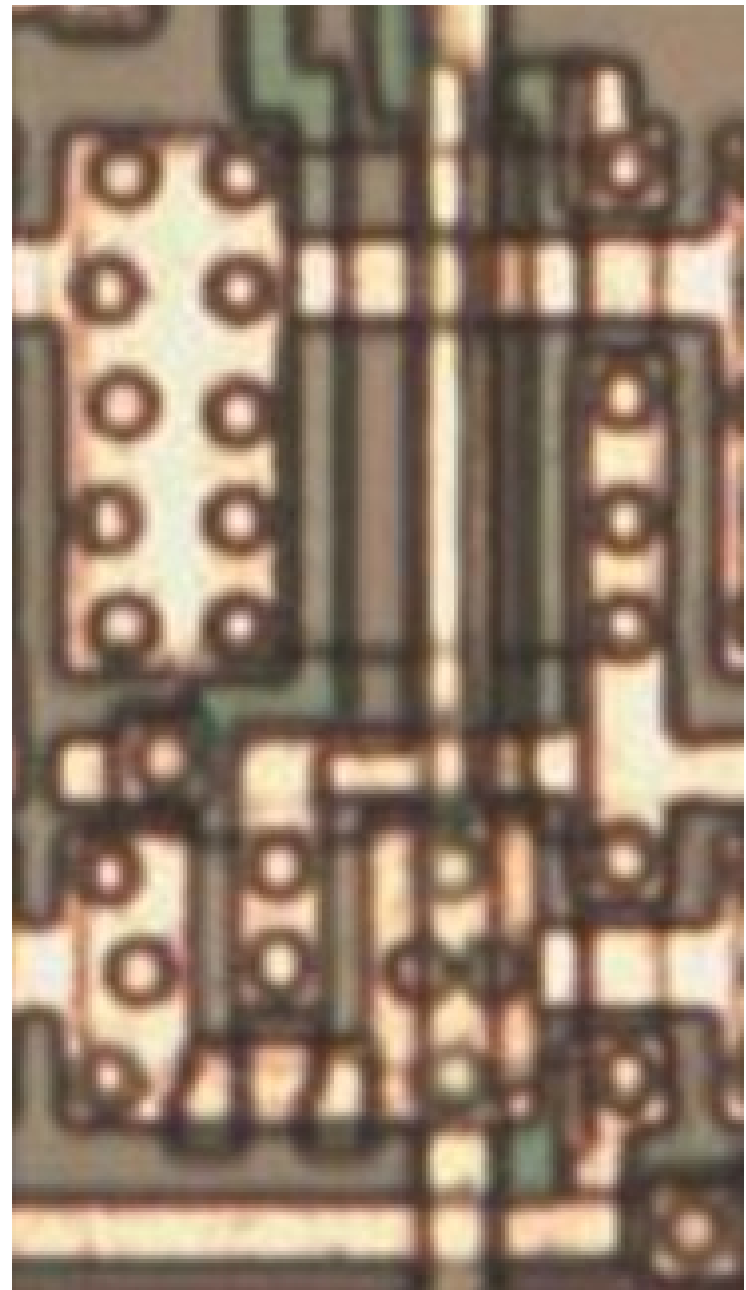
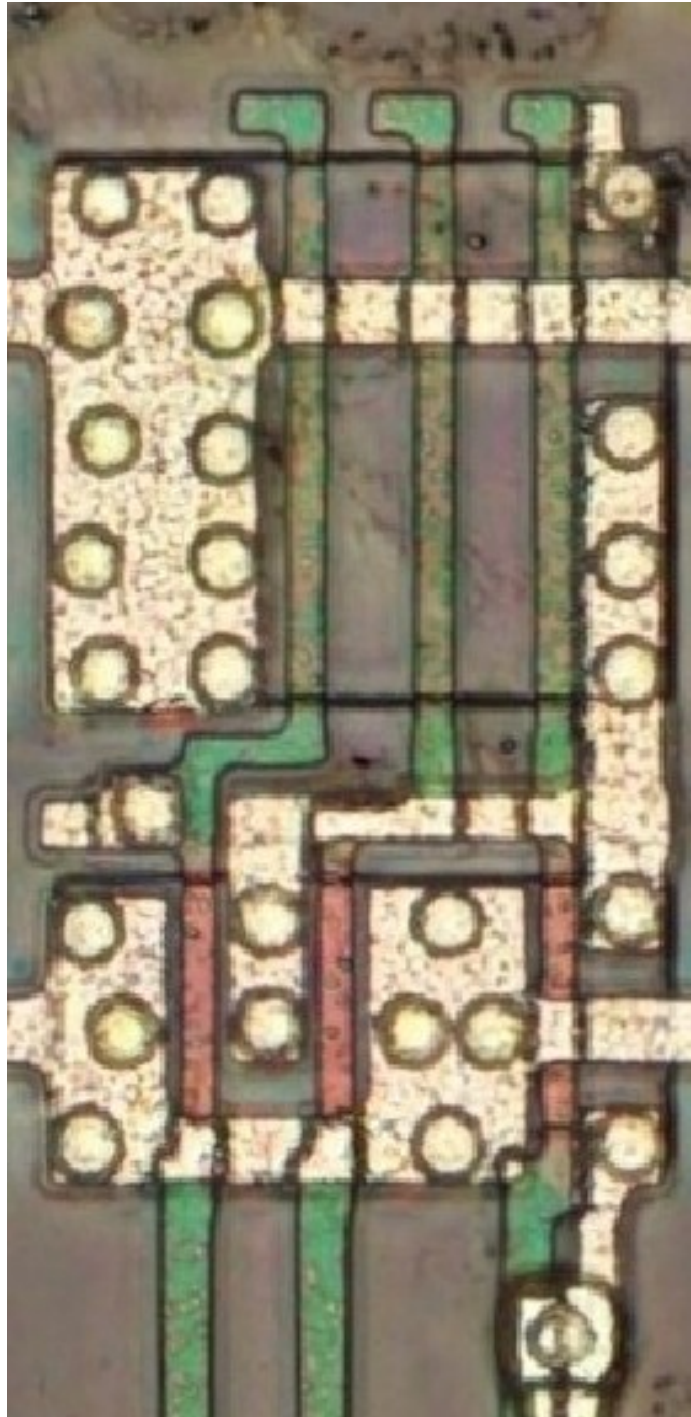
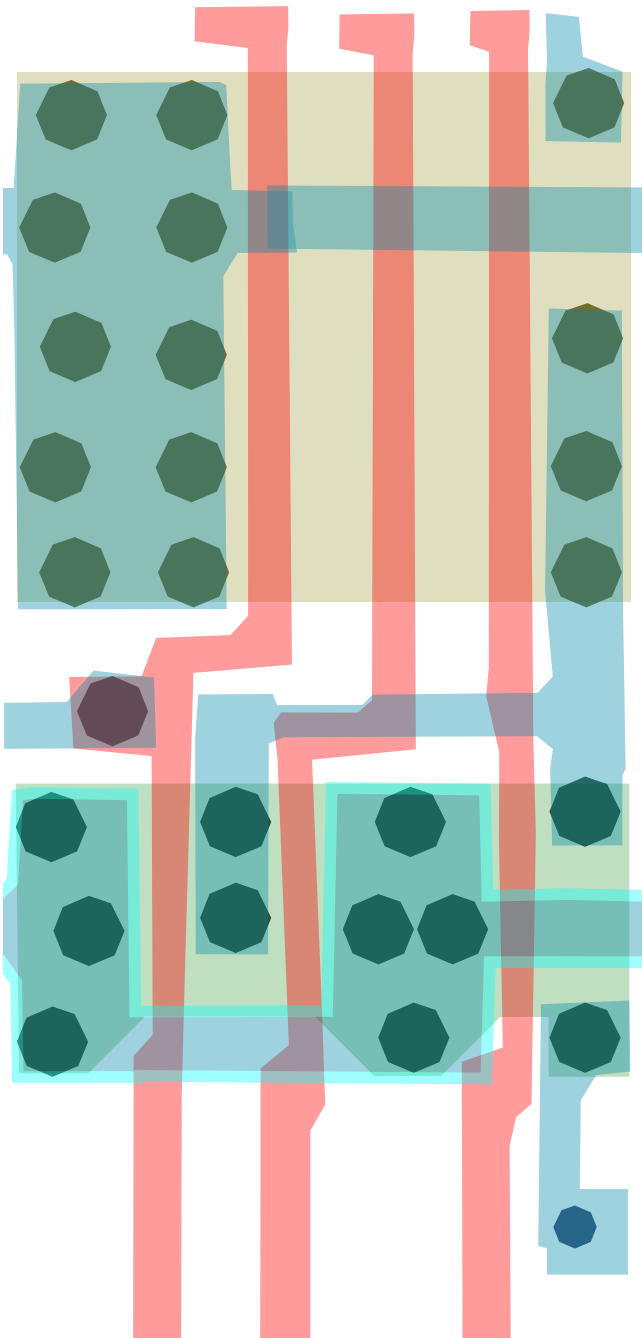


Same cell on
LA32
(MB87136A)

#970000ff

3-Input NOR

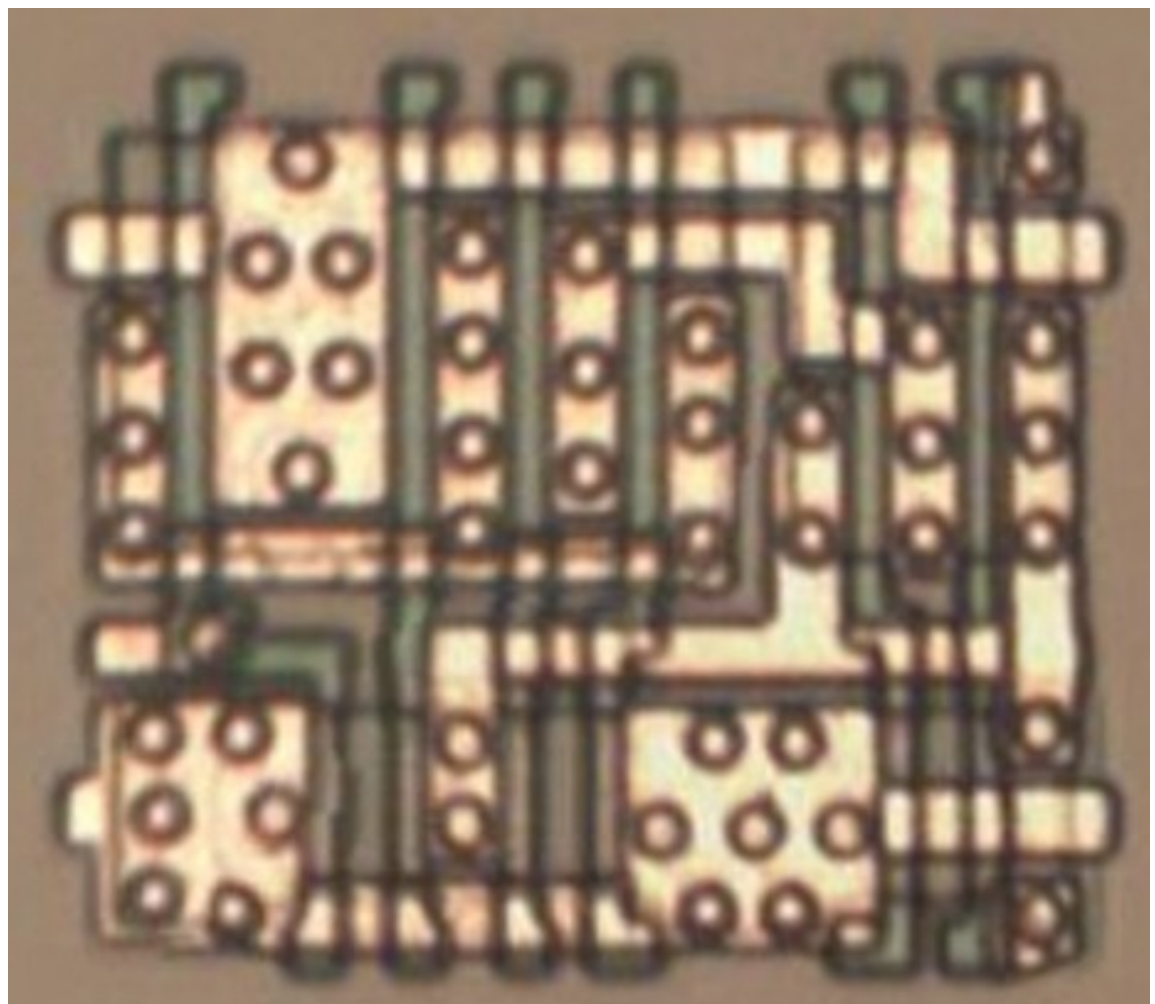
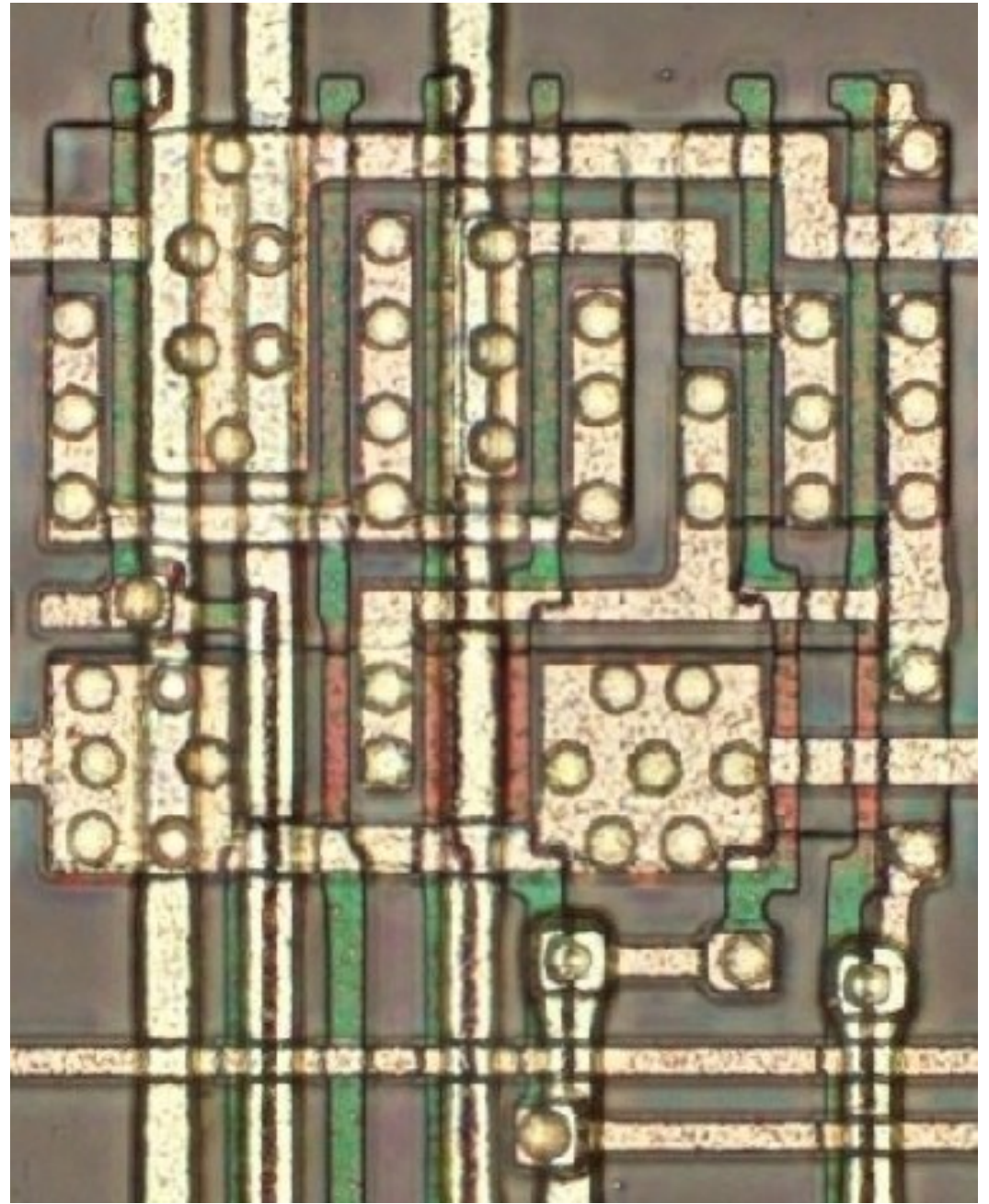
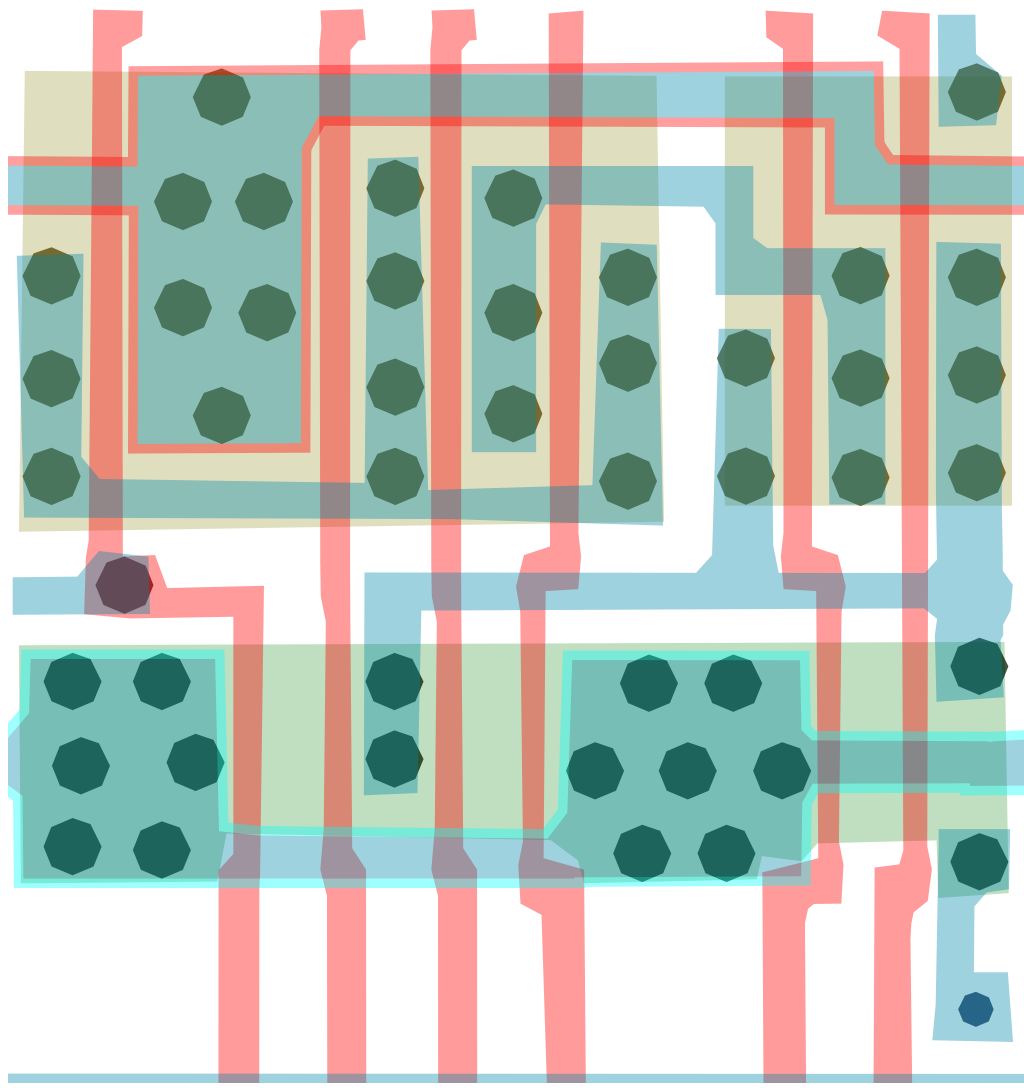
Cell 19



Same cell on
LA32
(MB87136A)

#ff0000ff

Cell type 20

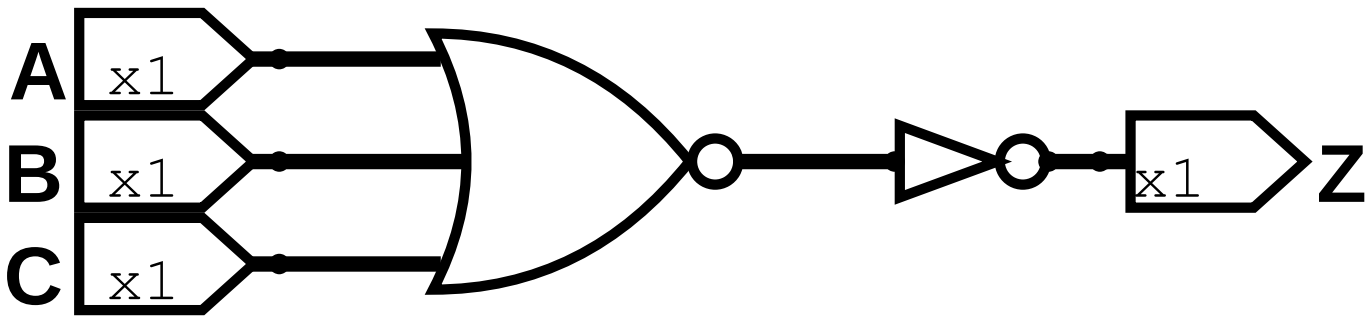


Same cell on
LA32
(MB87136A)

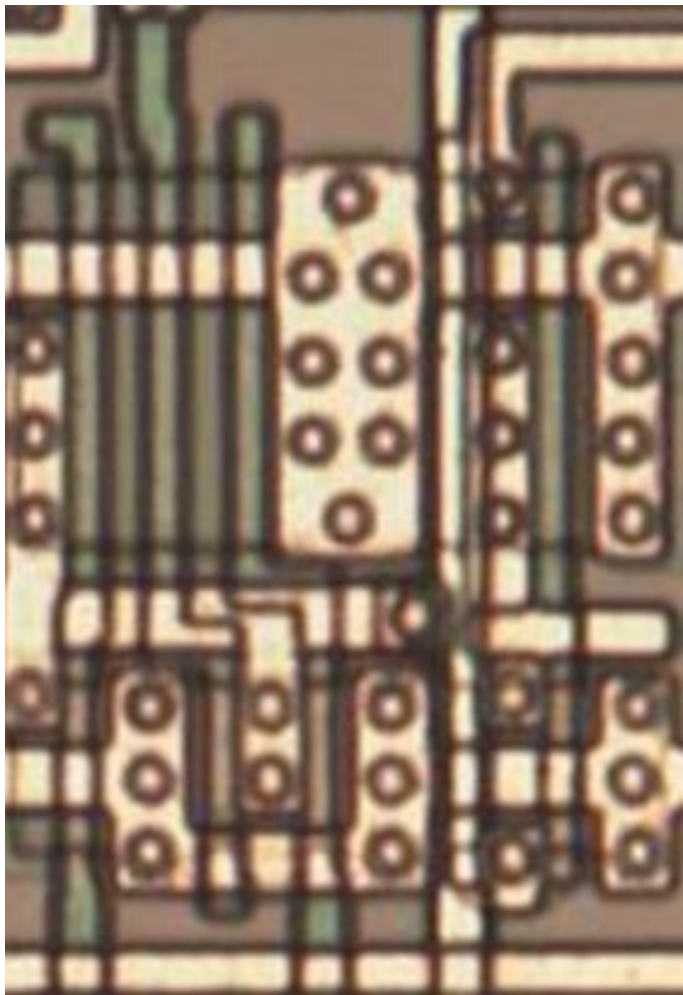
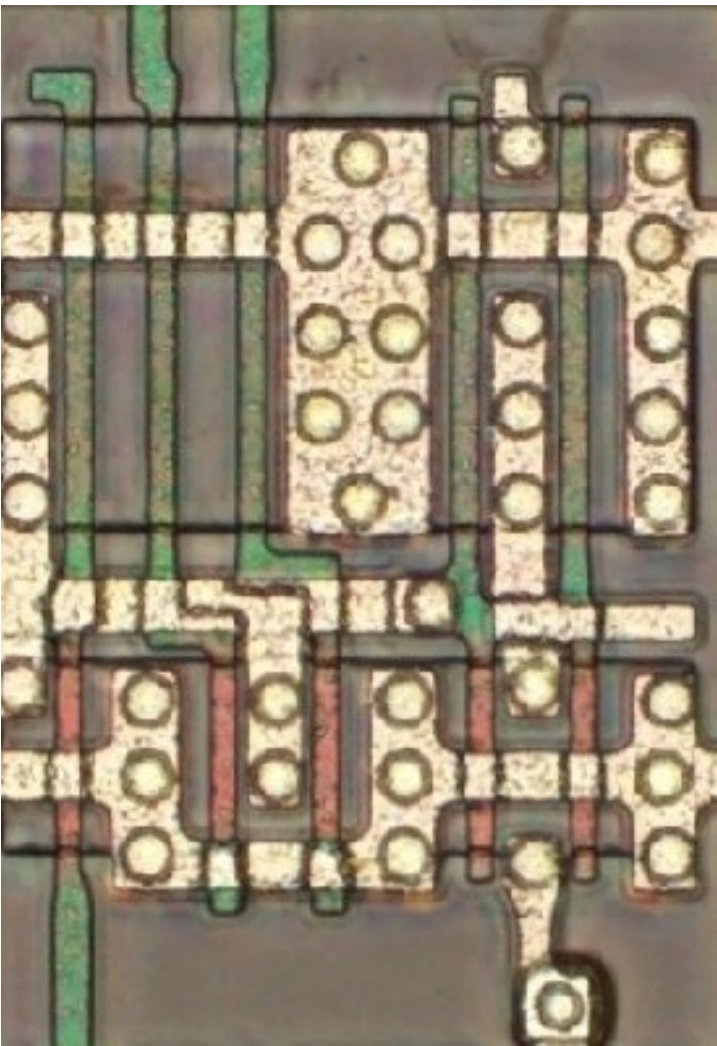
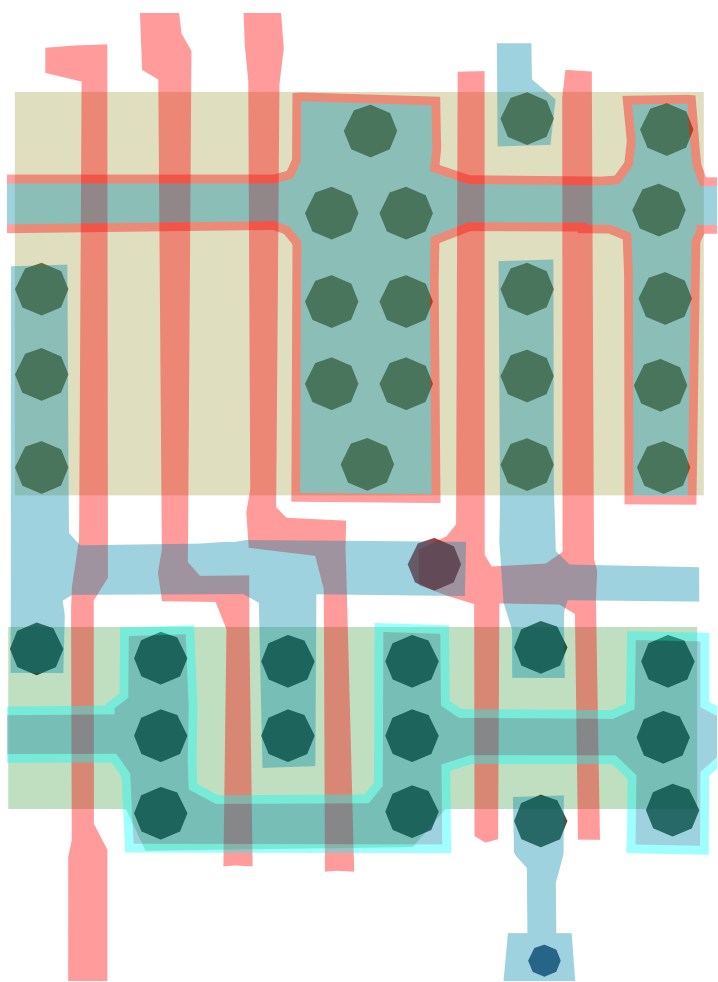
#218268ff

3-Input OR x1

Cell 21a



A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

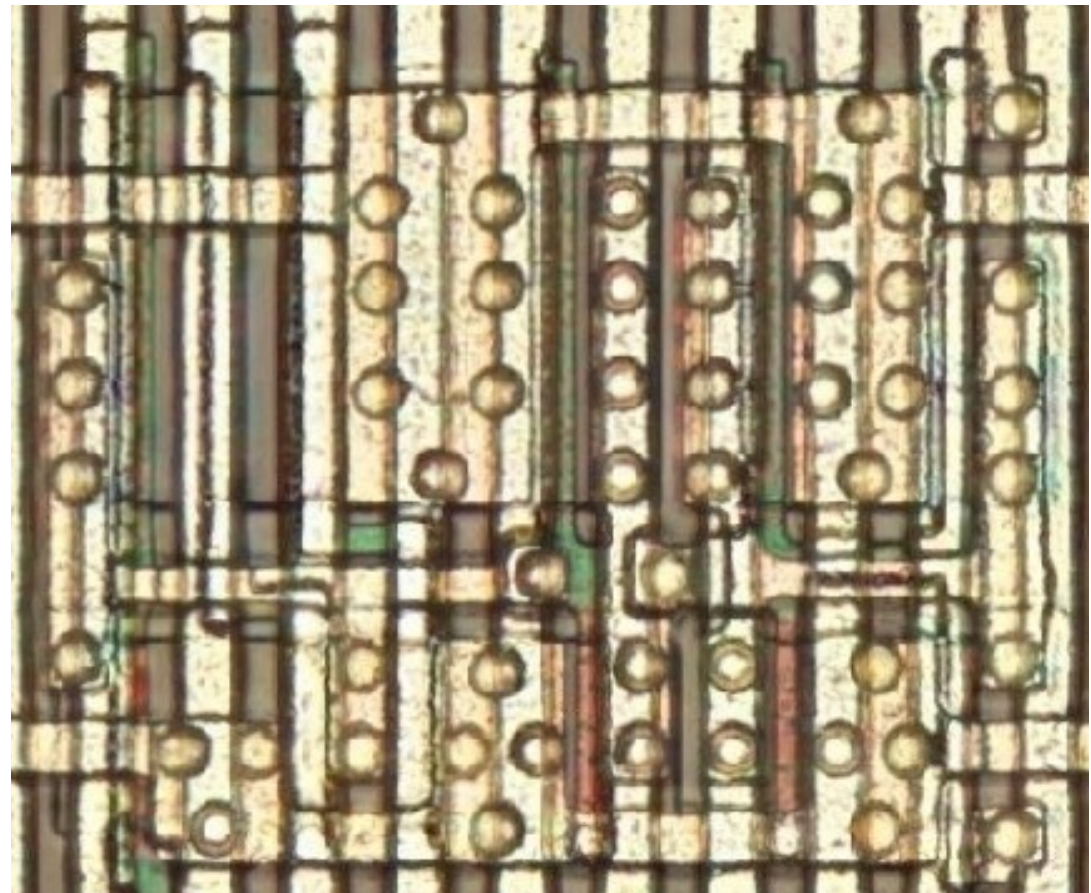
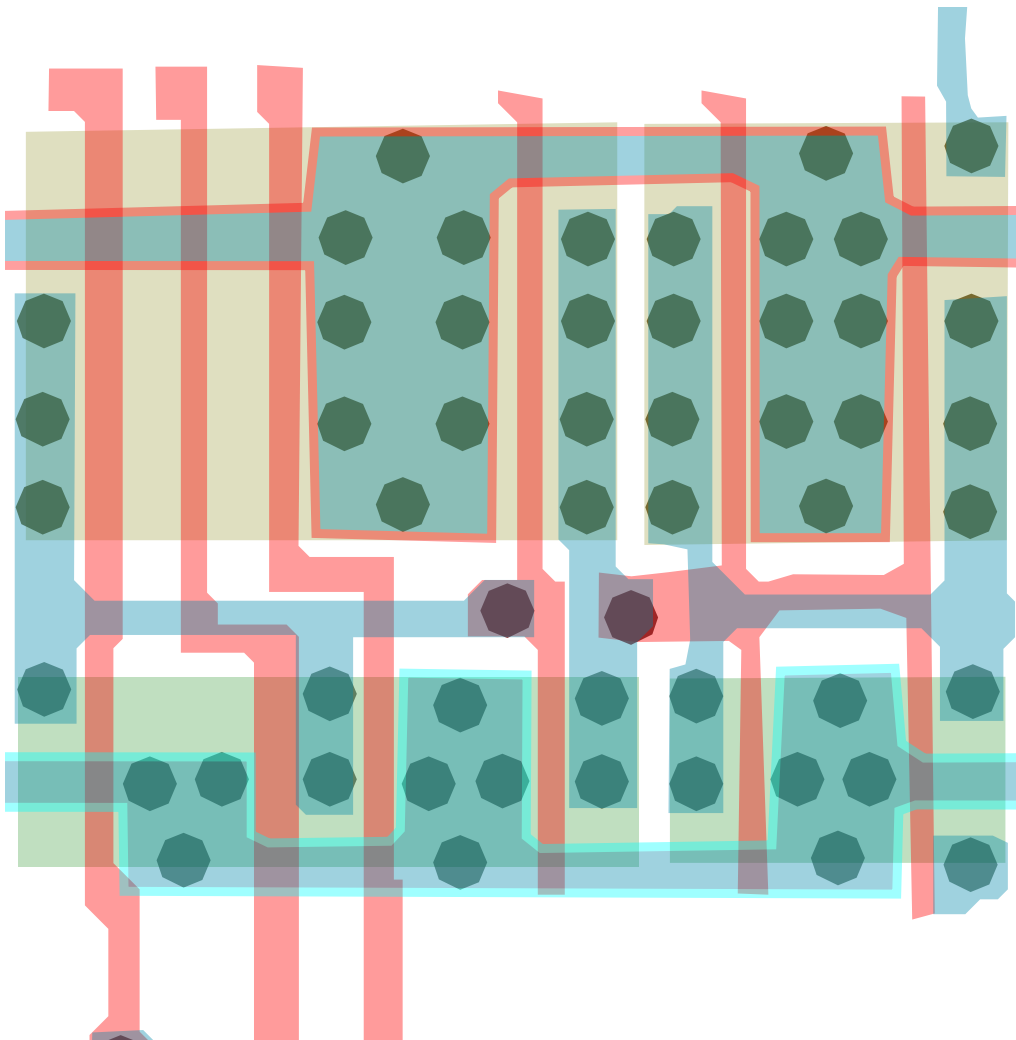
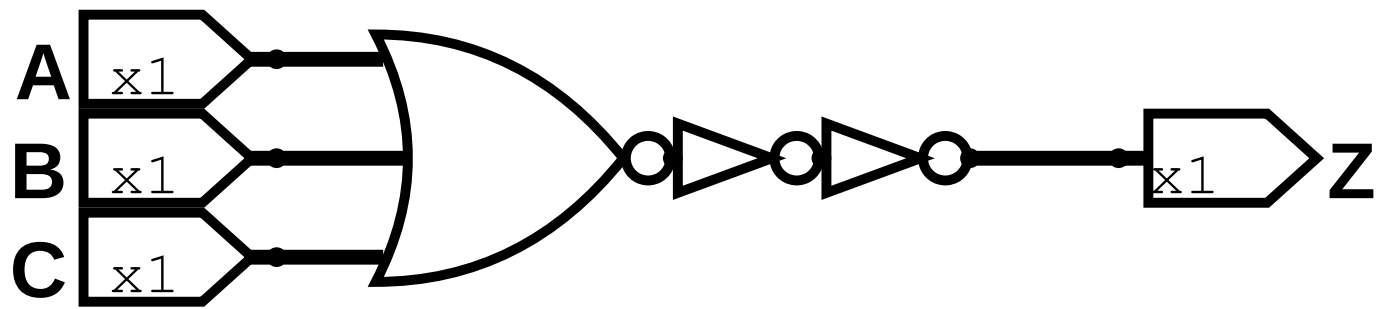


Same cell on
LA32
(MB87136A)

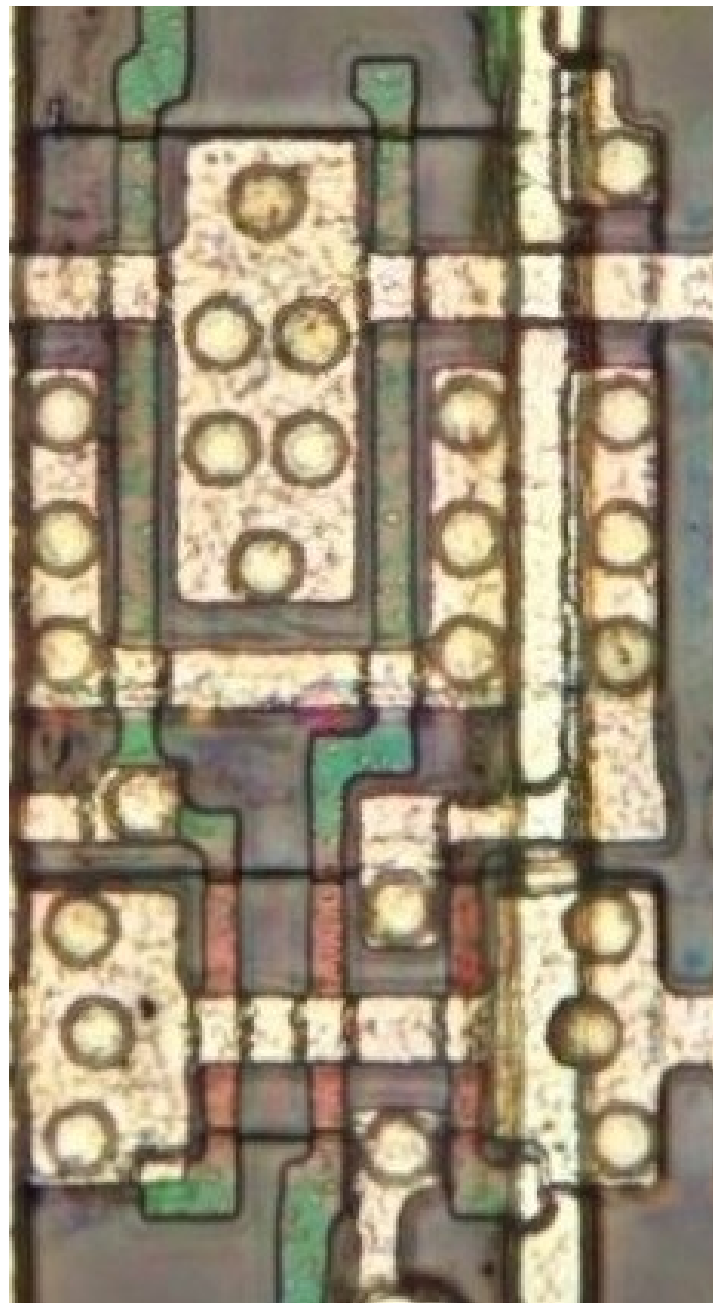
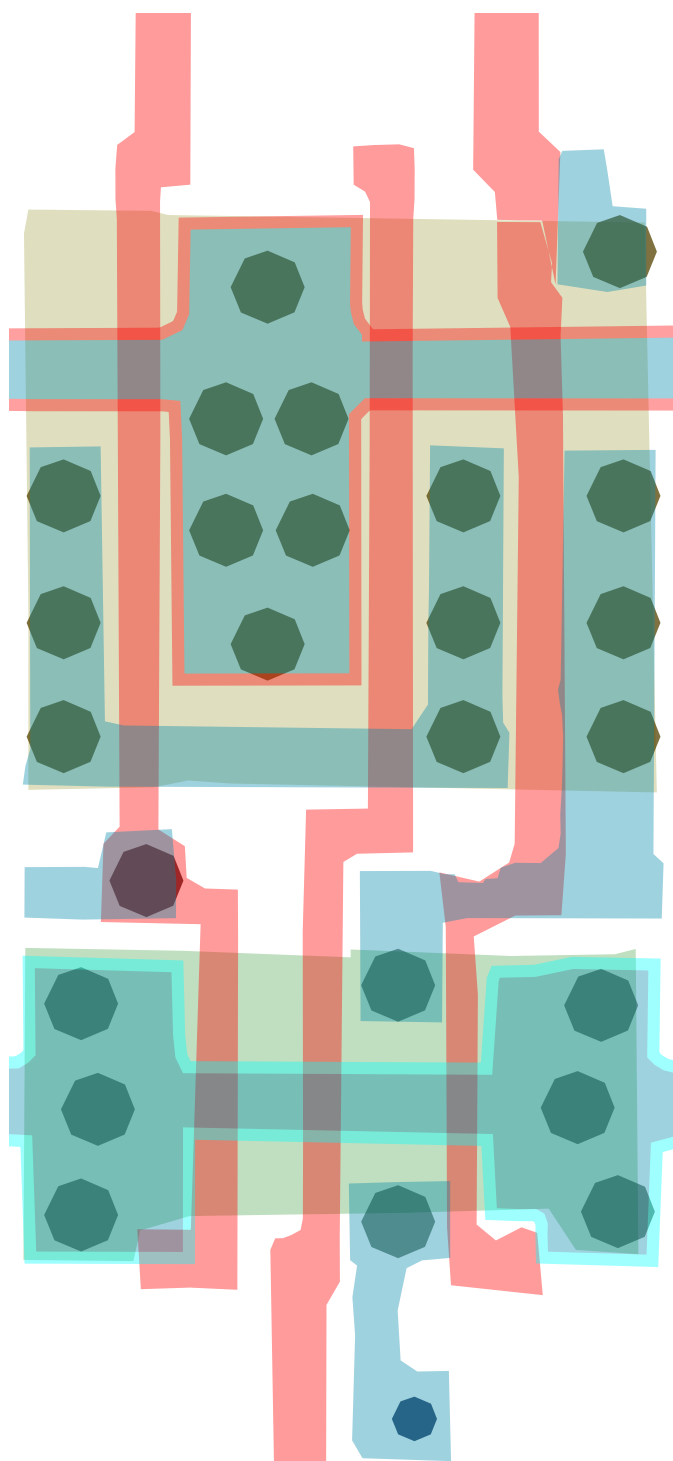
#115140ff

3-Input NOR

Cell 21b

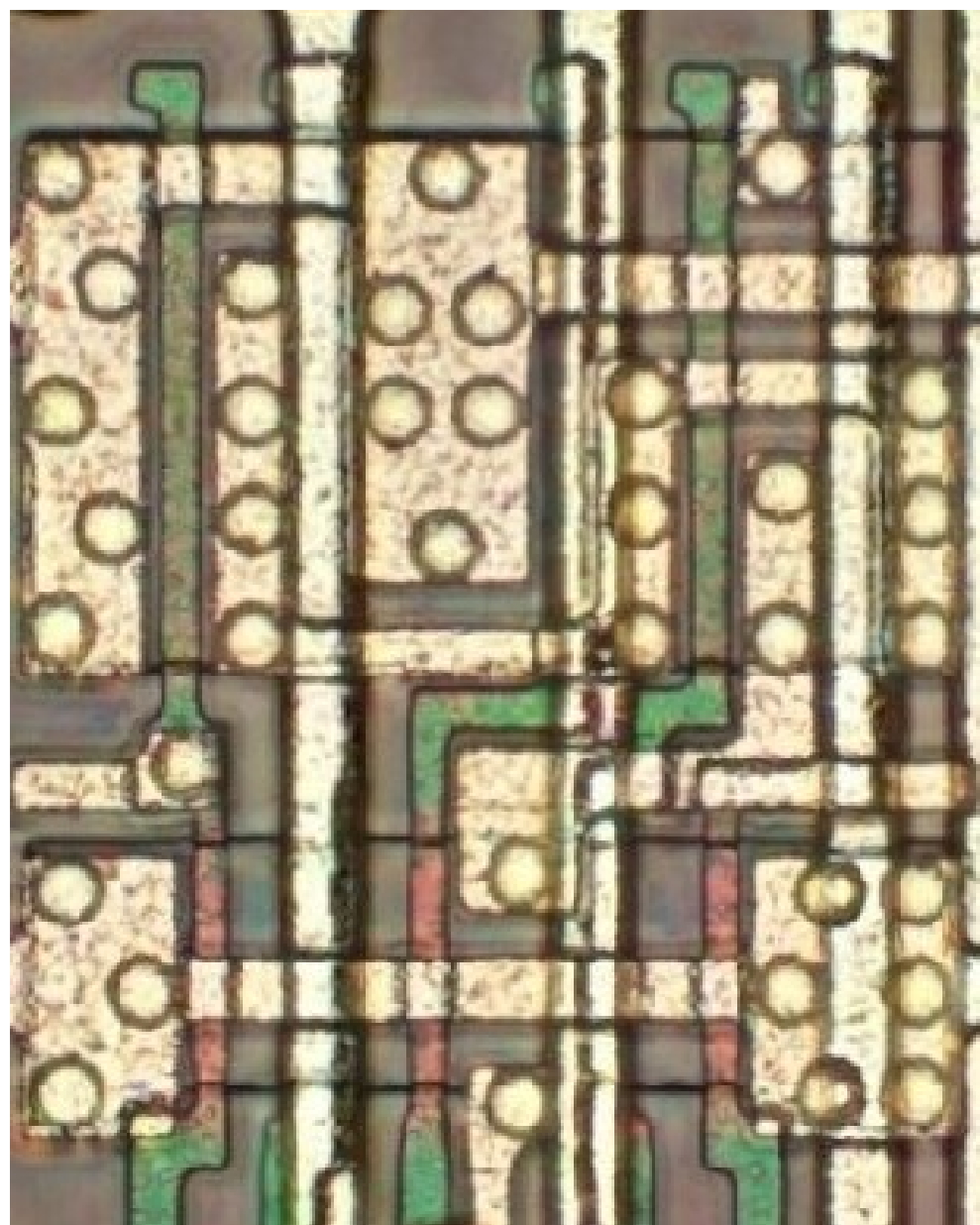
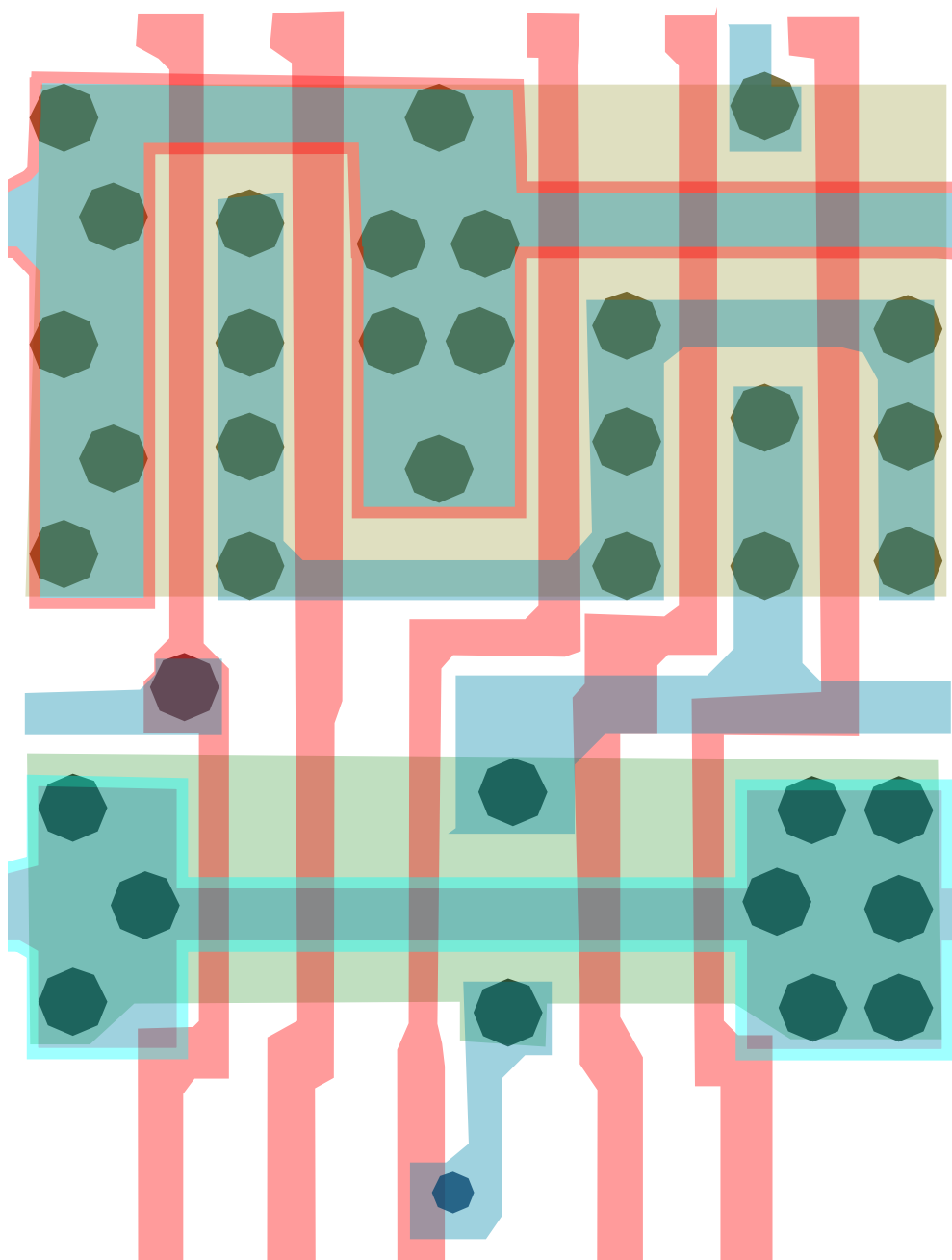


#d1796bff Cell type 24a



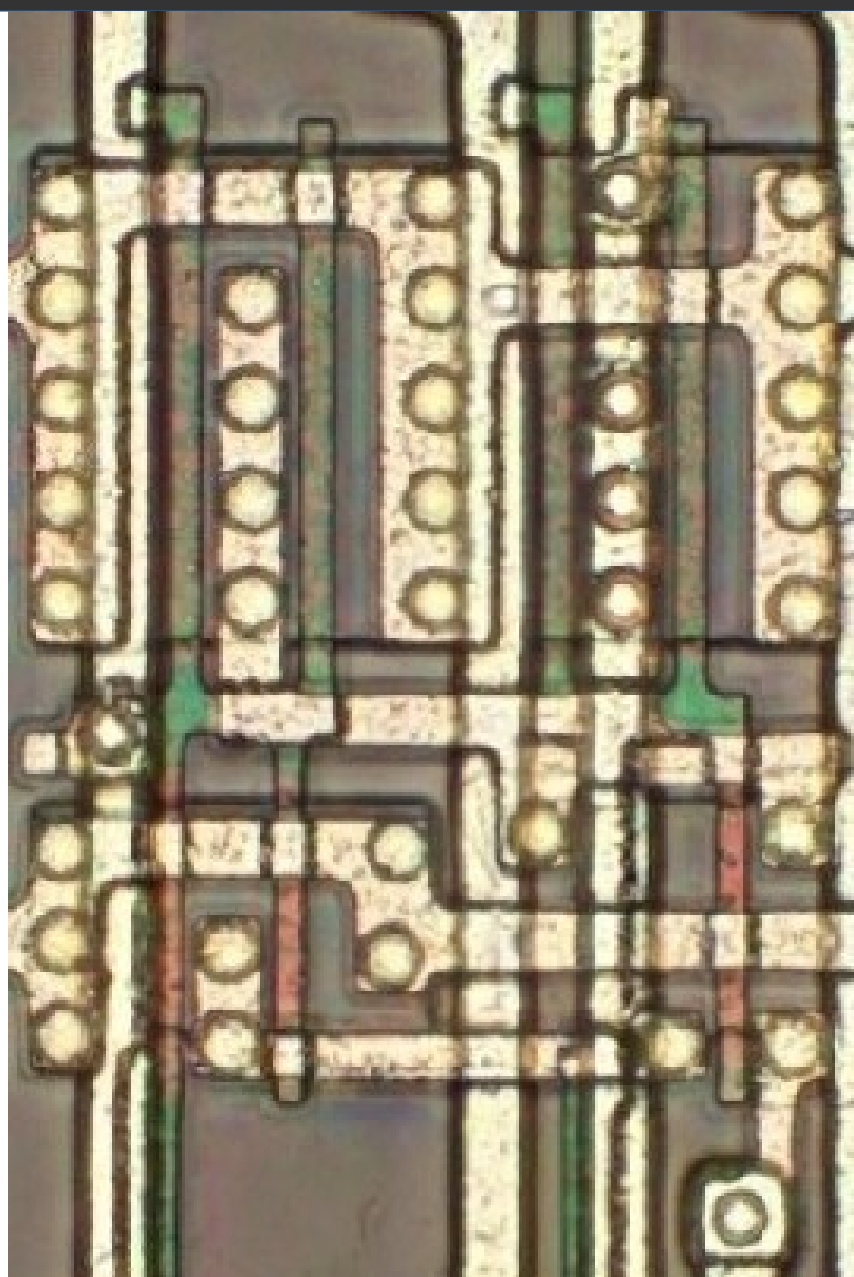
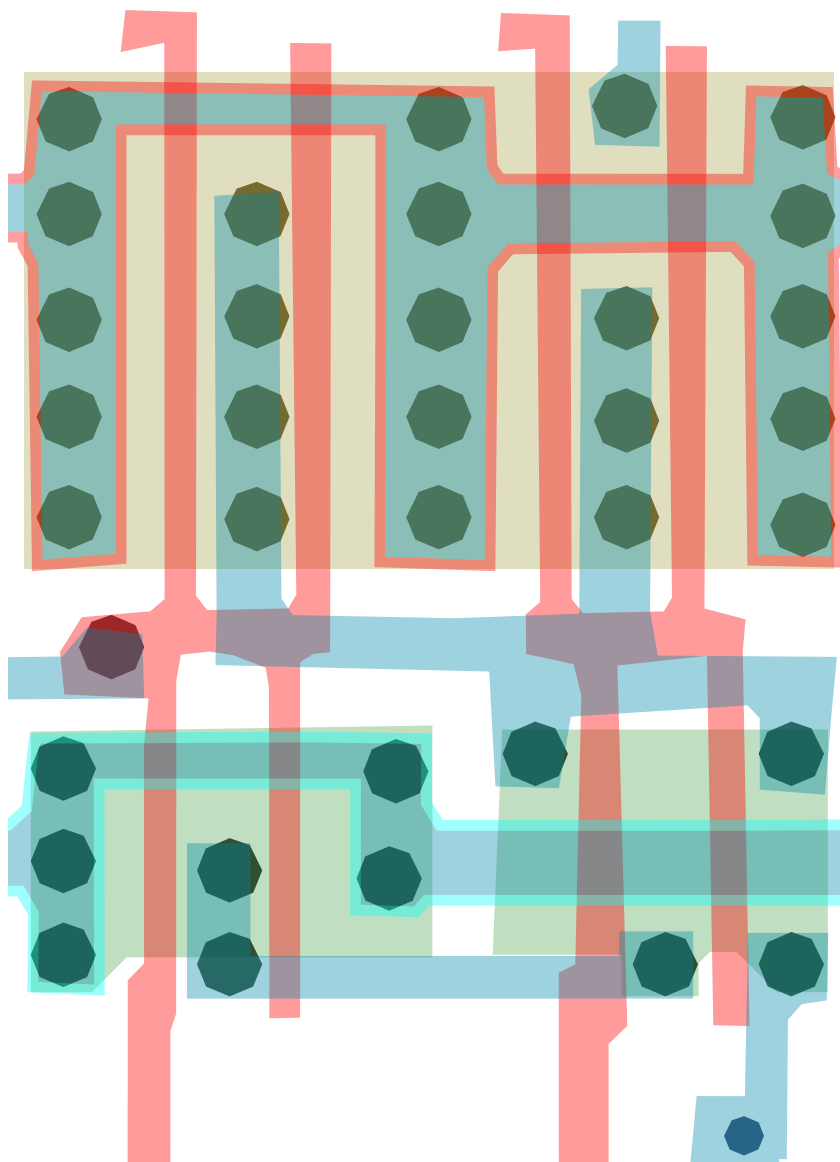
#7f8157ff

Cell type 24b



#ff9972ff

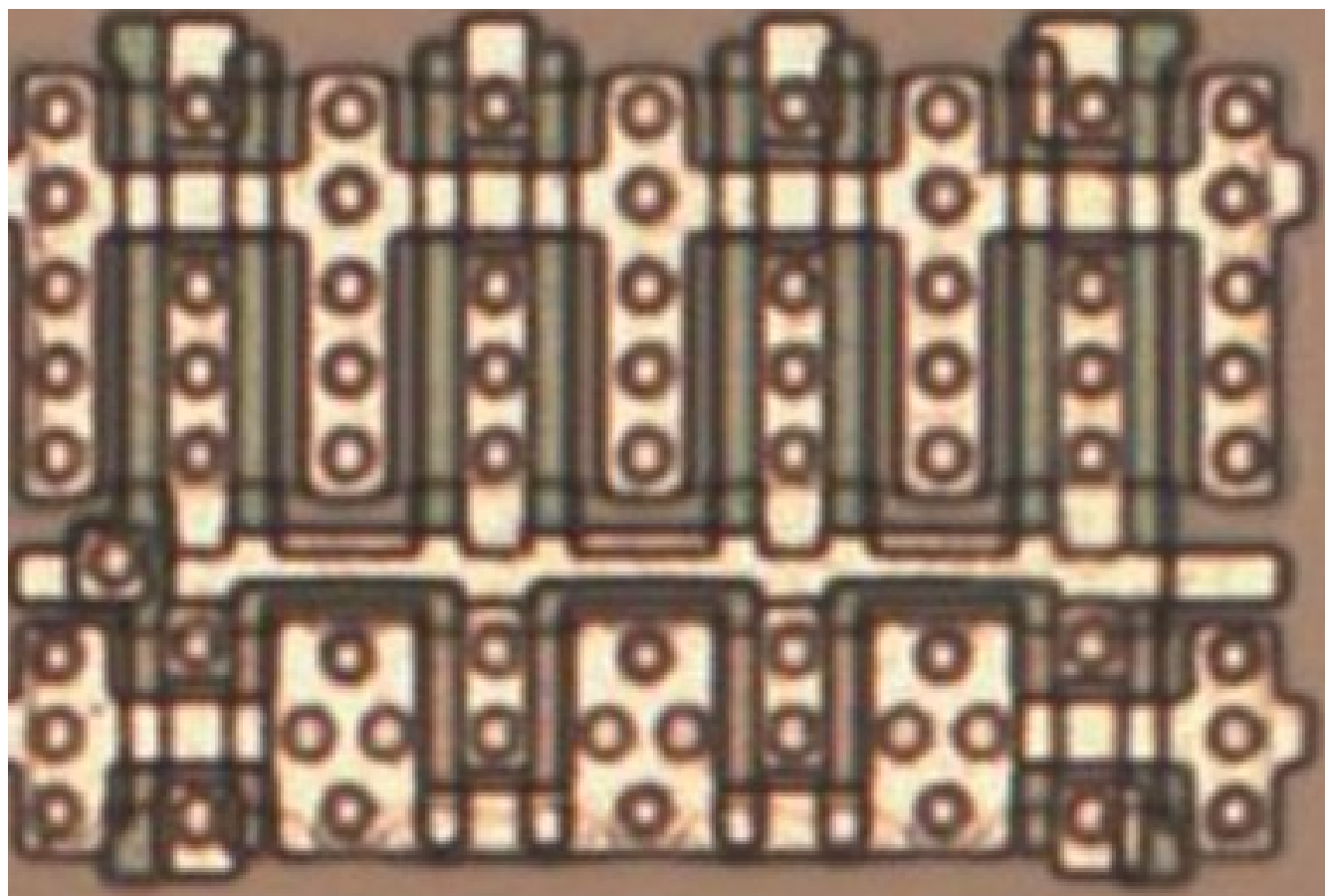
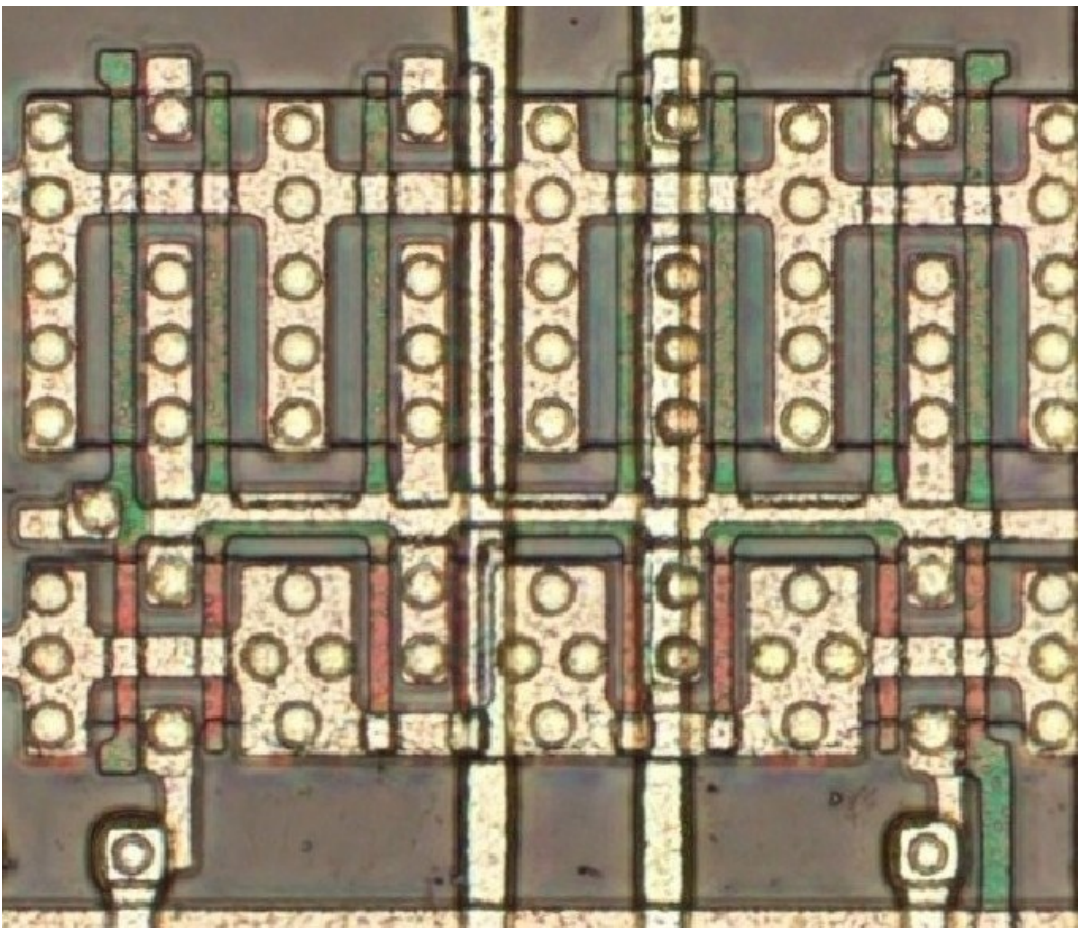
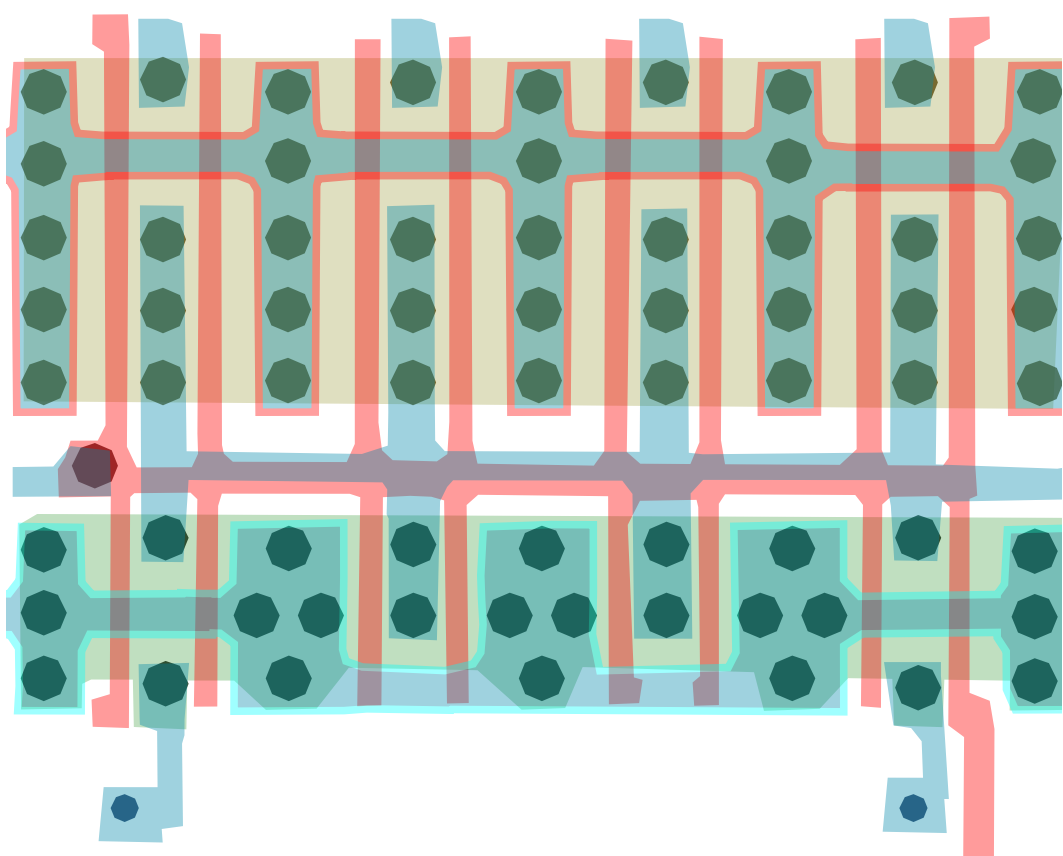
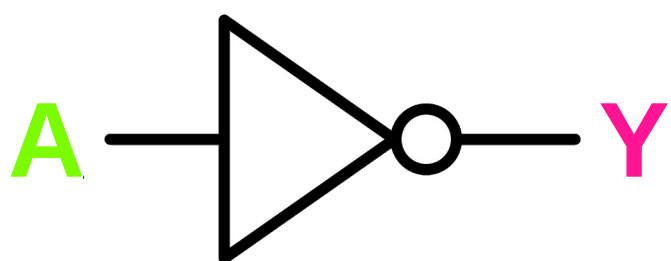
Cell type 25



#ffb94cff

Inverter/NOT x8

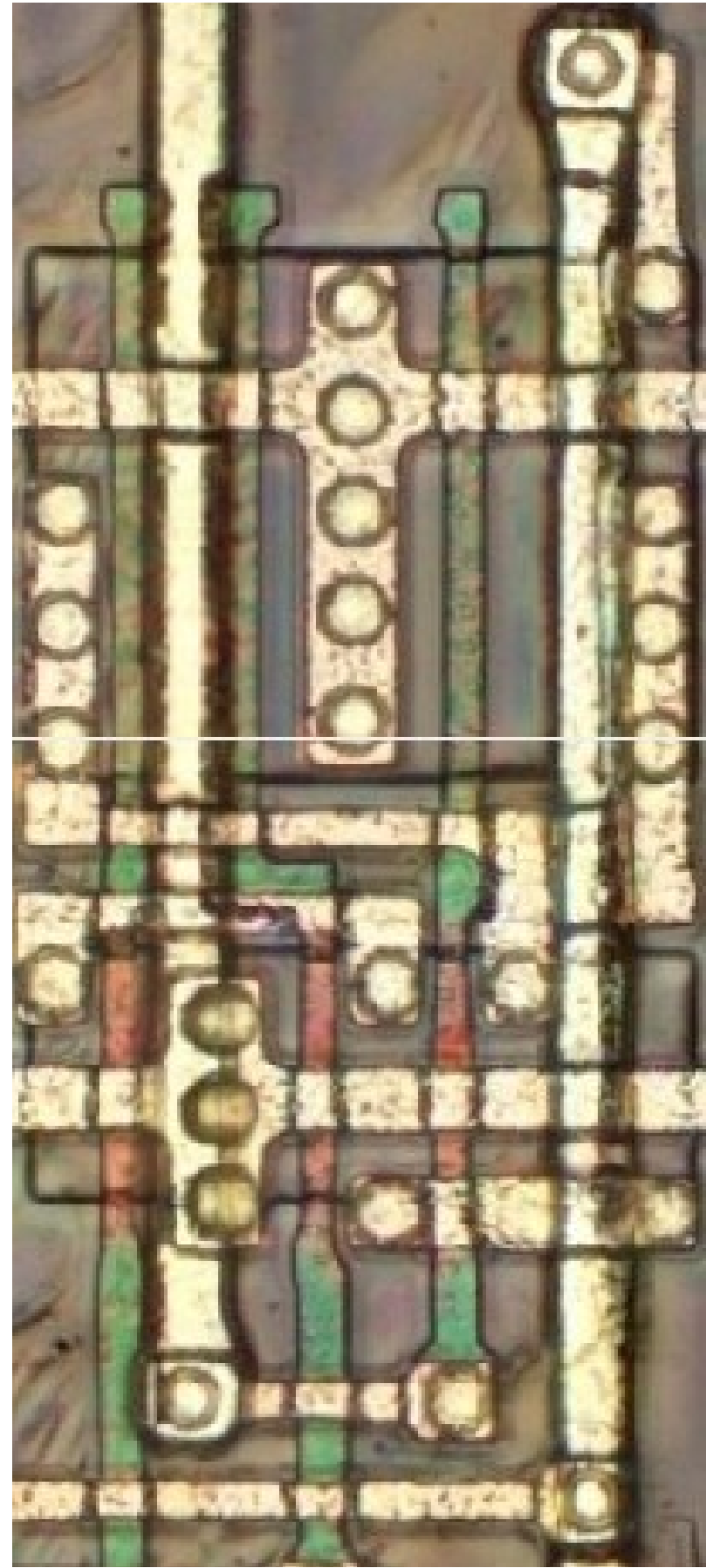
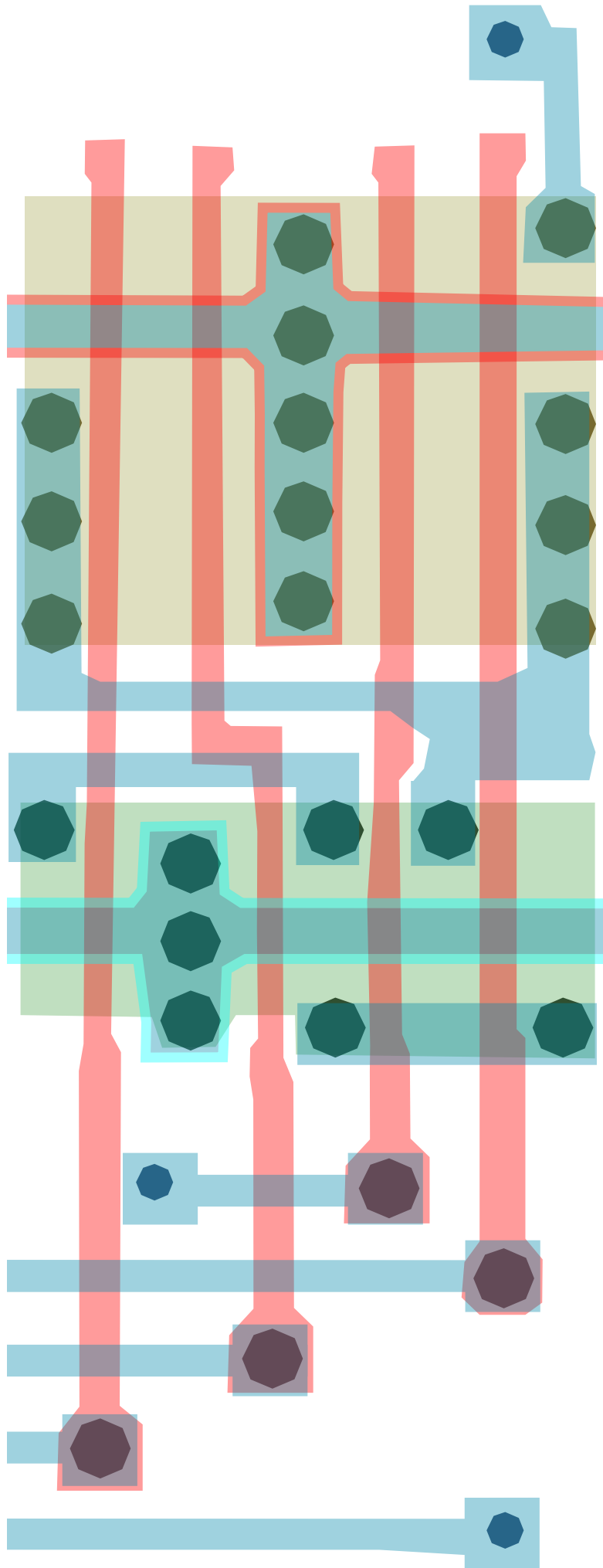
Cell 26



Same cell on
LA32
(MB87136A)

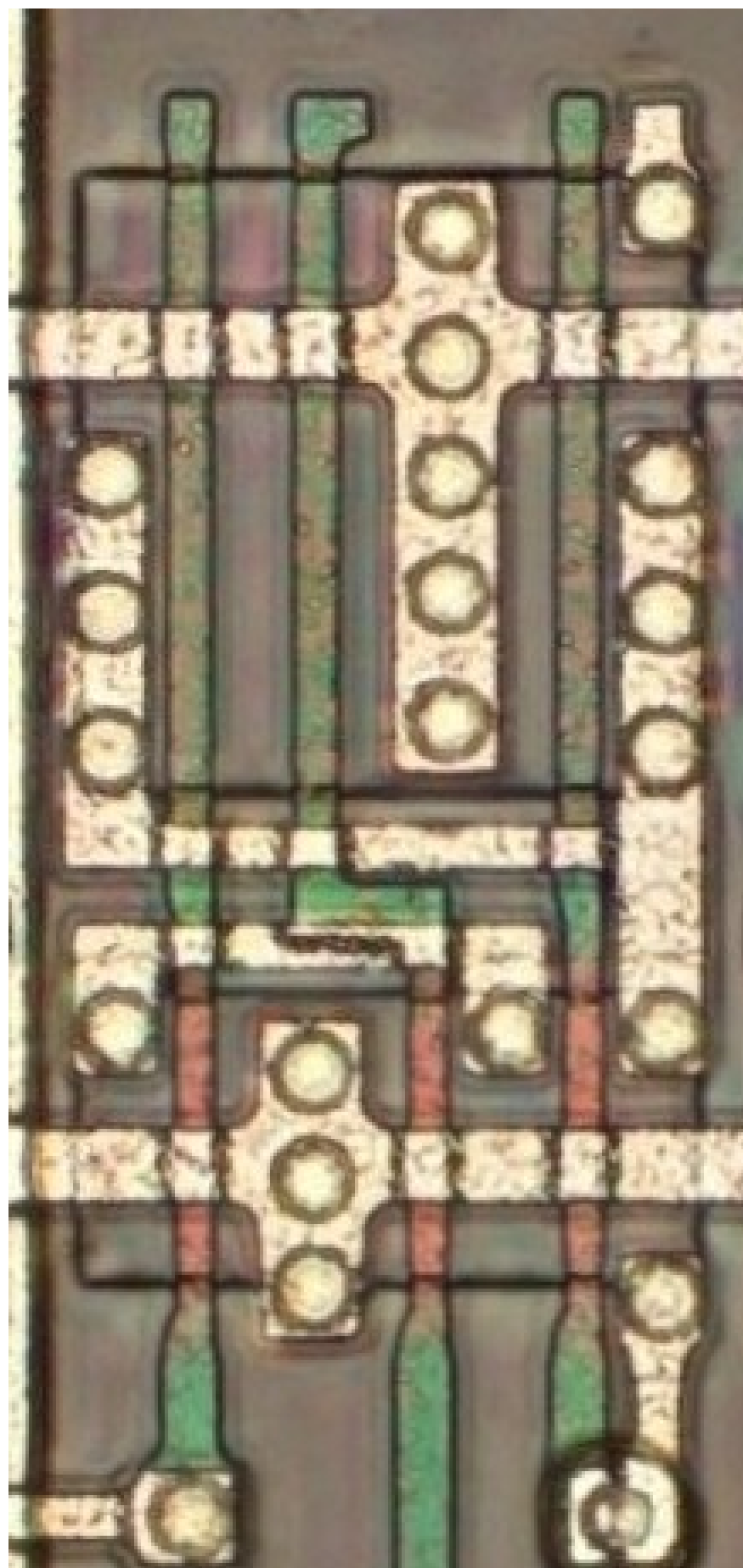
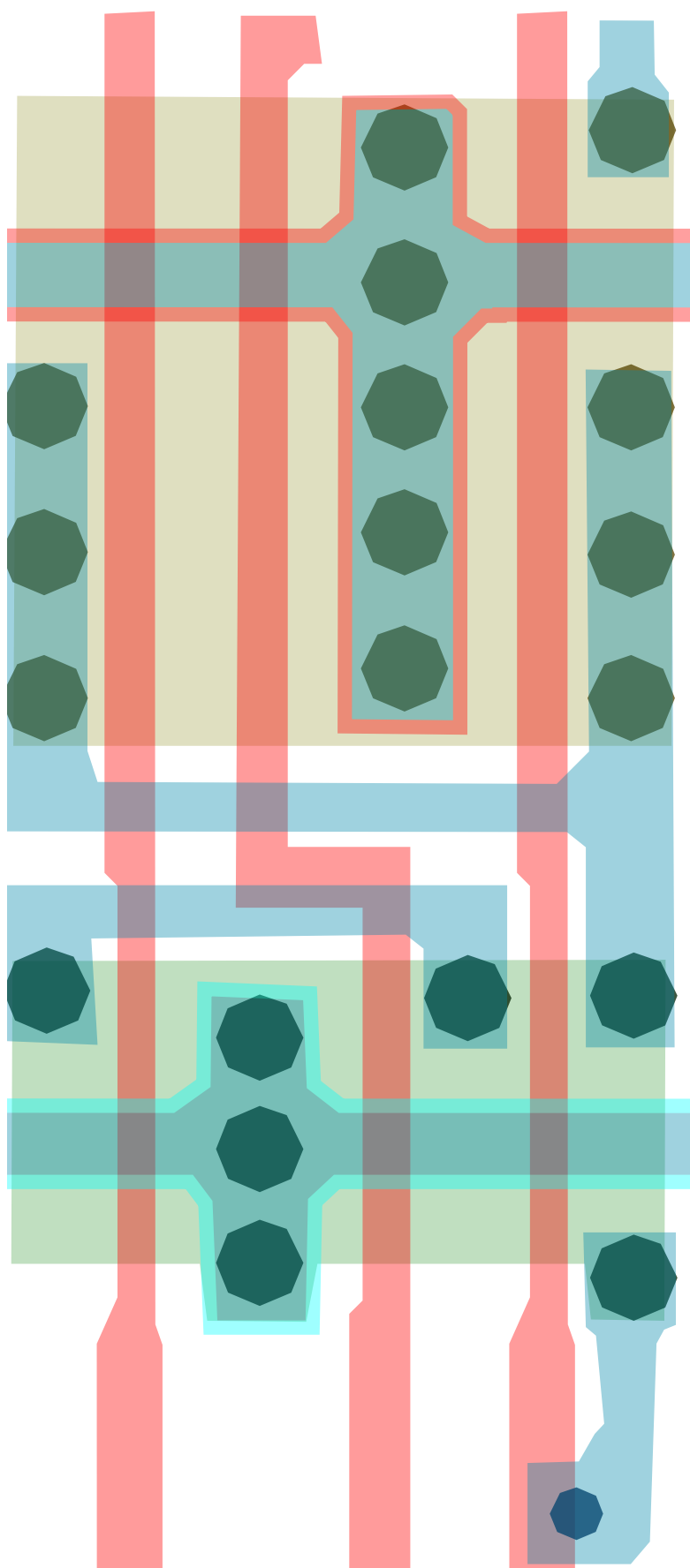
#bec1c3ff

Cell type 27a



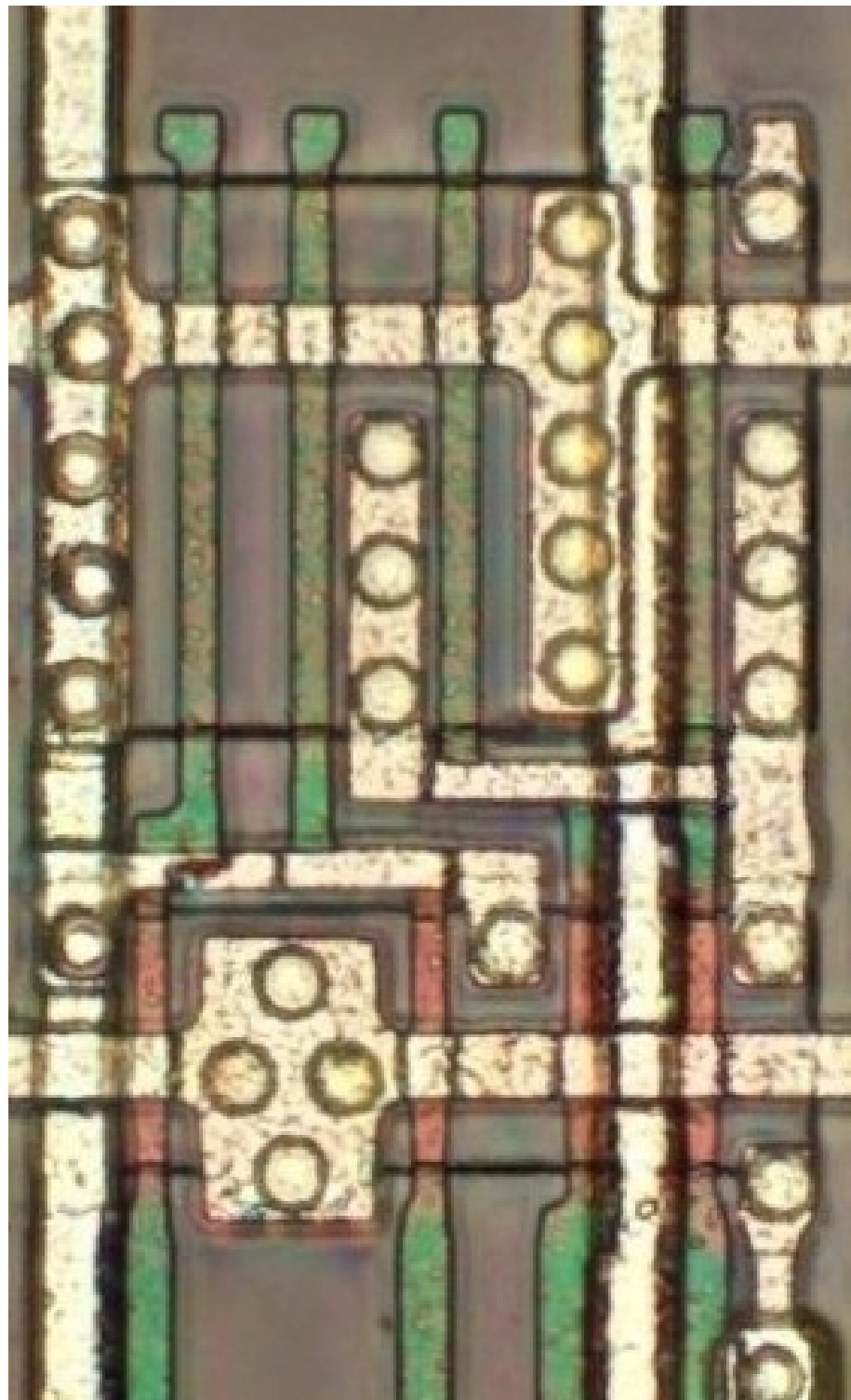
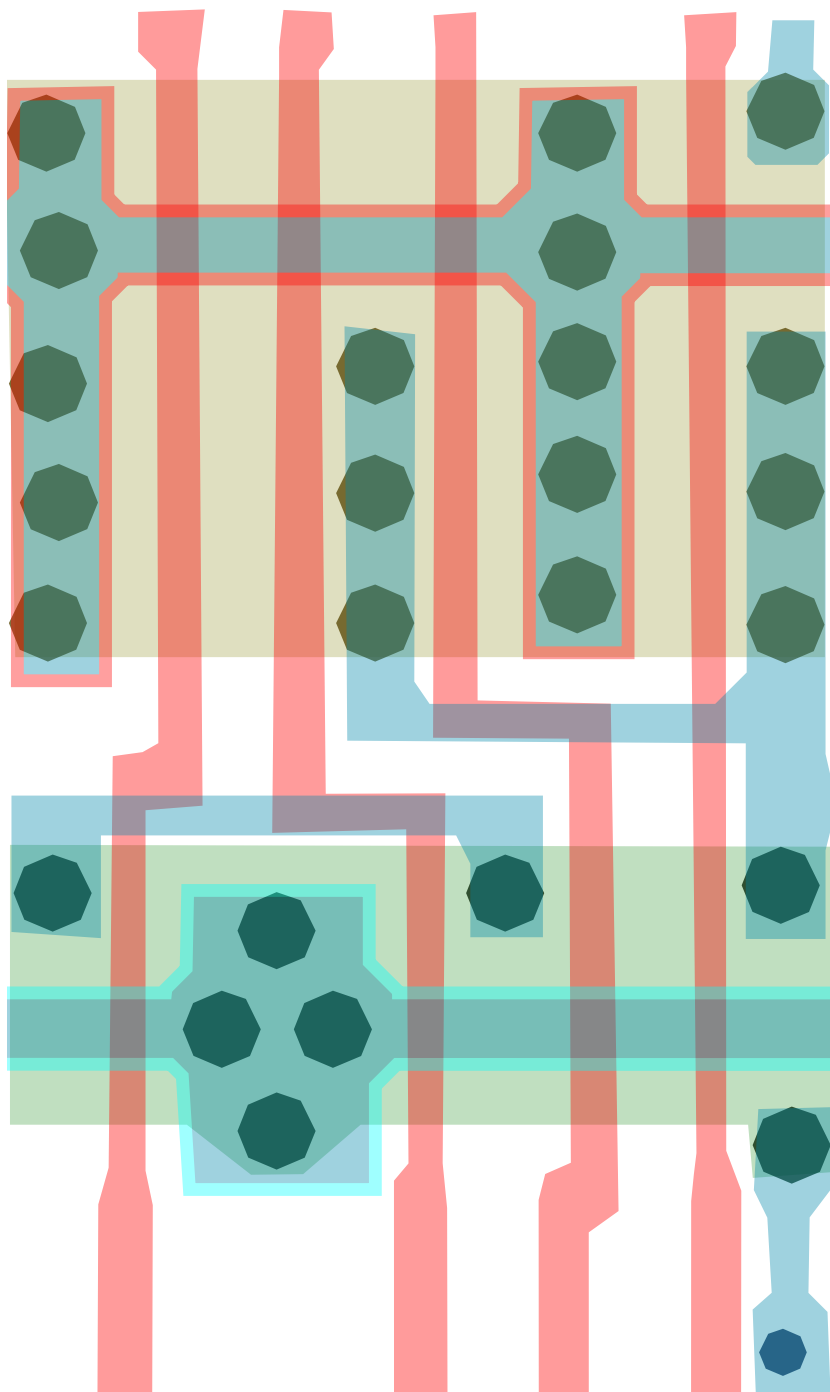
#787878ff

Cell type 27b



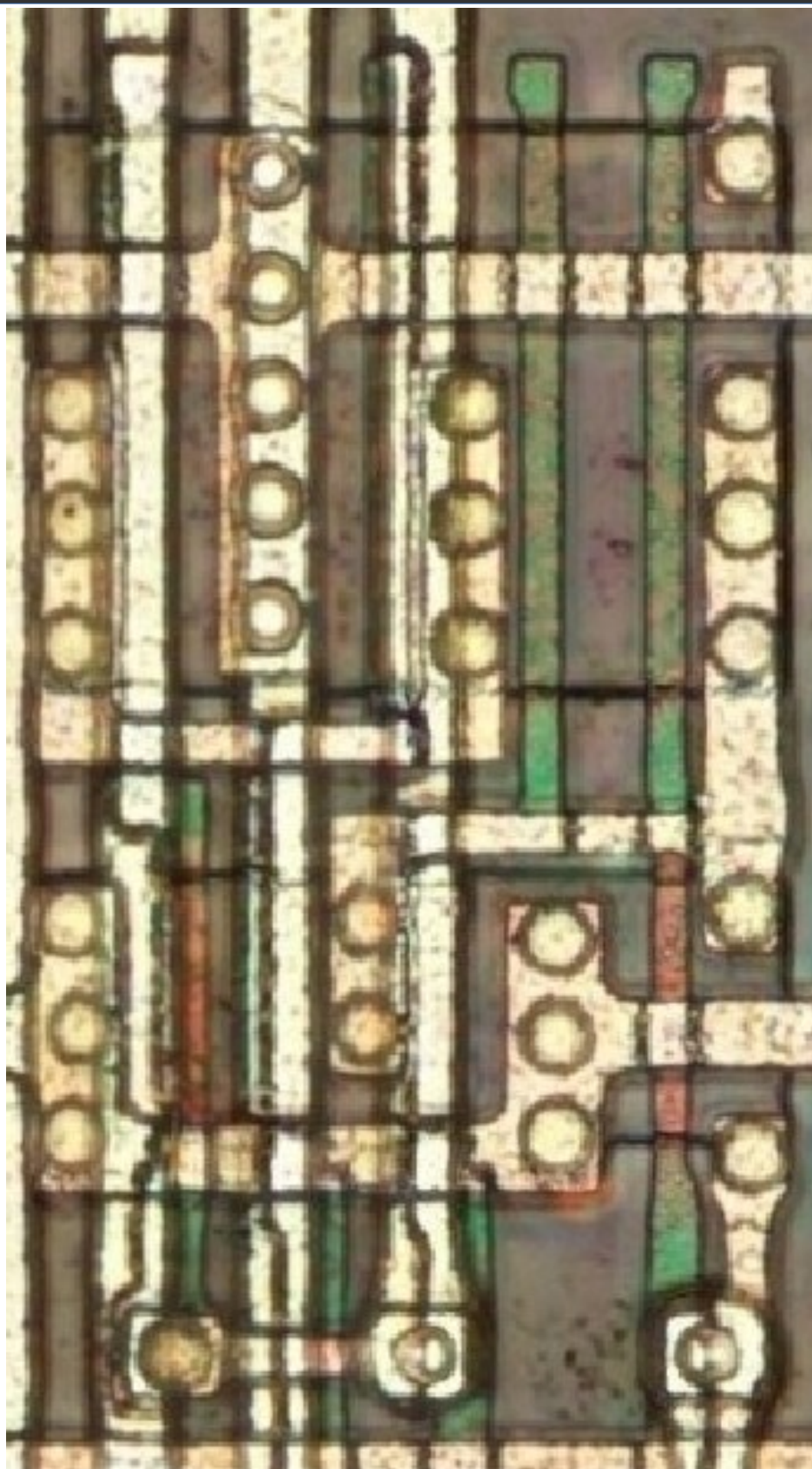
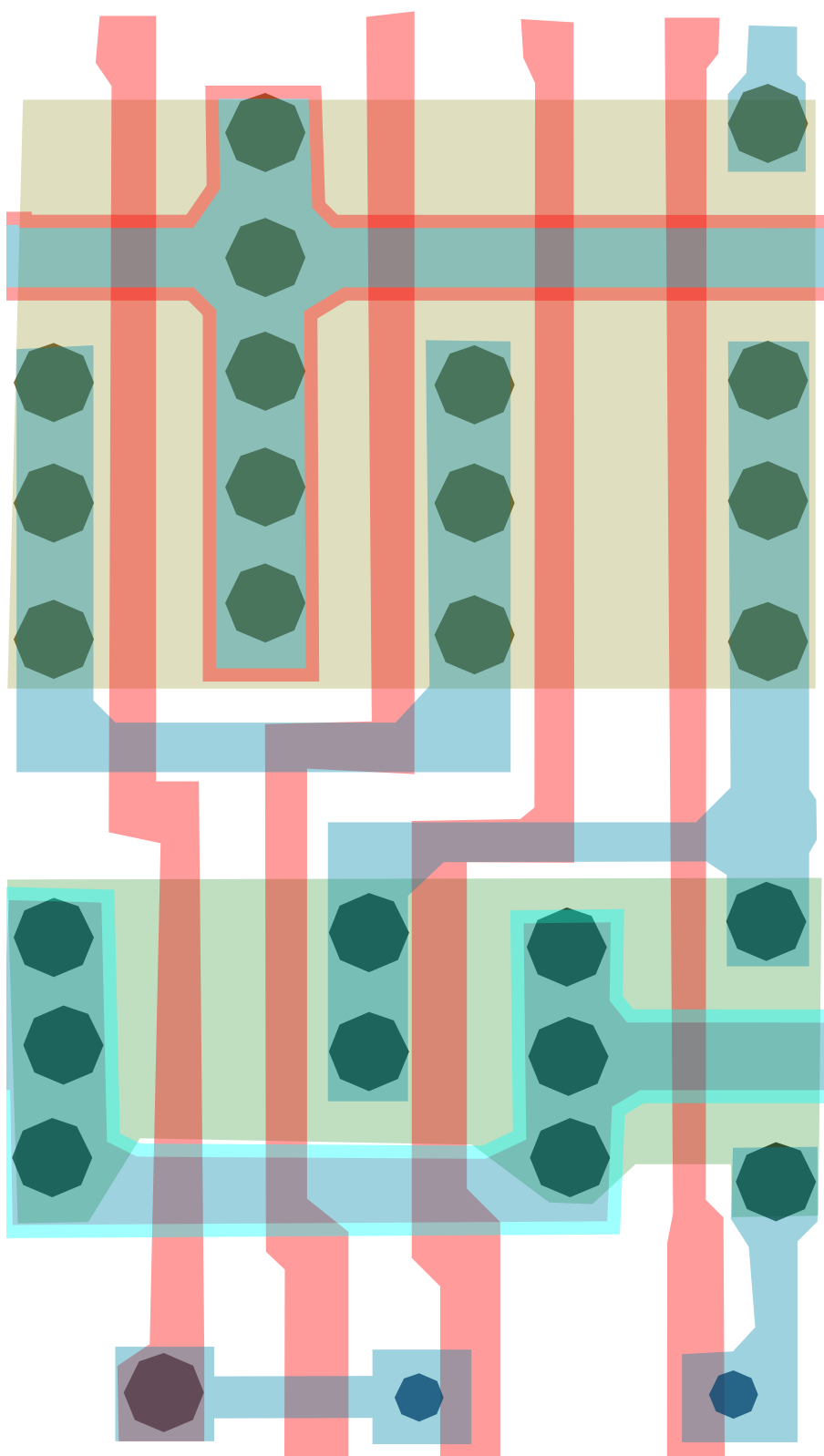
#5d5d5dff

Cell type 27C



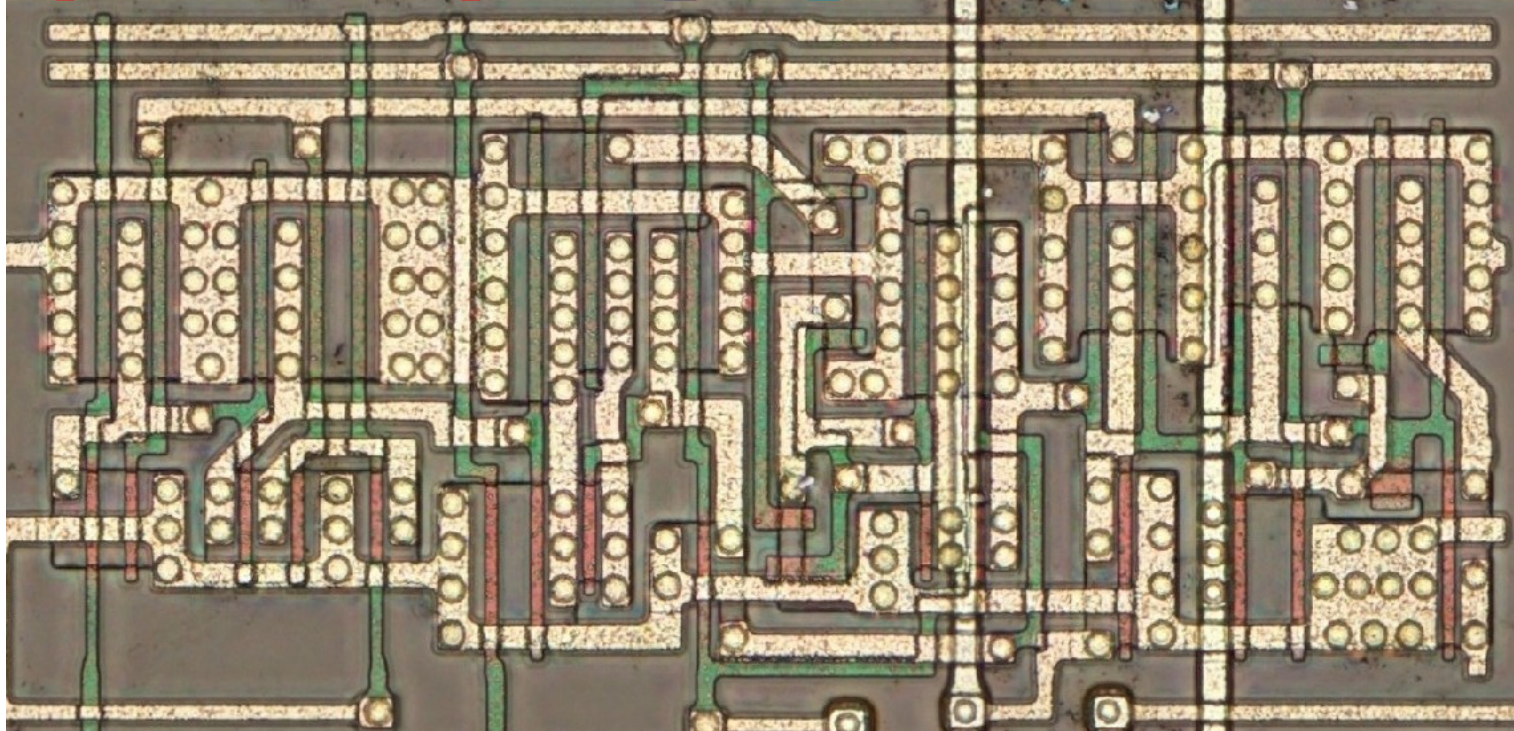
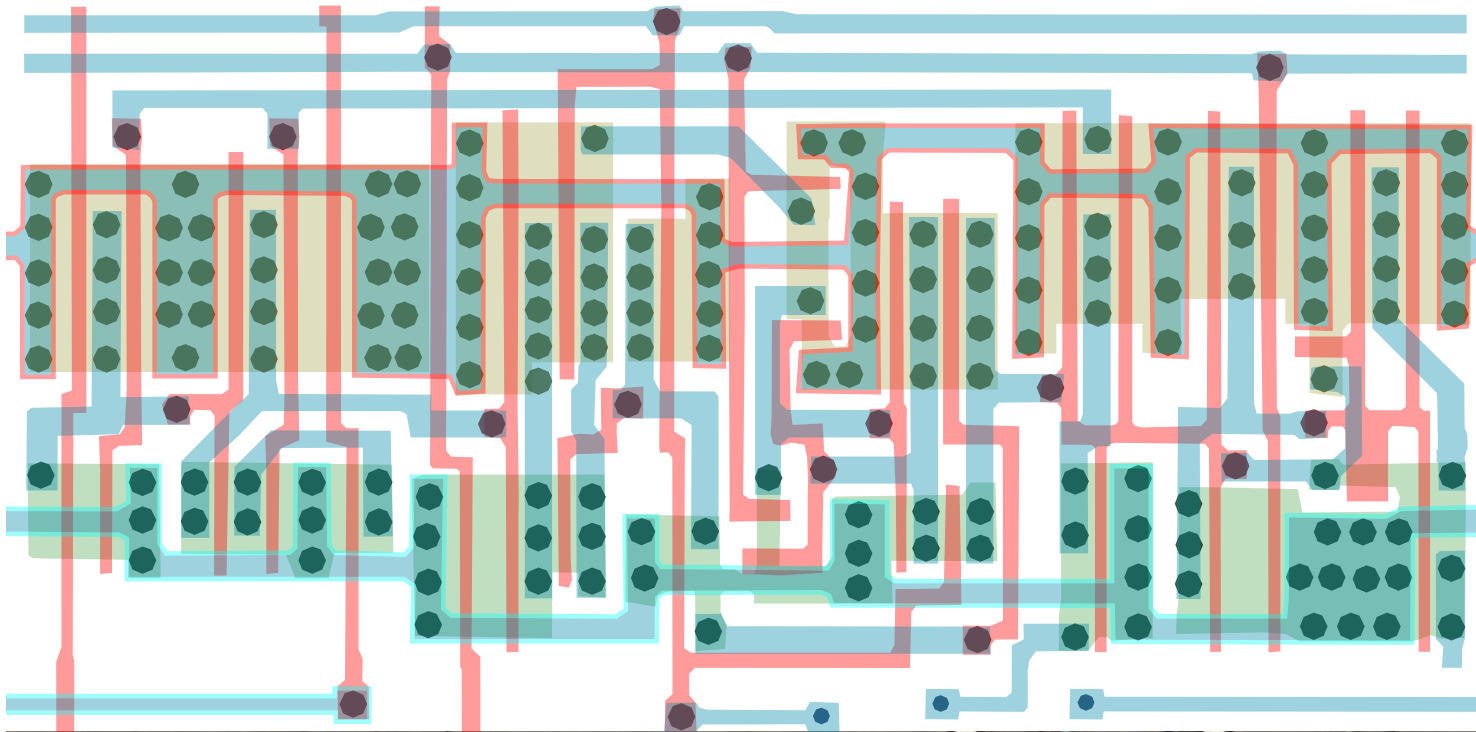
#668573ff

Cell type 27d

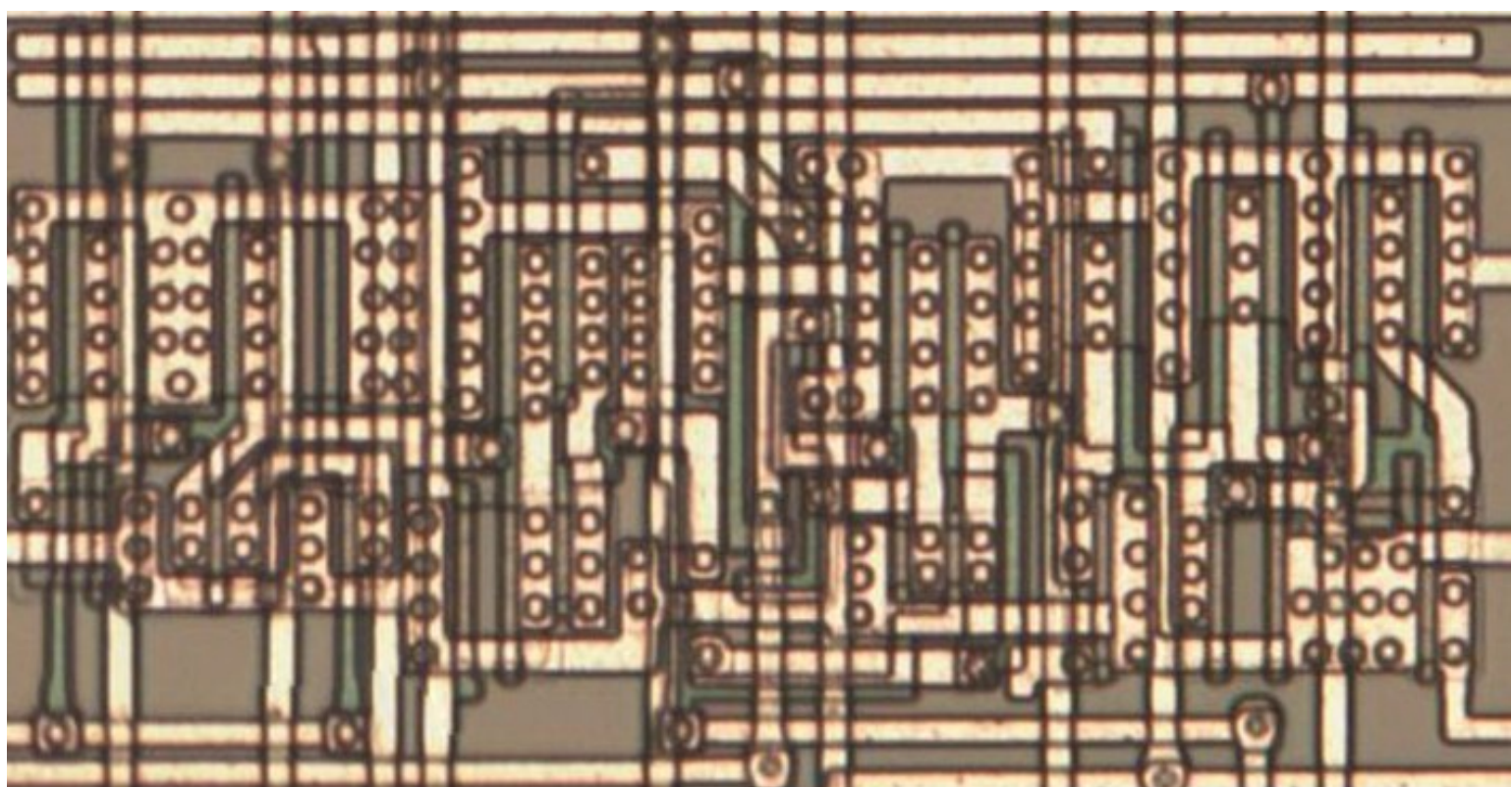


#00b75aff

Cell type 28

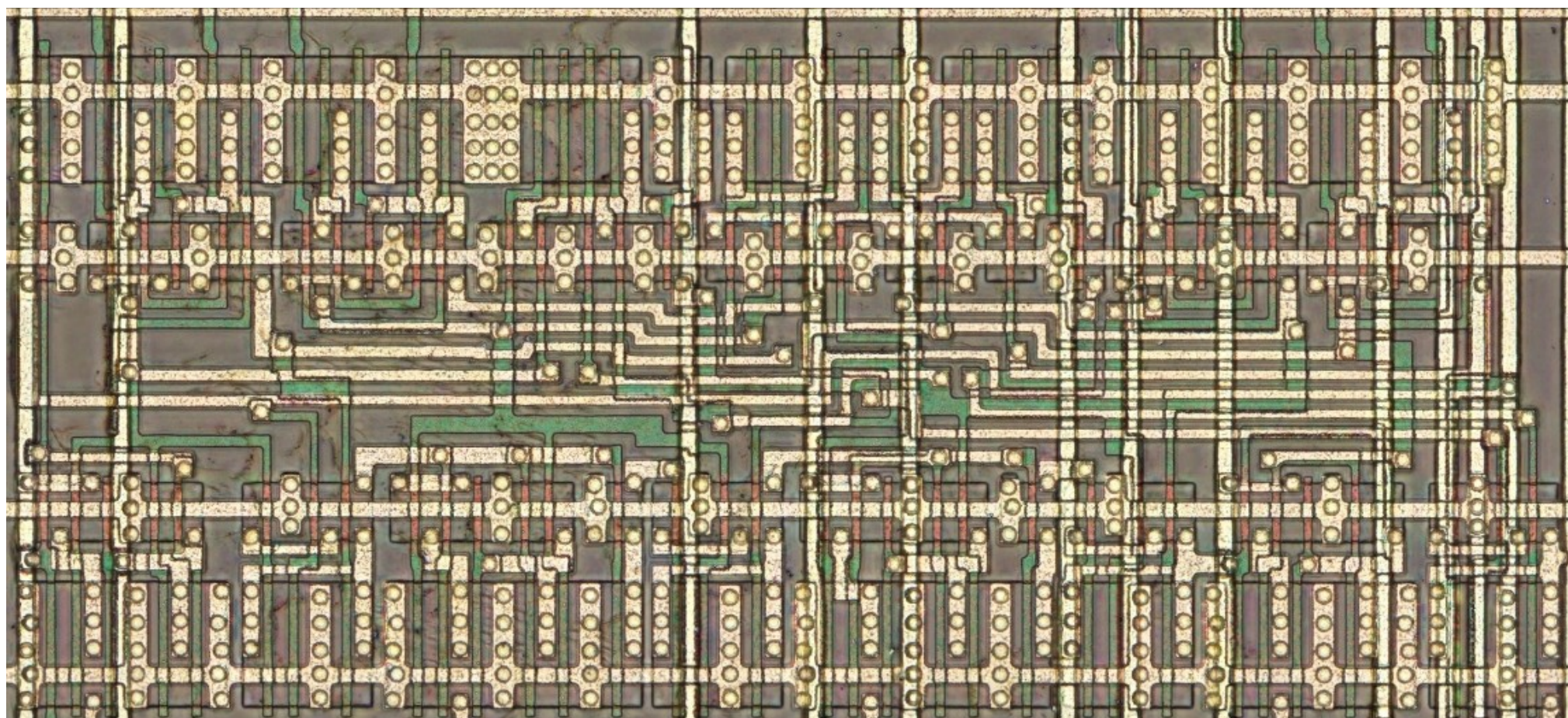
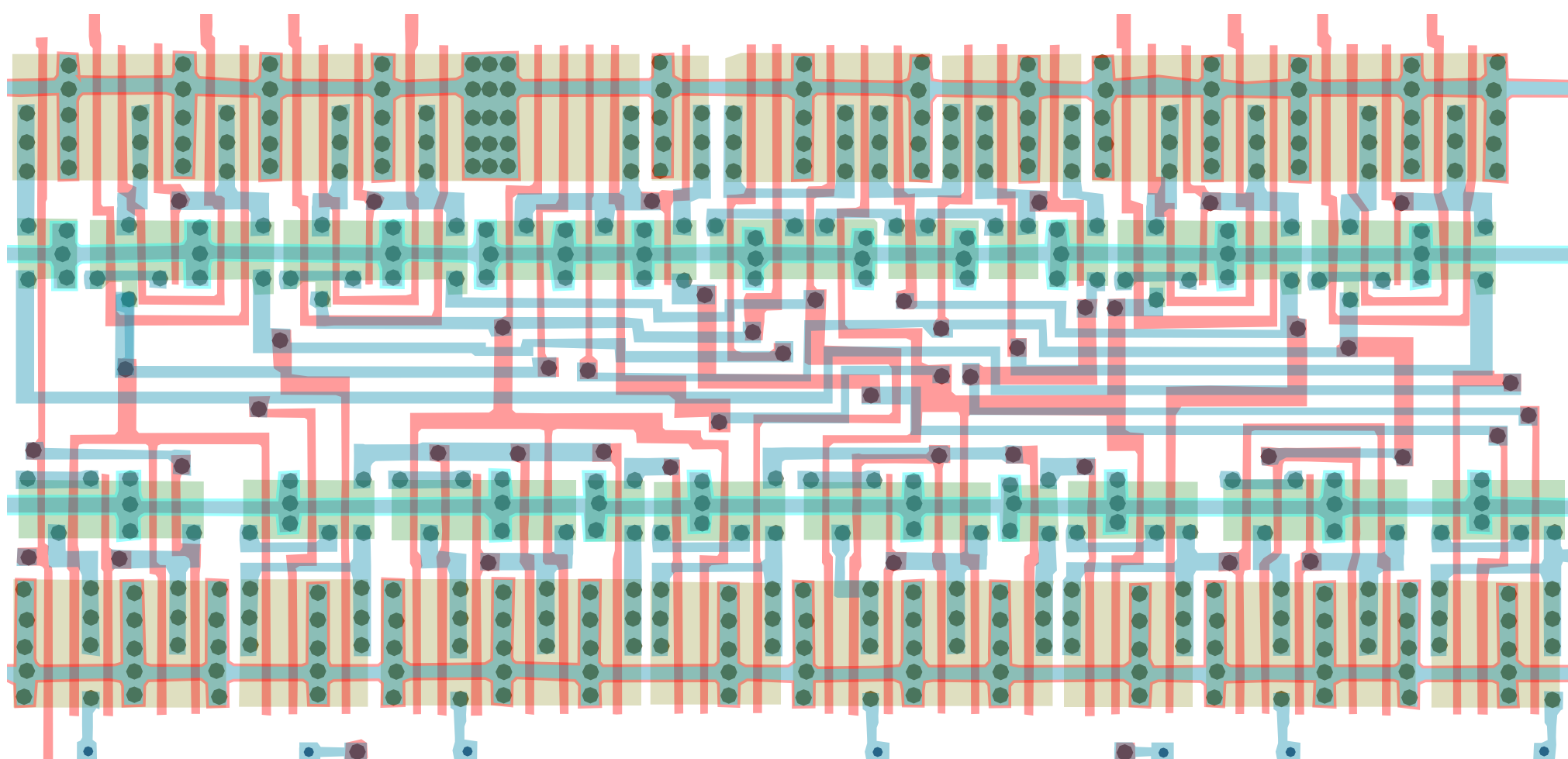


Same cell on
LA32
(MB87136A)



#d5546dff

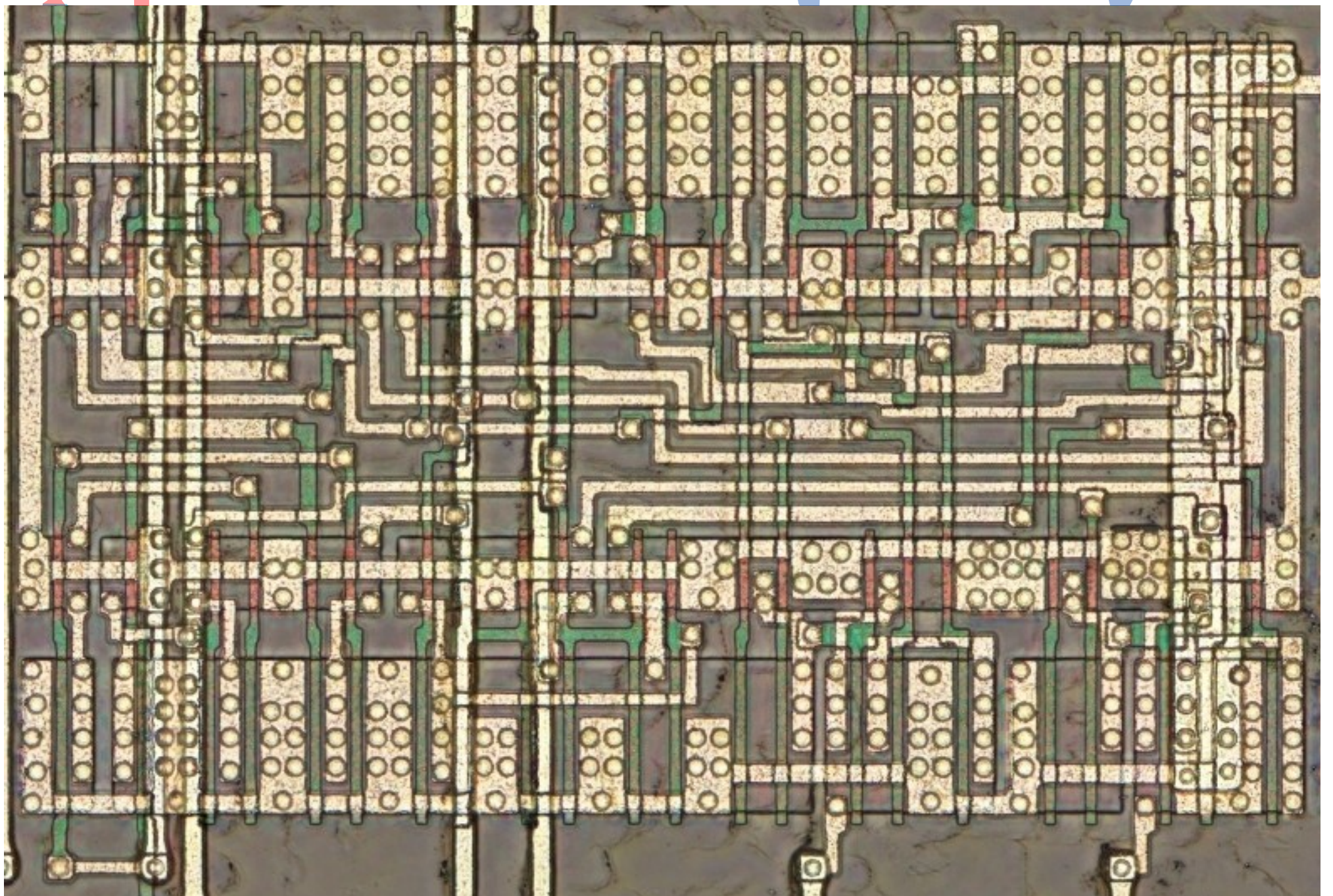
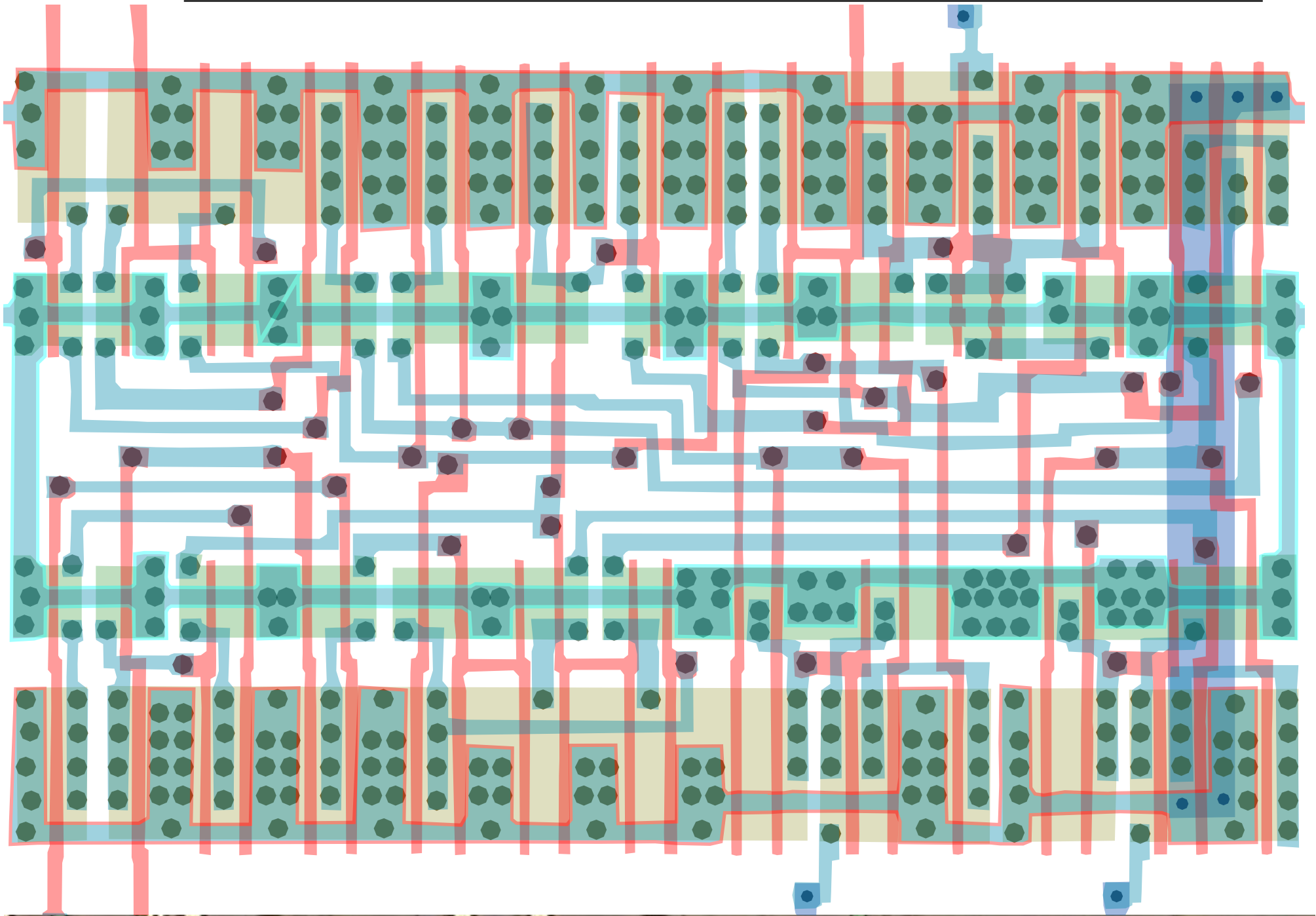
Cell 29



#a9546dff

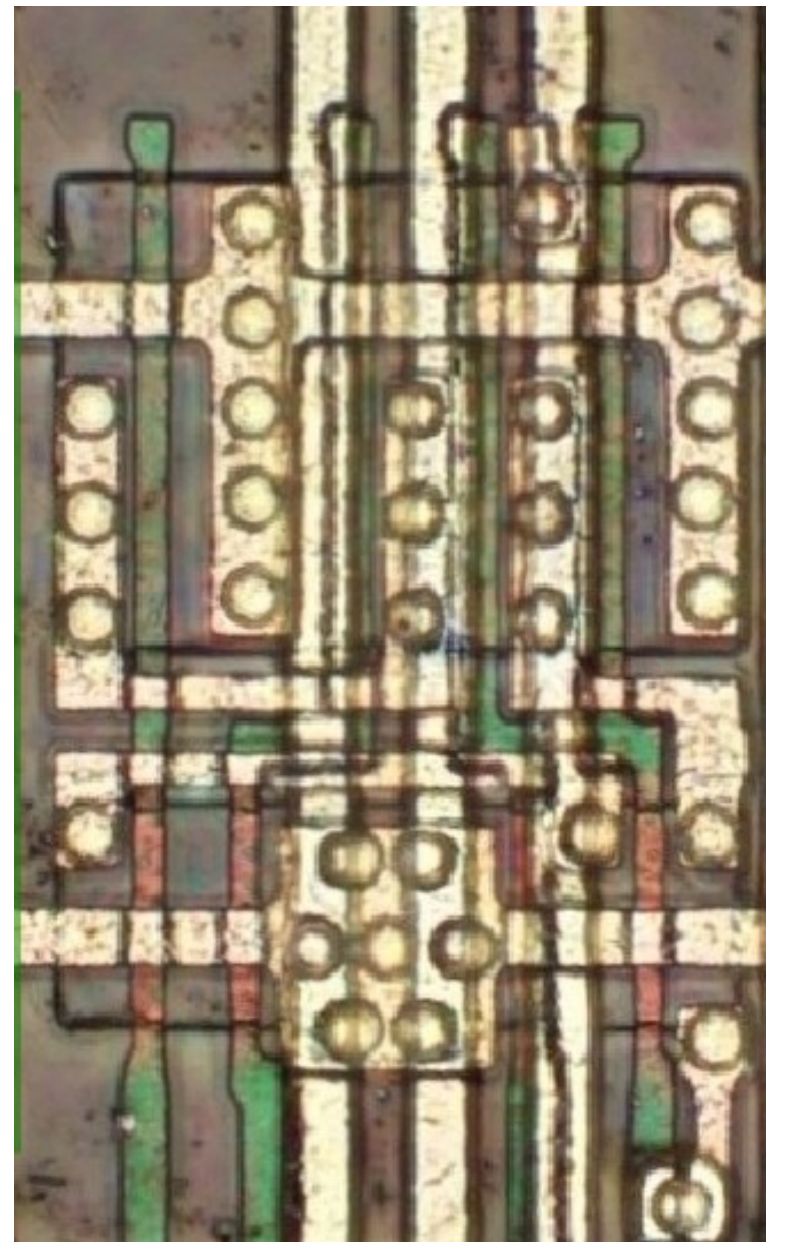
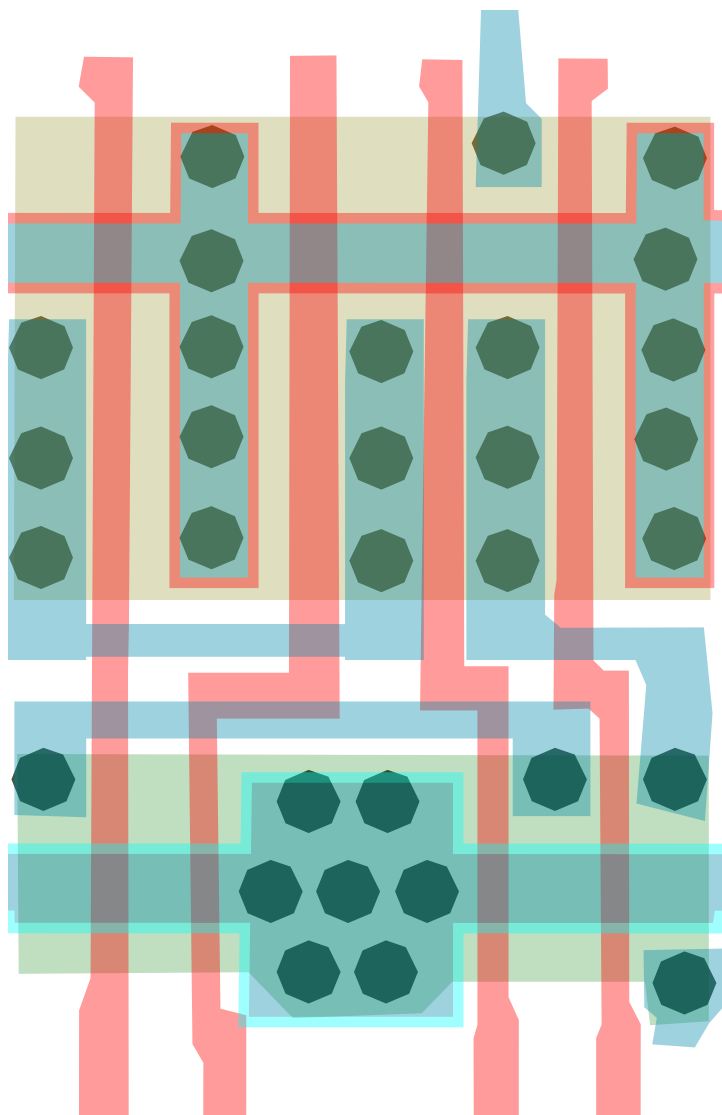
Cell 30

Standard cell only found in adder section of chip (as of now)



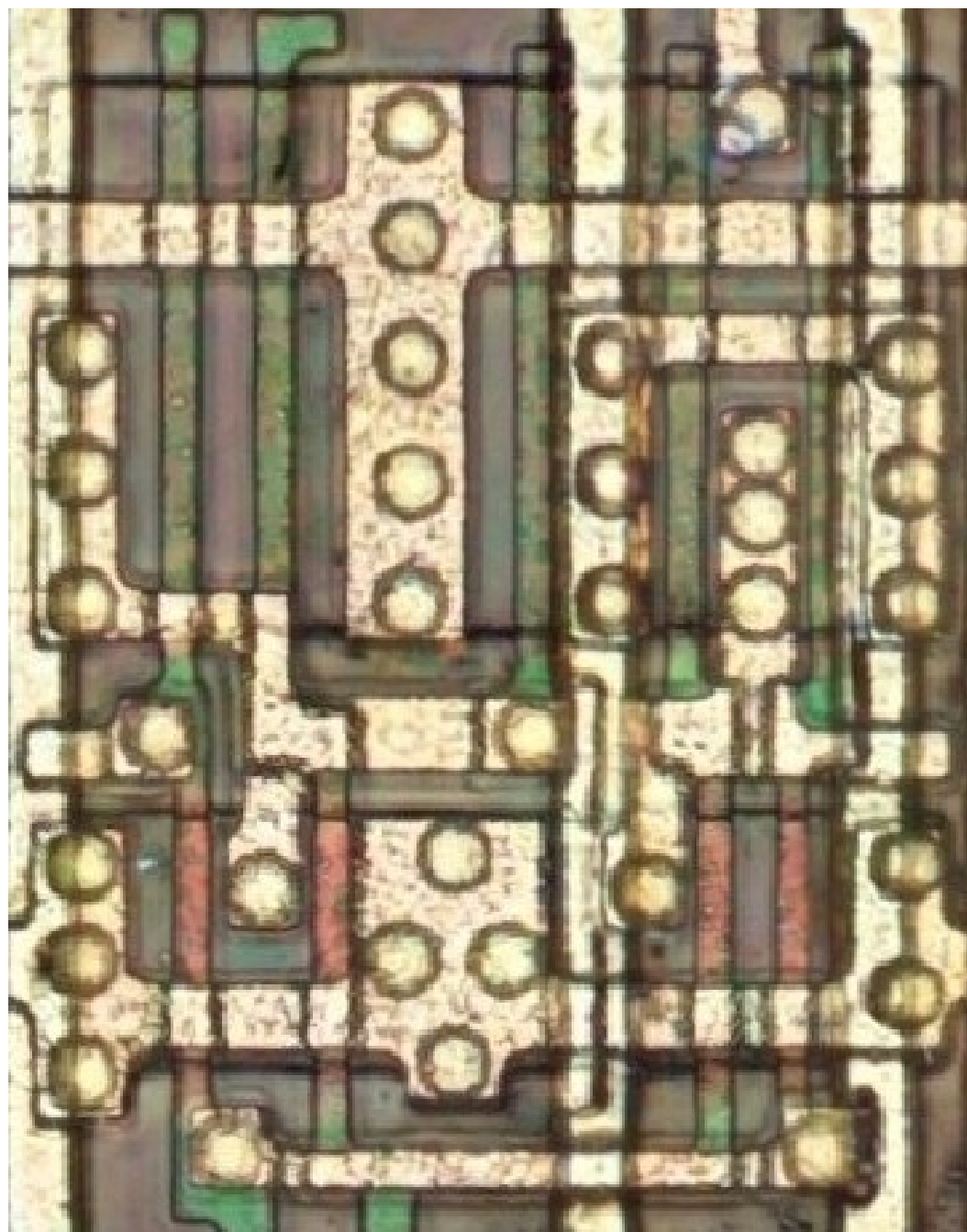
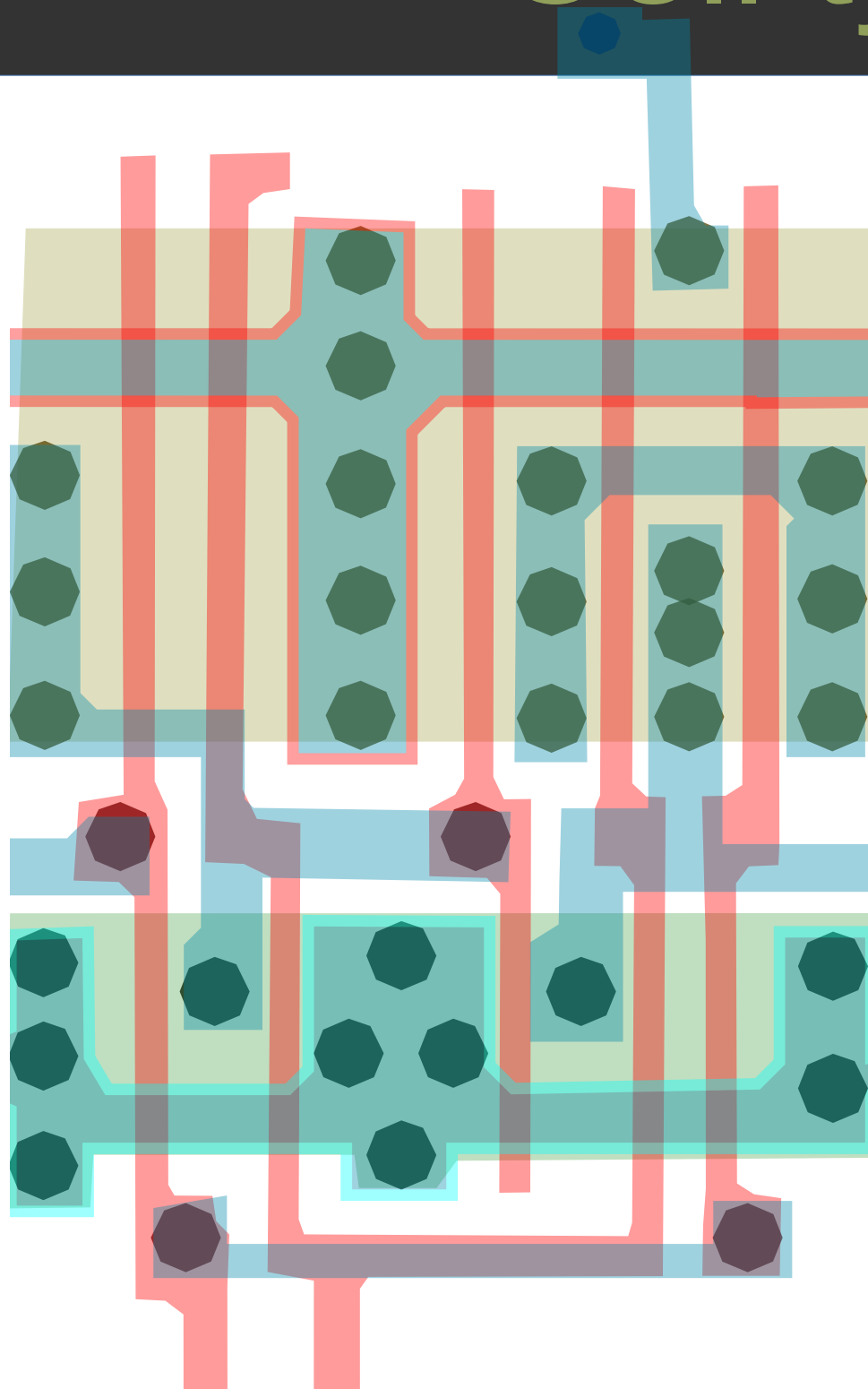
#ff4af8ff

Cell type 31



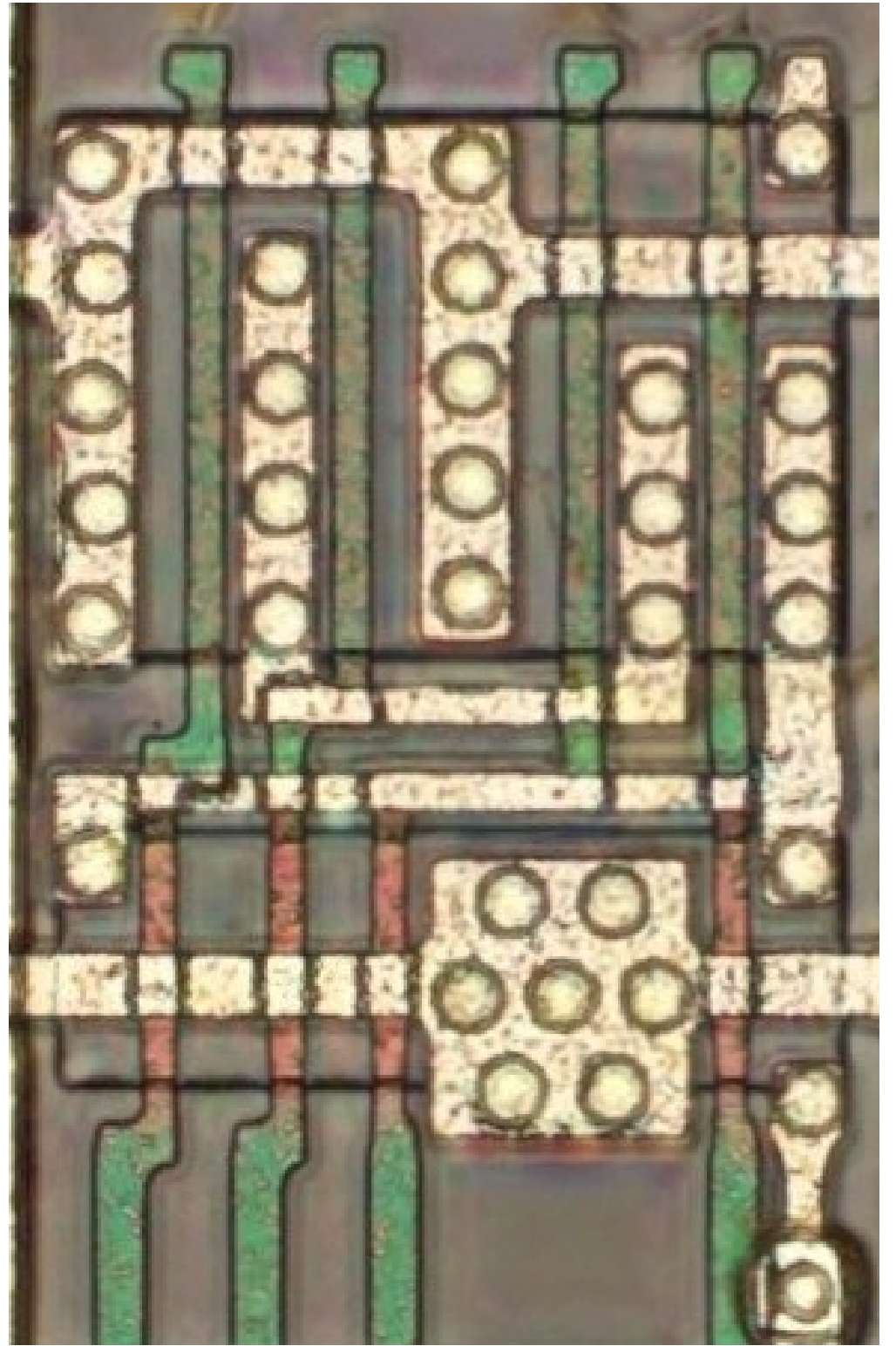
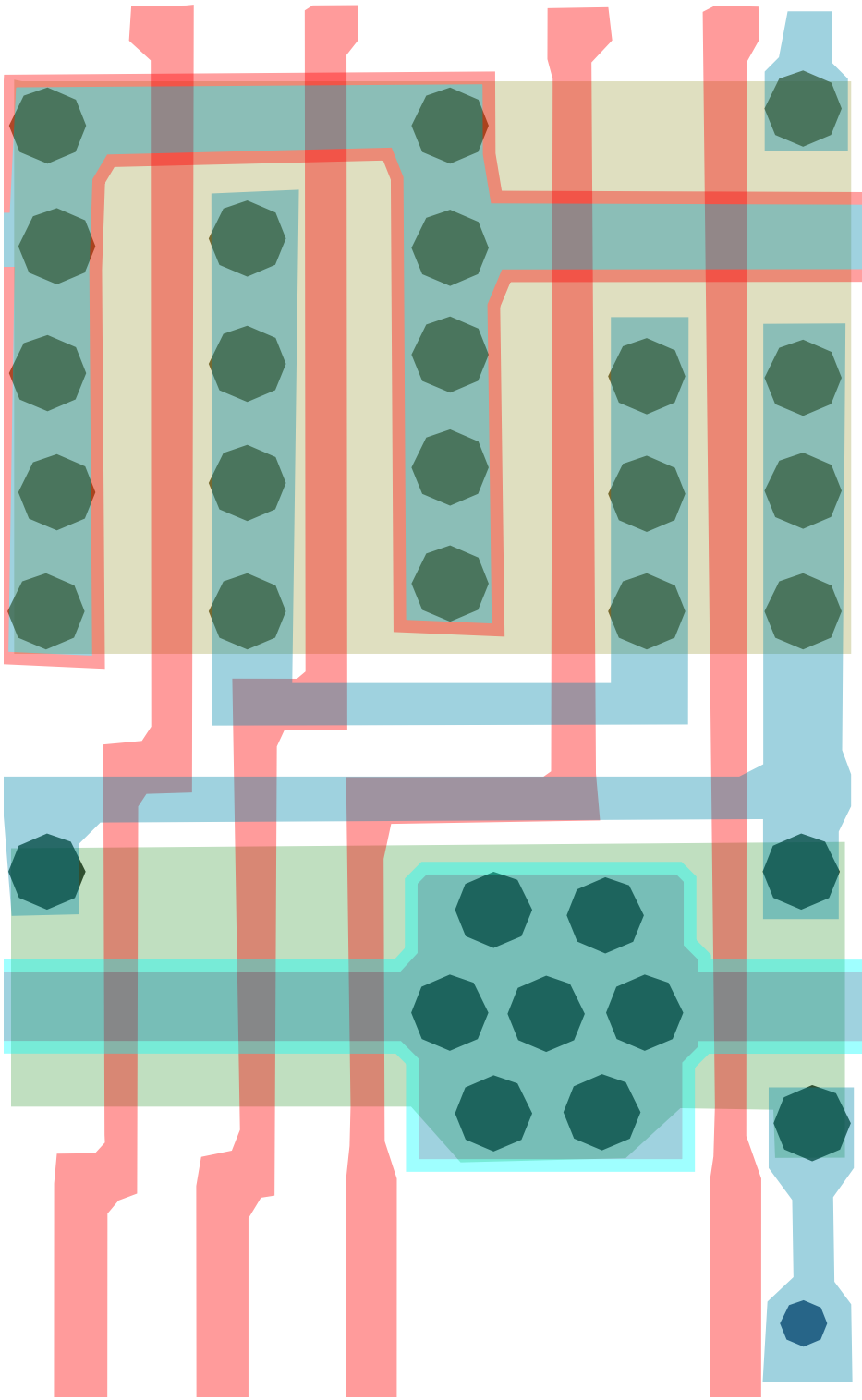
#90a05aff

Cell type 32



#57f330ff

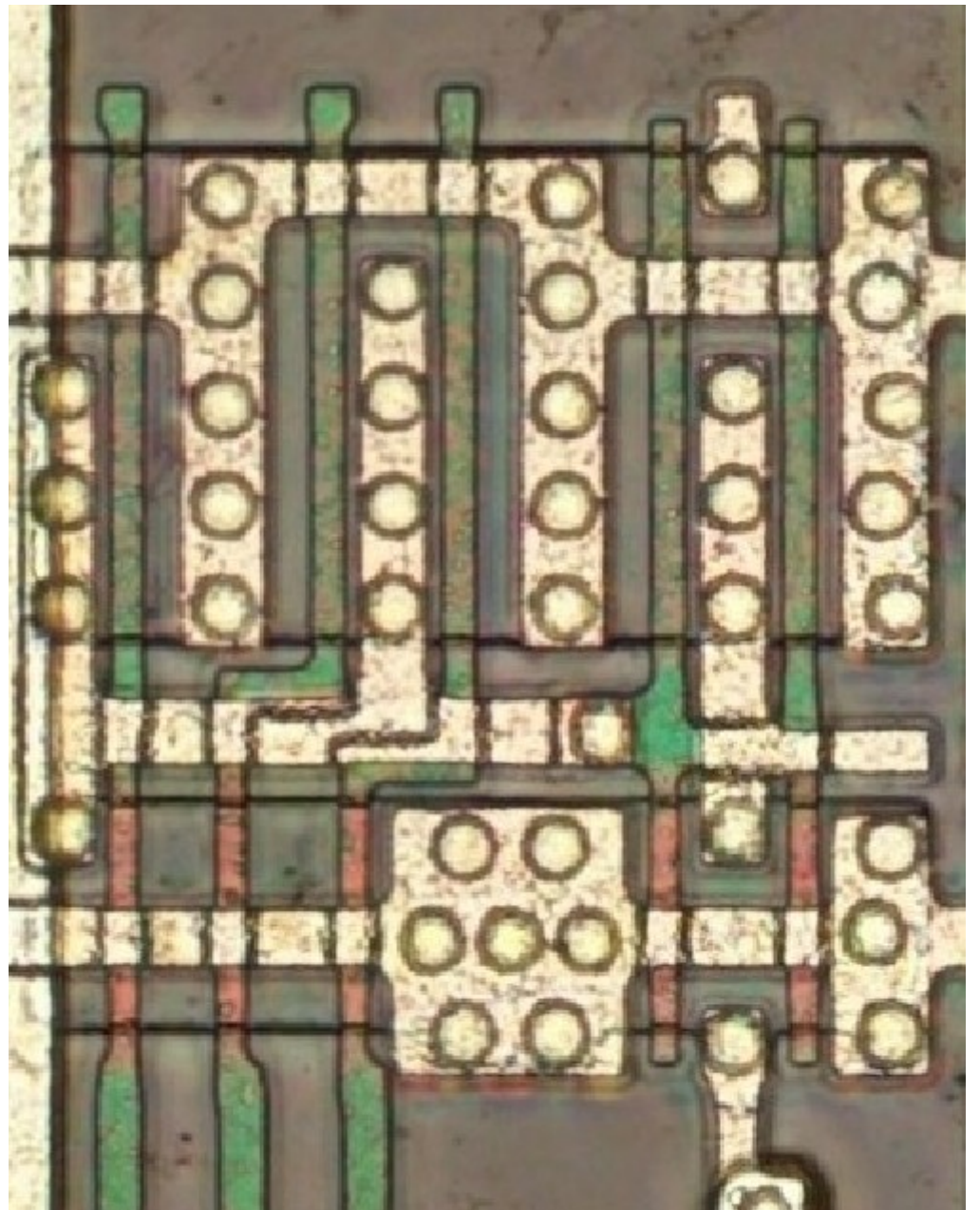
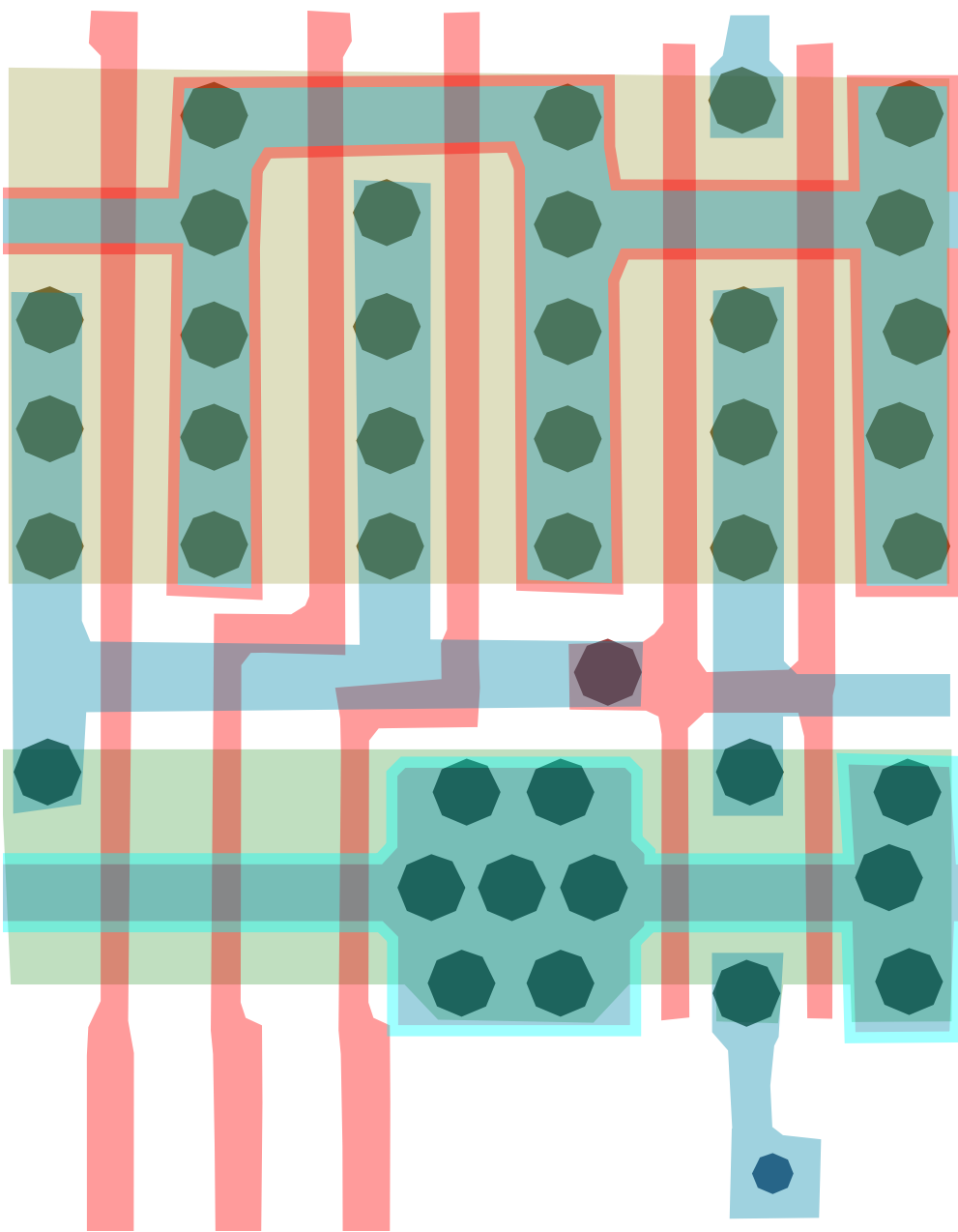
Cell type 33a



#77bd31ff

3-Input AND

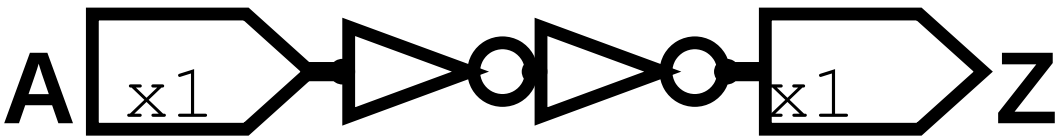
Cell 33b



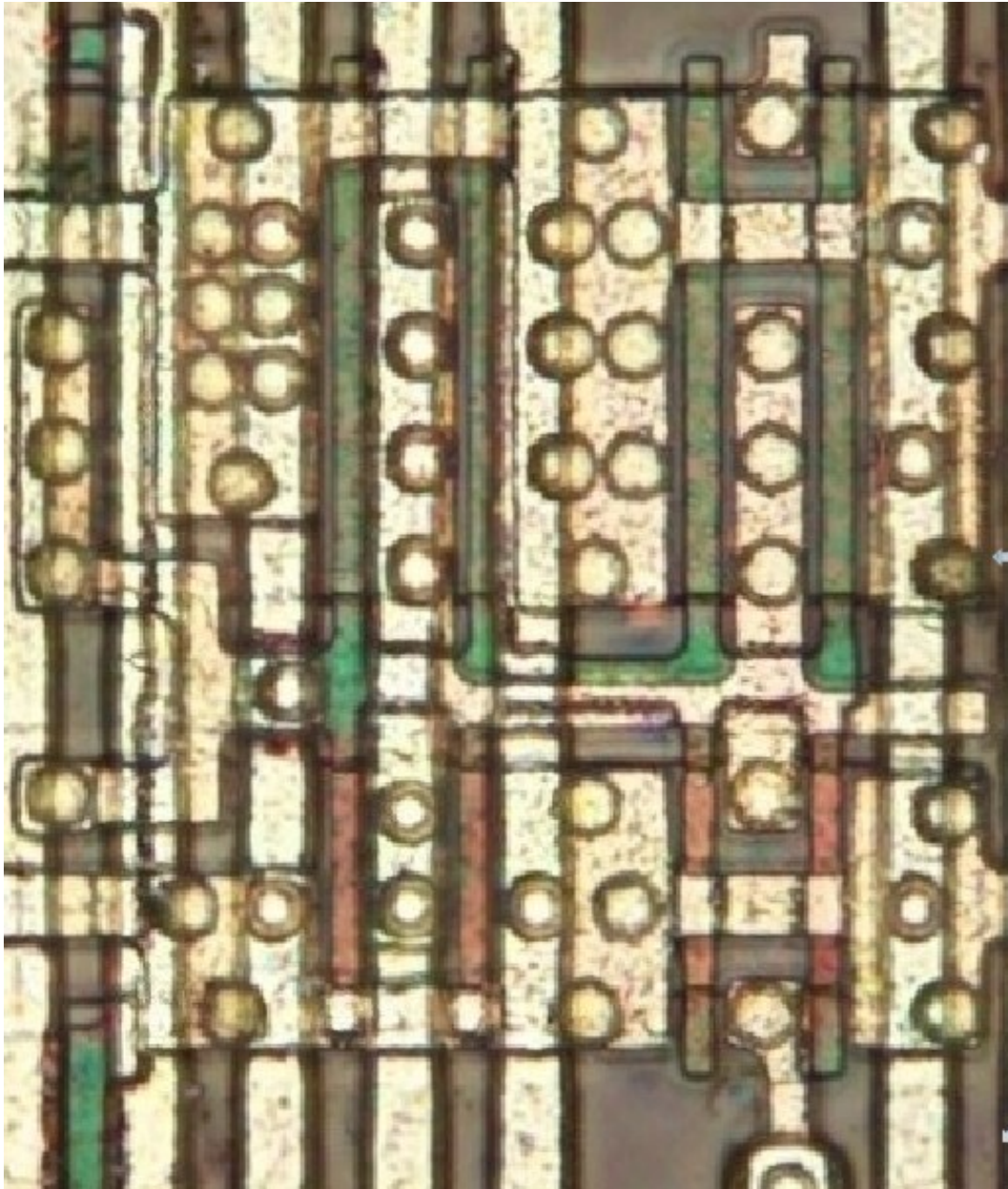
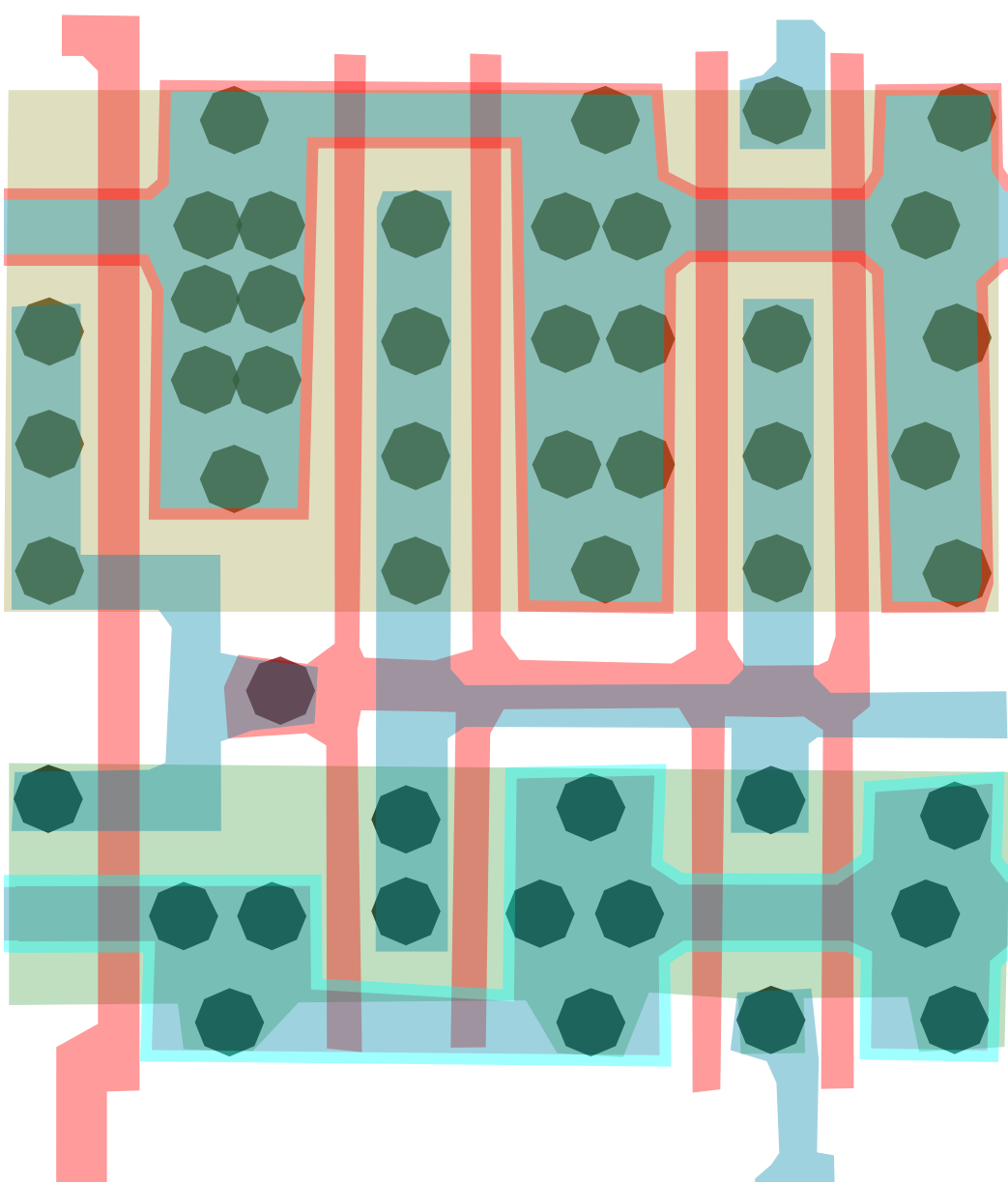
#5e58adff

Buffer

Cell 34

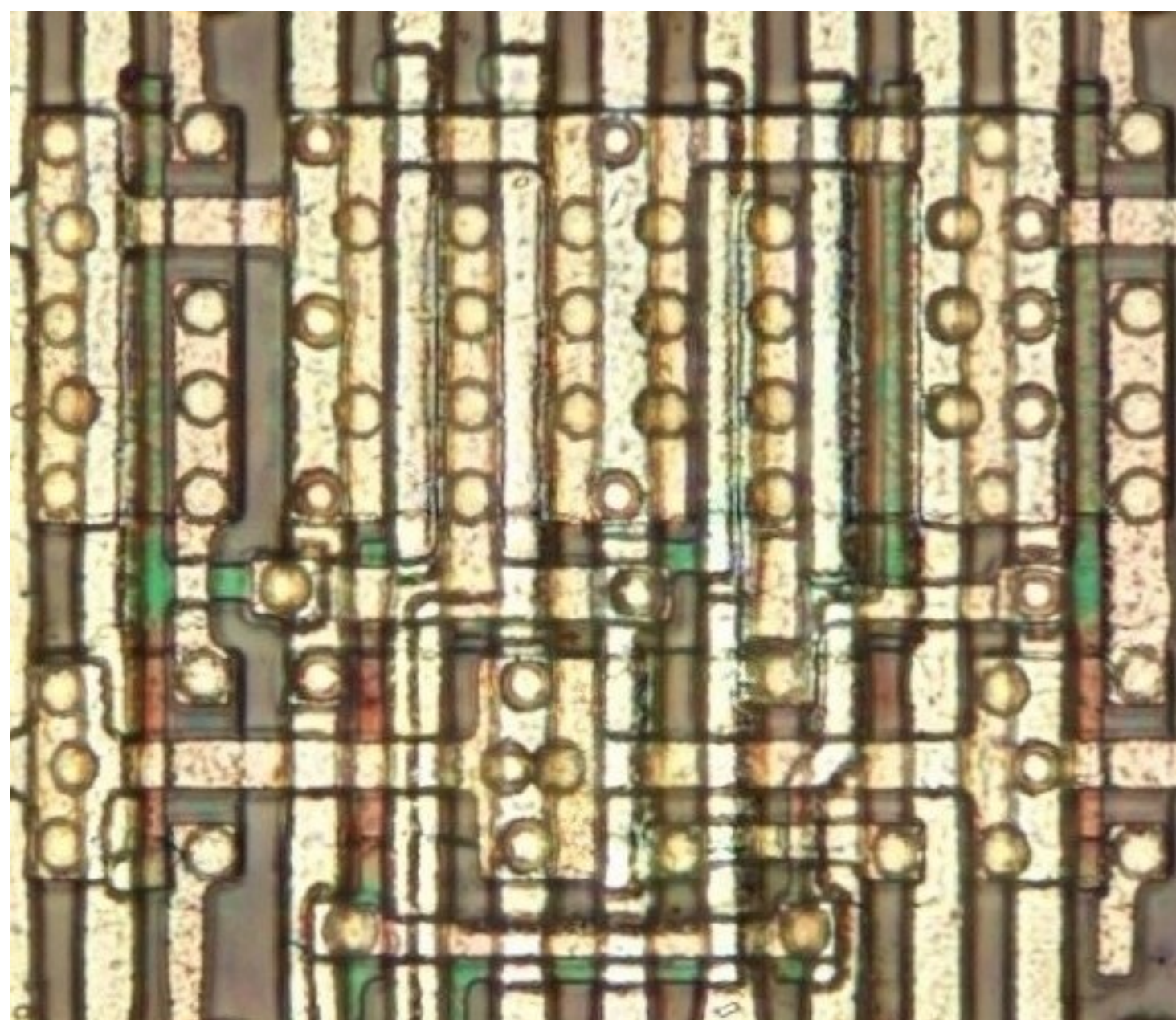
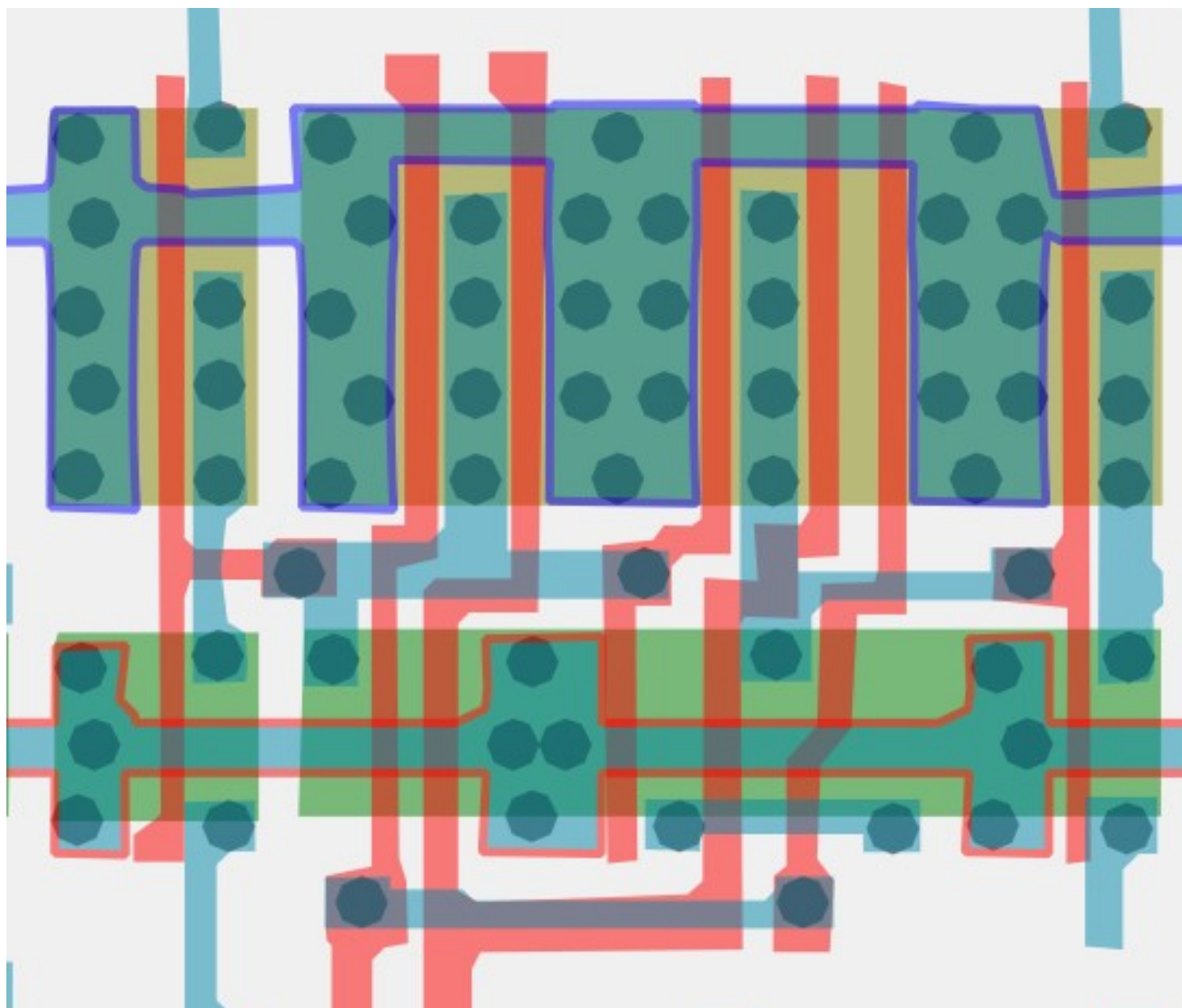


A	Z
0	0
1	1



#8581c6ff

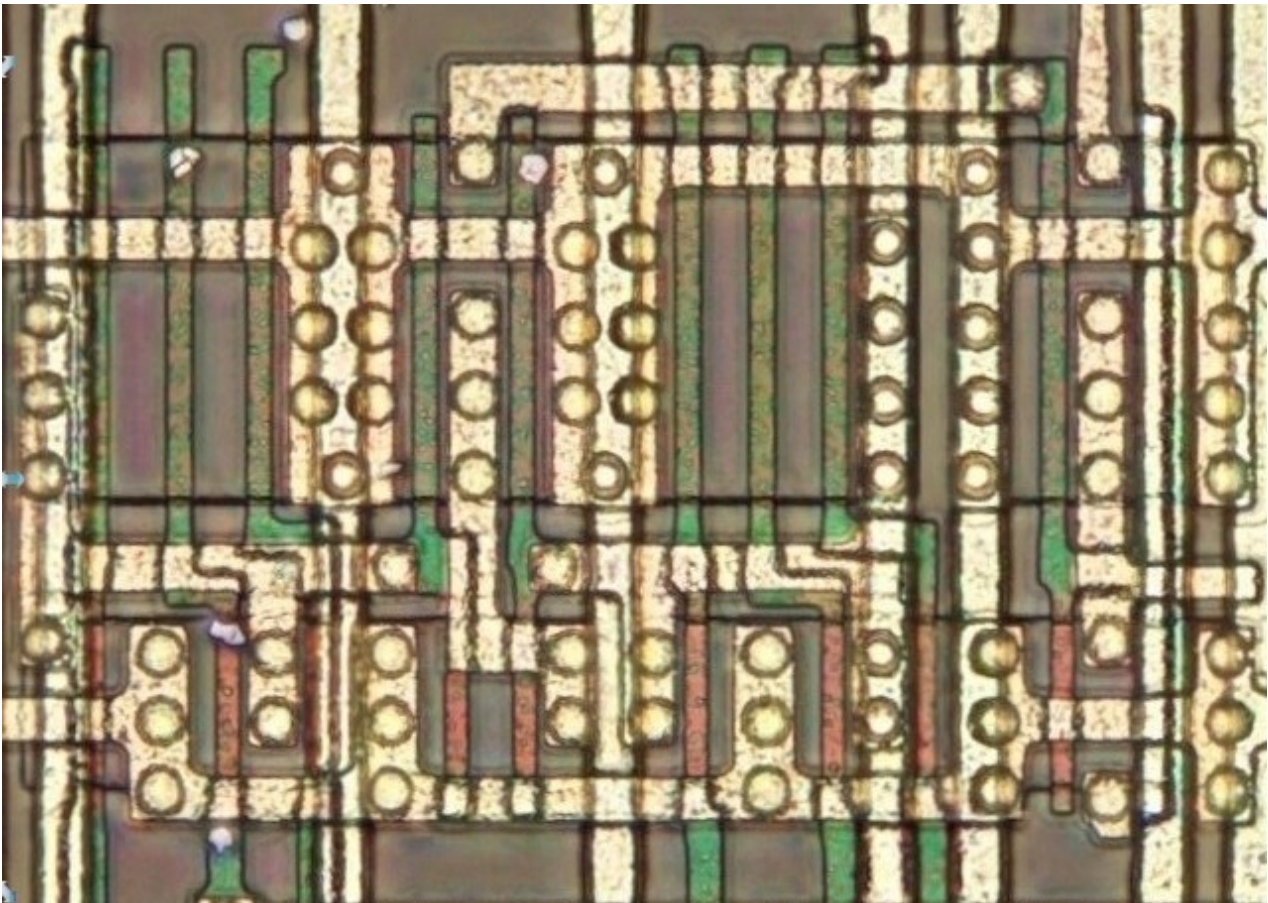
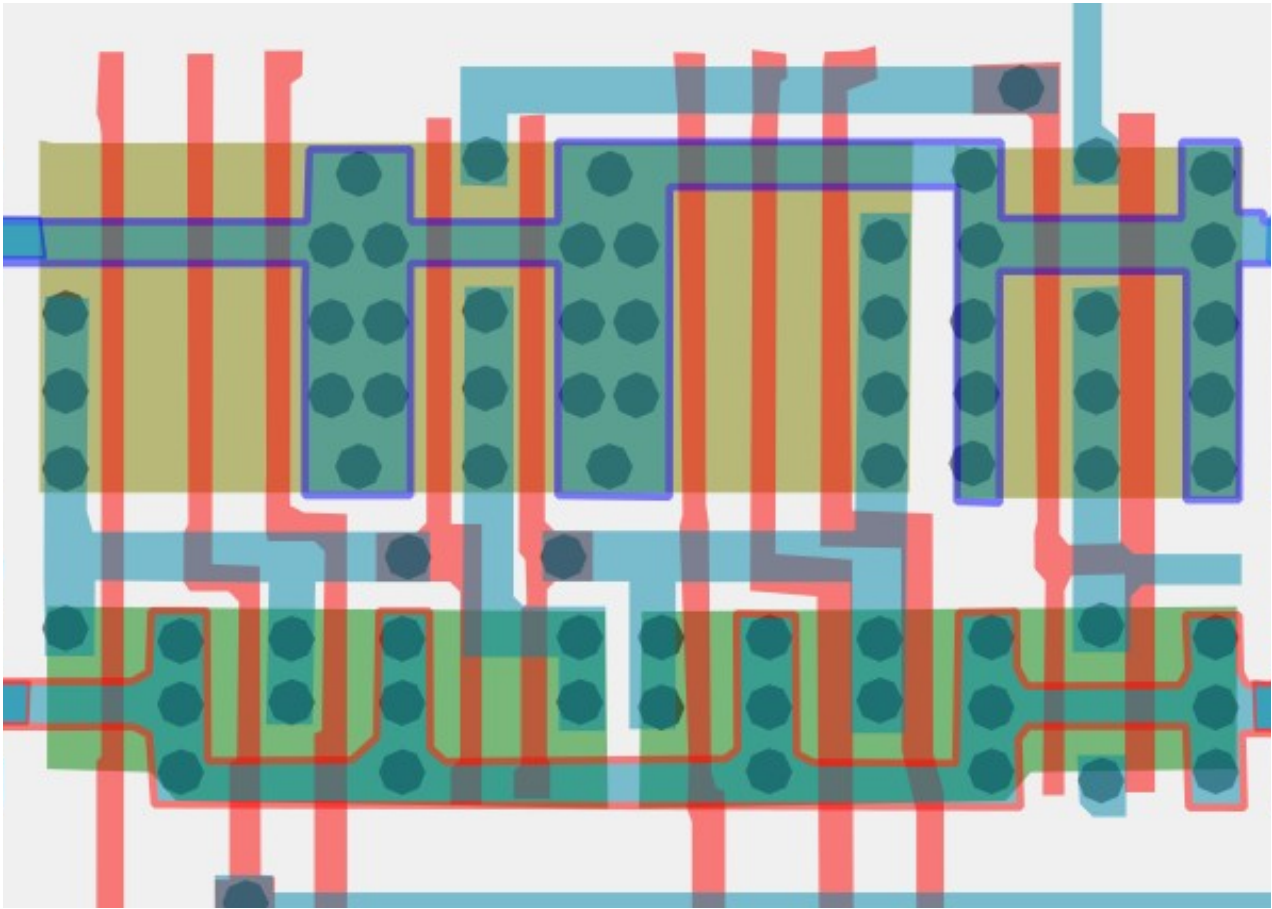
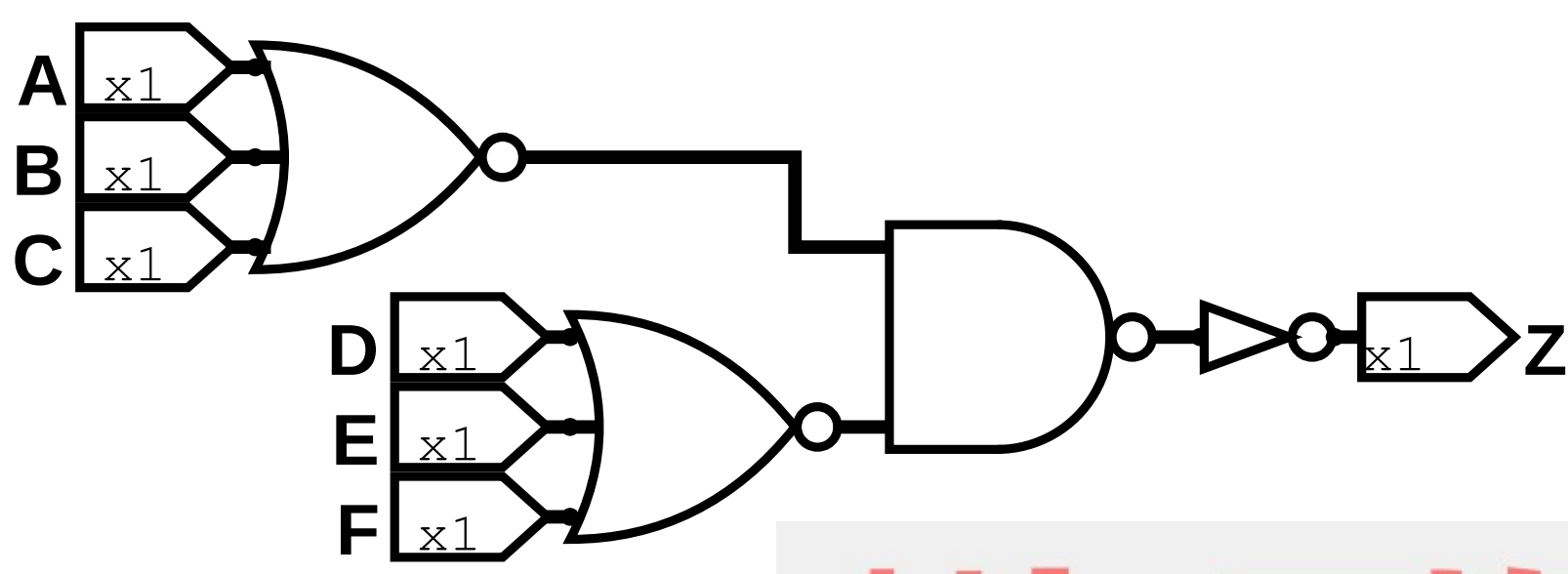
Cell type 35



#a09dd4ff

6-Input NOR

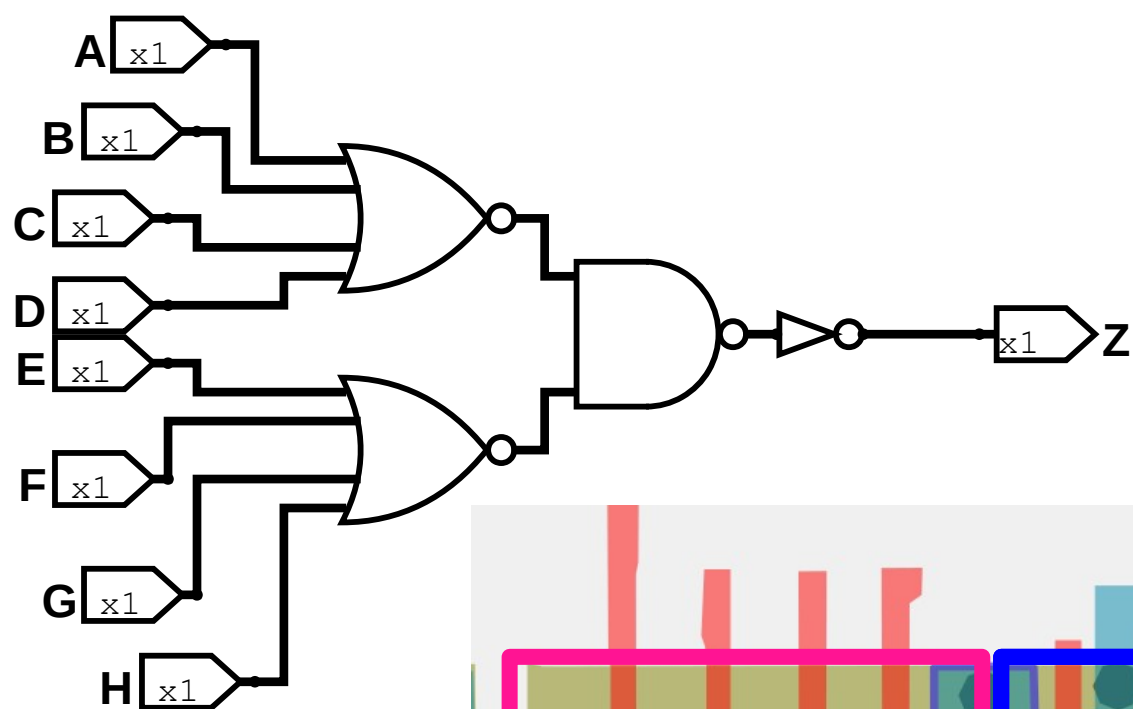
Cell 36a



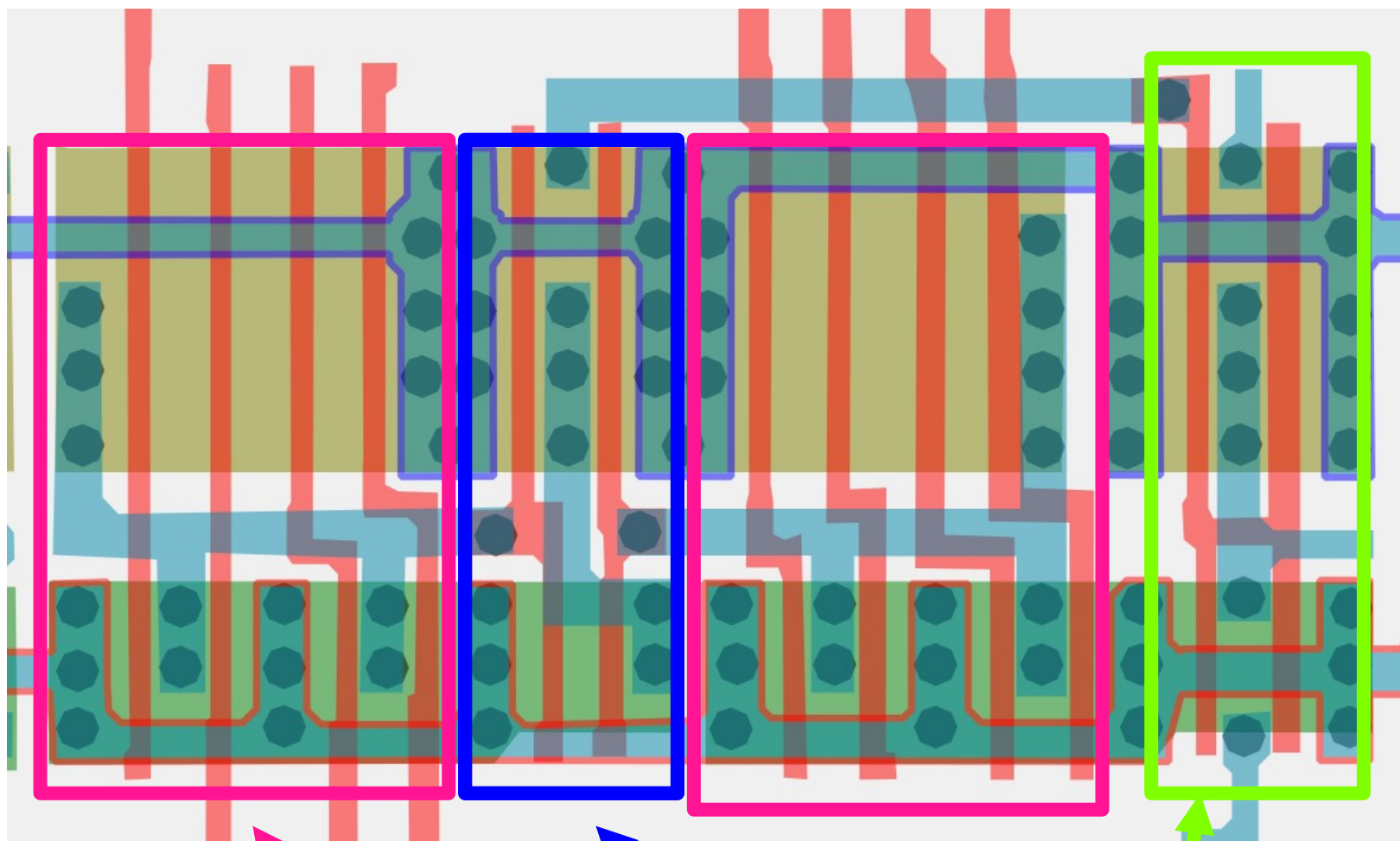
#b0aedbff

8-Input NOR

Cell 36b



I have verified
this one in
logisim



4 input NORs

2 input
NAND

Inverter
x2

